# Engineering AAV Tropism:

# DARPin-Mediated Retargeting Enables Specific Gene Delivery to Lymphocyte Subtypes

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Datum der Disputation:

*'The invention of CRISPR and the plague of COVID will hasten our transition to the third revolution of modern times. These revolutions arose from the discovery, beginning just over a century ago, of the three kernels of our existence:*

*the atom,*

*the bit*

*and the gene'*

- Walter Isaacson, The Code Breaker, 2021

Table of Content

Zusammenfassung....7

Summary....10

1 Introduction....13

1.1 Gene Therapy....13

1.2 Lymphocytes in Gene Therapy....15

1.3 T Cells....16

1.3.1 CD8: Structure, Isoforms, and Role in T-Cell Activation....17

1.3.2 CD8 T Cells in Gene Therapy....17

1.4 NK Cells....17

1.4.1 CD56: Structure, Function and Expression....18

1.4.2 NK Cells in Gene Therapy....19

1.5 Adeno-Associated Viral Vectors....19

1.5.1 AAV Serotypes and Capsid Engineering....22

1.6 Designed Ankyrin Repeat Proteins for AAV Retargeting....24

1.7 Aim....25

2 Results....26

2.1 CD56 Expression, Purification, and Preliminary Testing for Ribosome Display....26

2.1.1 Successful Expression and Purification of CD56....26

2.1.2 High Levels of Biotinylation....27

2.1.3 Finding the Optimal CD56 Concentration for Ribosome Display....28

2.1.4 Immobilized CD56 is Readily Detectable by Antibodies....28

2.2 Ribosome Display....29

2.3 Crude Lysate Screening....31

2.4 Binding Properties of Purified DARPins ..... 35
2.4.1 43H3 Exhibits Cross-Reactivity between Mouse and Human CD56 .... 37
2.5 DART-AAV Generation and Biophysical Characterisation ..... 38
2.5.1 DART-AAVs Retain an Unaltered AAV Core Structure ..... 40
2.6 CD8-AAVs are Highly Efficient on Target Receptor Positive Cell Lines 41
2.7 DART-AAVs Can Also Be Retargeted to Other T Cell Receptors ..... 43
2.8 Expanding Targets Beyond T Cells: Establishing CD56-AAV ..... 44
2.9 The DARPin-Retargeting Strategy is Adaptable to Other AAV Serotypes ..... 46
2.10 CD8-AAVs Selectively Transduce $CD8^+$ T-Cells *Ex Vivo.* ..... 47
2.10.1 CD4-AAV2 Demonstrates Lower Efficiency Compared to CD8-AAV2 . 50
2.10.2 CD8-AAV6 Achieves Potent Transduction While Preserving Cell Viability ..... 52
2.11 Optimization of CD56-AAV by Combining with NKp46-Specific DARPins ..... 54
2.12 Enhanced Transduction Efficiency with 2xCD56-AAV ..... 57
2.13 Translational Limitations of CD56-AAVs in Primary NK Cells ..... 59
2.14 The 43H3 DARPin Binds to Distal CD56 Epitope ..... 61
2.15 CD8-AAVs Outperform Unmodified AAV6 in Human Blood ..... 65
2.15.1 No Significant Difference in Resistance to Serum Inhibition ..... 67
2.16 CD8-AAVs Achieve Exceptional Selectivity in *In Vivo* ..... 67
2.16.1 T-Cell Activation is Crucial for CD8-AAV Transduction Efficiency ..... 70
3 Discussion ..... 73
3.1 Successful *De Novo* Selection of CD56-Specific DARPins ..... 73
3.1.1 Binding Properties of Purified DARPins ..... 75
3.2 Structural Implications of DARPin Incorporation into AAV Capsids ..... 76
3.3 Specific Transduction and Mechanistic Insights into DART-AAVs ..... 77
3.4 DART-AAV Retargeting Strategy is Compatible with AAV6 ..... 80

3.5 Enhanced Performance of DART-AAVs in Primary Cells ..... 81
3.6 Optimization of CD56-AAV ..... 83
3.7 CD56-AAVs: Major Challenges Remain ..... 87
3.8 CD8-AAVs Outperform Unmodified AAVs in Preclinical Settings ..... 88
3.9 Implications on AAV-based Gene Therapy of Lymphocytes ..... 89
3.9.1 CD8-AAVs: Precision Targeting for Next-Generation T Cell Gene Therapy... ..... 90

4 Material and Methods ..... 92

4.1 CD56 Expression, Purification and Pre-Tests for Ribosome Display ... 92
4.1.1 Molecular Cloning ..... 92
4.1.2 CD56 Protein Expression and Purification ..... 92
4.1.3 SDS-PAGE Analysis of Purified Protein ..... 93
4.1.4 Determination of Biotinylation ..... 93
4.1.5 Determination of Protein Amount Needed for Panning ..... 93
4.1.6 CD56 Antibody Binding ..... 94
4.2 Ribosome Display ..... 94
4.3 Crude Lysate Screening After Ribosome Display ..... 95
4.3.1 Production of *E. coli* Crude Lysates ..... 95
4.3.2 Validation of DARPin Expression ..... 95
4.3.3 Cellular Binding with Crude Lysates ..... 96
4.4 DARPin Expression and Purification ..... 96
4.4.1 Molecular Cloning ..... 96
4.4.2 DARPin Expression in *E. coli* ..... 96
4.4.3 DARPin Purification ..... 97
4.4.4 DARPin SDS-PAGE and Western Blots ..... 97
4.5 Generation of $CD56^+$ HeLa Cell Lines ..... 98
4.5.1 Molecular Cloning ..... 98

4.5.2 Generation of a VSV-Pseudotyped Lentiviral Vector ..... 98
4.5.3 Transduction and Selection of CD56$^{+}$ HeLa Cells ..... 98
4.6 DARPin Binding Assays ..... 99
4.6.1 Isolation und Cultivation of NK Cells ..... 99
4.6.2 NK Cell and HeLa DARPin Binding Assay ..... 99
4.7 Surface Plasmon Resonance Affinity Measurement ..... 99
4.8 DART-AAV Generation ..... 100
4.8.1 Molecular Cloning ..... 100
4.8.2 Production of AAVs ..... 101
4.9 Biophysicochemical Characterization of DART-AAVs ..... 102
4.9.1 AAV Genome Copy qPCR ..... 102
4.9.2 Western Blotting ..... 102
4.9.3 Transmission and Cryo Electron Microscopy ..... 102
4.10 Functional Characterisation of DART-AAVs ..... 103
4.10.1 *In Vitro* Transduction Experiments with Cell Lines ..... 103
4.10.2 *Ex Vivo* Transduction of Primary NK Cells ..... 104
4.10.3 *Ex Vivo* Transduction of Activated Human PBMC ..... 104
4.10.4 Transduction Inhibition by Human Serum ..... 105
4.10.5 Whole Blood Transduction Assay ..... 105
4.10.6 *In Vivo* Gene Transfer of CD8-DART-AAVs ..... 105
4.11 Immunostaining and Flow Cytometry ..... 106
4.12 ZMB Products Used ..... 107
4.13 Primers ..... 108
4.14 Flow Cytometry Antibodies ..... 109
4.15 Analysis, Visualization and Writing ..... 109
4.15.1 Illustrations ..... 109
4.15.2 Writing ..... 109

4.15.3 DARPin Sequence Alignment ..... 109
4.15.4 Protein and AAV Structure Visualizations ..... 109
4.15.5 Statistics and Graphing ..... 110
5 References ..... 111
6 Abbreviations ..... 127
7 List of Figures and Tables ..... 130
7.1 Figures ..... 130
7.2 Tables ..... 131
8 Curriculum vitae ..... 132
8.1 Awards & Scholarships ..... 133
9 List of publications ..... 134
10 Acknowledgments ..... 135
11 Declaration of Collaboration ..... 137
12 Declaration of an Oath ..... 139

## Zusammenfassung

Die Gentherapie hat in den letzten Jahrzehnten insbesondere durch den Einsatz von Adeno-assoziierten Viren als Vektoren (AAV) bedeutende Fortschritte gemacht. Diese Vektoren zeichnen sich durch geringe Immunogenität und ein günstiges Sicherheitsprofil aus und haben sich vor allem bei der Behandlung genetischer Erkrankungen wie Hämophilie, Muskeldystrophien und retinalen Dystrophien bewährt. Wie bei jedem innovativen therapeutischen Ansatz bleiben jedoch Herausforderungen bestehen – insbesondere die Optimierung der Vektorproduktion und die Minimierung potenzieller *Off-Target*-Effekte. Zuvorderst ist hier das Risiko unerwünschter Genexpression in für die Therapie irrelevanten Geweben zu nennen, die zu Immunreaktionen oder Toxizität führen kann. Gewebespezifische Gentherapie bietet das Potenzial, diese Risiken zu verringern, indem sie die Genübertragung auf therapierelevante Zelltypen beschränkt. Besonders interessant ist dies bei Erkrankungen, bei denen nur bestimmte Zellpopulationen betroffen sind – etwa bei Krebs oder Autoimmunerkrankungen. In solchen Fällen eröffnet die gezielte Modifikation einzelner Immunzelltypen neue therapeutische Möglichkeiten: Bei Tumorerkrankungen kann die präzise Genmodifikation zytotoxischer $CD8^+$ T-Zellen deren antitumorale Aktivität verstärken, während bei chronischen Entzündungen eine selektive Modulation von NK-Zellen therapeutisch wirksam sein kann. Die gezielte Modifikation der AAV-Kapsidproteine ist ein innovativer Ansatz, um spezifische Zelloberflächenmoleküle zu adressieren. Diese Dissertation beschreibt die erstmalige Erzeugung und Analyse von Designed Ankyrin Repeat Proteins (DARPins)-retargetierten AAV-Vektoren (DART-AAVs), die humane Lymphozyten gezielt transduzieren. Im Fokus stehen dabei zwei neu entwickelte Vektorklassen: CD8-DART-AAVs zur Genübertragung in $CD8^+$ T-Zellen und CD56-DART-AAVs für die spezifische Transduktion von NK-Zellen.

DARPins sind synthetische Proteine, die mit hoher Affinität spezifische Zielmoleküle binden können. In dieser Studie wurden DARPins in eine hypervariable Region (HVR) des GH2/GH3-Loop des VP1-Kapsidproteins der AAV-Serotypen 2 und 6 integriert. Diese Region wurde ausgewählt, da sie gut zugänglich ist und die Insertion großer Polypeptide in diesem Beriech die grundlegende Kapsidstruktur nicht beeinträchtigt. Die modifizierten Kapside wurden einer Vielzahl biochemischer Analysen unterzogen und durch hochauflösende Kryo-Elektronenmikroskopie konnte gezeigt werden, dass die grundlegende Kapsidstruktur unverändert blieb. In Zellkulturen und komplexen primären Zellmischungen, einschließlich humanem Vollblut, zeigten die CD8-DART-AAVs eine außergewöhnlich hohe Spezifität und Effizienz. Bis zu 80 % der $CD8^+$ T-Zellen wurden erfolgreich transduziert, während CD8-negative Zellen unbeeinträchtigt blieben. Unmodifizierte AAV2- und AAV6-Vektoren konnten hingegen nicht zwischen

Zellpopulationen unterscheiden und wiesen keine Selektivität für $CD8^+$ Zellen auf. Besonders deutlich wurde die Überlegenheit der CD8-DART-AAVs in humanem Vollblut, wo sie nicht nur gezielte Transduktion, sondern auch hohe Genübertragungsraten aufwiesen. Zwar transduzierte AAV6 isolierte und aktivierte T-Zellen effizient, seine Aktivität in Vollblut blieb jedoch deutlich hinter der von CD8-DART-AAVs zurück. Auch in humanisierten Mäusen, waren die CD8-DART-AAVs, unmodifizierten AAVs überlegen. Nach einer einzigen systemischen Injektion erreichten sie eine nahezu absolute Selektivität für $CD8^+$ T-Zellen, wobei die Gentransferrate bis zu 80 % betrug, wenn die T-Zellen vor Transplantation *ex vivo* aktiviert wurden. In nicht aktivierten T-Zellen war die Effizienz geringer, konnte jedoch durch eine *in vivo*-Konditionierung mit IL-2 und Anti-CD3-Antikörpern signifikant gesteigert werden.

Es wurde gezeigt, dass beide Serotypen, AAV2 und AAV6, durch DARPin-Modifikation erfolgreich auf $CD8^+$ T-Zellen targetiert werden können. Obwohl AAV6 natürlicherweise besser für die Transduktion von hämatopoetischen Zellen geeignet ist, erzielten beide Serotypen nach CD8-*Retargeting* vergleichbare Transduktionseffizienzen. Auffällig war zudem, dass die häufig beobachtete Zytotoxizität von unmodifiziertem AAV6 bei Verwendung von CD8-DART-AAV6 signifikant reduziert war. $CD8^+$ T-Zellen tolerierten eine um ein Vielfaches höhere Vektordosis, ohne dass negative Effekte auf die Zellviabilität auftraten.

Darüber hinaus untersuchte diese Dissertation die Machbarkeit von CD56-targetierten AAV-Vektoren für eine NK-Zell-spezifischen Gentransfer. Zu diesem Zweck wurden neuartige DARPins gegen CD56 (NCAM1) mithilfe von Ribosomen-Display *de novo* selektiert, funktionell charakterisiert und gezielt in den GH2/GH3-Loop des VP1-Kapsidproteins von AAV2 integriert. Um die Bindungseigenschaften und die Transduktionseffizienz weiter zu steigern, wurde zusätzlich eine bi-DARPin-Version (2xCD56) entwickelt, bei der zwei identische CD56-DARPins über einen flexiblen Peptidlinker innerhalb eines VP1-Moleküls gekoppelt wurden. Diese Vektoren zeigten deutlich verbesserte Bindungseigenschaften und Transduktionseffizienz in der klinisch relevanten NK-92-Zelllinie, verglichen mit der mono-DARPin-Variante und unmodifizierten AAV6-Vektoren. Während sich dieser Ansatz in der NK-92-Zelllinie als erfolgreich erwies, blieb die Transduktion primärer NK-Zellen eine Herausforderung.

Diese Studie leistet einen fundamentalen Beitrag zur Entwicklung rational targetierter AAV Kapside, indem sie die Vorteile eines rationalen Kapsiddesigns durch Peptidinsertion in HVRs im Vergleich zu bibliotheksbasierten Selektionsansätzen aufzeigte. Die Möglichkeit, Targeting-Liganden wie DARPins modular auszutauschen, ermöglicht eine schnelle

Anpassung von AAVs an verschiedene Immunzell-Subtypen, was ihr translationales Potenzial erheblich verbessert.

Im Rahmen dieser Arbeit konnte gezeigt werden, dass CD8α und CD56 als funktionelle Andockstellen für AAV-Vektoren genutzt werden können. Die Interaktion der CD8-DARPins mit der extrazellulären Domäne von CD8α ermöglichte eine hochselektive und effiziente Genübertragung in $CD8^+$ T-Zellen, ohne CD8-negative Zellpopulationen zu beeinträchtigen. Damit wurde eine zelltyp-spezifische Transduktion über ein definiertes Oberflächenmolekül realisiert, das nicht nur für therapeutische Anwendungen, sondern auch für grundlagenwissenschaftliche Fragestellungen relevant ist.

Zudem lieferten die verbesserten Eigenschaften der 2xCD56-Vektoren wertvolle Hinweise auf einen Aviditätseffekt der DART-AAVs sowie auf eine mögliche Rezeptorvernetzung durch die Bindung an homodimeres CD56 auf der Zelloberfläche. Die geringe Transduktionseffizienz in primären NK-Zellen deutet jedoch auf zelltypspezifische Barrieren im AAV-Verarbeitungspfad hin – etwa Unterschiede in den Endozytosemechanismen, in der Expression relevanter Eintrittsrezeptoren oder in der Effizienz des endosomalen Escape-Prozesses.

Im Hinblick auf die Anwendung stellt diese Arbeit einen bedeutenden Fortschritt in der Entwicklung immunzell-spezifischer AAV-Vektoren dar. CD8-DART-AAVs wurden als leistungsfähige Plattform zur zielgerichteten Modifikation von T-Zellen etabliert und bilden damit die Grundlage für zukünftige *in vivo*-Gentherapien, wie etwa zur vereinfachten Erzeugung von CAR-T-Zellen direkt im Patienten, wodurch aufwendige und kostenintensive *ex vivo*-Prozesse entfallen könnten.

Gleichzeitig wurden wichtige Grundlagen für die weiterführende Entwicklung sowohl NK-Zell- als auch $CD8^+$ T-Zell-spezifischer AAVs gelegt. Die gewonnenen Erkenntnisse zur CD56-Bindung, zur optimalen Ligandenanordnung und zu zelltypspezifischen Transduktionsunterschieden bei NK-Zellen sowie zur CD8-vermittelten Spezifität bei T-Zellen liefern wertvolle Ansatzpunkte für weitere Optimierungsstrategien – beispielsweise durch gezielte Modifikation der Kapsidstruktur, kombinatorische *Targeting*-Ansätze und den Einsatz verbesserter regulatorischer Elemente wie gewebespezifischer Promotoren.

Neben dem therapeutischen Nutzen bieten die hier entwickelten Vektoren auch ein vielversprechendes Werkzeug für die immunologische Grundlagenforschung, indem sie eine gezielte Genmodifikation definierter Immunzellsubtypen ermöglichen. Dadurch eröffnen sich neue Perspektiven für das Verständnis von T- und NK-Zellfunktionen und deren Rolle bei Tumorimmunität, Infektionen und Autoimmunerkrankungen.

## Summary

Gene therapy has made significant advances in recent decades, particularly through the use of adeno-associated viral vectors (AAVs). These vectors are well-established tools, characterized by low immunogenicity and a favourable safety profile. They have proven particularly effective in the treatment of genetic disorders such as haemophilia, muscular dystrophies, and retinal dystrophies. However, like all emerging therapeutic approaches, key challenges remain—most notably the optimization of vector production, the assurance of long-term safety, and the reduction of off-target effects. A particularly critical issue is the risk of undesired gene expression in non-target tissues, which can lead to side effects such as immune responses or toxicity.

Cell type-specific gene transfer is particularly relevant in diseases where only certain immune cell populations are affected—such as cancer or autoimmune disorders. In these cases, the targeted modification of specific immune cell types opens up new therapeutic opportunities: in cancer, precise genetic modification of cytotoxic $CD8^+$ T cells can enhance their antitumor activity, whereas in chronic inflammation, selective modulation of NK cells may provide therapeutic benefit.

This dissertation focuses on the development of immune cell-specific AAV-based gene therapy vectors, with a particular emphasis on $CD8^+$ T cells and NK cells. The core strategy involved the rational engineering of AAV capsids through insertion of high-affinity designed ankyrin repeat proteins (DARPins) into the GH2/GH3 loop of the VP1 protein, enabling receptor-specific retargeting.

A major achievement of this work was the successful development of CD8 and CD56-targeted AAVs by insertion of a CD8 or CD56-specific DARPins into a hypervariable region (HVR) within the GH2/GH3 loop of the capsid protein 1 (VP1) of AAV2 and AAV6. This resulted in precise and efficient transduction of their respective cell types. Remarkably, the capsid core structure was unaltered as confirmed by high-resolution cryogenic electron microscopy (cryo-EM). In complex primary cell mixtures, including donor blood or systemic injections into humanized mice, the CD8-targeted AAVs were by far superior to unmodified AAV2 and AAV6 in terms of selectivity, target cell viability and gene transfer rates. Up to 80 % of $CD8^+$ T cells were successfully transduced, while CD8-negative cells remained unaffected. In contrast, unmodified AAV2 and AAV6 vectors could not distinguish between cell populations and showed no selectivity for $CD8^+$ cells. The superiority of CD8-DART-AAVs was particularly evident in whole blood, where both selective transduction and high gene transfer rates were achieved. Although AAV6 efficiently transduced isolated, activated T cells, its performance in whole blood was clearly inferior to that of CD8-DART-AAVs. *In*

*vivo*, up to 80 % of activated $CD8^+$ T cells were hit upon a single vector injection into conditioned humanized mice. Selectivity for $CD8^+$ T cells was close to absolute. Transduction of resting T cells was less efficient but could be significantly improved by IL-2 and anti-CD3 *in vivo*-conditioning.

It was demonstrated that both AAV2 and AAV6 can be successfully retargeted to $CD8^+$ T cells through DARPin modification. While AAV6 naturally exhibits superior transduction efficiency in hematopoietic cells, AAV2 achieved comparable results when specifically engineered for CD8 targeting.

Furthermore, the cytotoxicity commonly associated with high-dose AAV6 administration was significantly reduced using CD8-DART-AAVs. T cells transduced with CD8-DART-AAVs tolerated significantly higher doses without any detectable negative impact on cell viability, highlighting the potential of this approach for safe and efficient gene delivery.

In parallel, CD56-DART-AAVs were developed to target NK cells. Novel CD56-binding DARPins were identified via ribosome display, and both mono- and bivalent (2xCD56) formats were inserted into the GH2/GH3 loop of AAV2. The 2xCD56 design, featuring two DARPins linked within a single VP1 protein, significantly enhanced transduction of NK-92 cells, likely due to avidity effects and engagement of homodimeric CD56. However, primary NK cell transduction remained limited, suggesting cell-type-specific barriers in AAV processing, such as endocytosis, receptor density, or endosomal escape efficiency.

This study provides a fundamental contribution to the development of rationally targeted AAV capsids by demonstrating the advantages of peptide insertion HVRs over library-based selection strategies. The ability to modularly exchange targeting ligands such as DARPins allows rapid adaptation of AAVs to different immune cell subsets, significantly improving their translational potential.

In addition, this study demonstrated that CD8α and CD56 can serve as a functional attachment receptor for AAV vectors. The interaction between CD8-specific DARPins and the extracellular domain of CD8α enabled highly selective and efficient gene transfer into $CD8^+$ T cells without affecting CD8-negative cell populations. This established cell type-specific transduction through a well-defined surface molecule—an approach relevant not only for therapeutic applications but also for fundamental immunological research.

Furthermore, the enhanced performance of the 2xCD56 vectors provided important insights into the avidity effects of DART-AAVs and suggested potential receptor clustering upon engagement with homodimeric CD56 on the cell surface. However, the limited transduction efficiency in primary NK cells indicated cell-type-specific barriers in the AAV processing

pathway—such as differences in endocytosis mechanisms, expression levels of relevant entry receptors, or the efficiency of endosomal escape.

From an application standpoint, this work marks a significant advance in the development of immune cell-specific AAV vectors. CD8-DART-AAVs were established as a robust platform for the targeted genetic modification of T cells and lay the groundwork for future *in vivo* gene therapies—such as the in-body generation of CAR-T cells, which could eliminate complex and costly *ex vivo* manufacturing steps.

At the same time, important groundwork was laid for further development of both NK cell- and $CD8^+$ T cell-specific AAVs. Insights gained into CD56 and CD8 targeting, optimal ligand configuration, and cell-type-specific differences in transduction offer valuable starting points for continued optimization—such as capsid redesign, combinatorial targeting, and the use of improved regulatory elements like tissue-specific promoters.

In addition to therapeutic applications, the vectors developed here offer a promising tool for basic immunology by enabling precise genetic modification of defined immune cell subtypes. This opens new avenues for understanding the roles of T and NK cells in tumour immunity, infection, and autoimmune diseases.

# 1 Introduction

## 1.1 Gene Therapy

Modern medicine is undergoing an unprecedented revolution. For the first time in human history, we are no longer limited to merely managing symptoms or supporting the body's natural healing processes—we now have the ability to target the root causes of disease at the genetic level. This transformation is being driven by the rapidly advancing field of gene therapy.

Gene therapy allows us to replace, modify, overexpress, or silence specific genes within our DNA, providing the potential to cure hereditary disorders, correct genetic defects, and even combat complex diseases like cancer at their molecular origin. What was once the realm of science fiction is now becoming reality, as researchers harness cutting-edge genetic engineering tools such as CRISPR, transposons, viral vectors, lipid nanoparticles (LNPs), and polymer-based gene delivery systems to develop precise, efficient, and personalized treatments, and in some cases, even curative therapies [1–3]. These viral and non-viral delivery platforms are revolutionizing gene therapy, enabling the correction of genetic defects, targeted cancer therapies, and regenerative medicine approaches with unprecedented precision.

This paradigm shift in medicine fundamentally redefines the way we approach disease, offering not just treatment, but the possibility of permanent cures for conditions once thought to be untreatable.

As of February 2025, the European Medicines Agency (EMA) has granted marketing authorization for fifteen gene therapy medicinal products (GTMPs) under the advanced therapy medicinal products (ATMPs) framework (Table 1).

Among these, six therapies are chimeric antigen receptor (CAR) T-cell therapies, which are *ex vivo* gene therapies where patients' T cells are genetically engineered to target specific cancer cells. These therapies have revolutionized the treatment landscape for hematologic malignancies, providing personalized and highly effective immunotherapy options [4].

Additionally, five of the approved *in vivo* gene therapies utilize adeno-associated viral vectors (AAVs) to deliver therapeutic genes directly into target cells within the patient's body. These AAV-based therapies have been particularly successful in treating monogenic disorders, such as inherited retinal diseases, spinal muscular atrophy, and haemophilia [5].

**Table 1: Market authorized GTMPs in the EU.**

| Trademark (INN) | Indication | *Ex vivo* or *in vivo* | Transducing agent | Transgene |
|---|---|---|---|---|
| Abecma (Idecabtagen vicleucel) | Multiple myeloma | *Ex vivo* | Lentiviral vector | αBCMA-CAR |
| Breyanzi (Lisocabtagen maraleucel) | B cell lymphoma | *Ex vivo* | Lentiviral vector | αCD19-CAR |
| Carvykti (ciltacabtagen autoleucel) | Multiple myeloma | *Ex vivo* | Lentiviral vector | αBCMA-CAR |
| Casgevy (Exagamglogene autotemcel) | beta thalassaemia and sickle cell disease | *Ex vivo* | Electroporation (CRISPR-Cas9) | - |
| Hemgenix (Etranacogen dezaparvovec) | Haemophilia A | *In vivo* | AAV5 | hFIX-Padua |
| Imlygic (Talimogen laherparepvec) | Melanoma | *In vivo* | Attenuated HSV-1 | GM-CSF |
| Kymriah (Tisagenlecleucel) | Acute lymphatic leukaemia<br>B cell lymphoma | *Ex vivo* | Lentiviral vector | αCD19-CAR |
| Libmeldy (Atidarsagene autotemcel) | Metachromatic leukodystrophy | *Ex vivo* | Lentiviral vector | ARSA |
| Luxturna (Voretigen Neparvovec) | Retinal dystrophy | *In vivo* | AAV2 | hRPE65 |
| Roctavian (Valoctocogen Roxaparvovec) | Haemophilia B | *In vivo* | AAV5 | hFVIII-SQ |
| Strimvelis | ADA-SCID | *Ex vivo* | Retroviral vector | ADA |
| Tecartus (brexucabtagene autoleucel) | Mantle cell lymphoma | *Ex vivo* | γ-Retroviral vector | αCD19-CAR |
| Upstaza (Eladocagene exuparvovec) | DDC deficiency | *In vivo* | AAV2 | DDC |
| Yescarta (Axicabtagen-Ciloleucel) | B cell lymphoma | *Ex vivo* | Retroviral vector | αCD19-CAR |
| Zolgensma (Onasemnogen-Abeparvovec) | SMA | *In vivo* | AAV9 | SMN |

Currently, AAV-based *in vivo* gene therapies and CAR T-cell therapies dominate the gene therapy market, highlighting the need for a deeper exploration of AAVs as a versatile *in vivo* gene delivery tool and immune cells, particularly T cells, as prime targets for genetic modification.

## 1.2 Lymphocytes in Gene Therapy

Lymphocytes, a key subset of the hematopoietic system, originate from hematopoietic stem cells (HSCs) in the bone marrow and play a central role in adaptive immunity. They are broadly categorized into T cells, B cells, and natural killer (NK) cells, each with distinct immune functions (Figure 1). T cells mediate cellular immunity, B cells produce antibodies for humoral immunity, and NK cells provide innate immune surveillance against infections and malignancies.

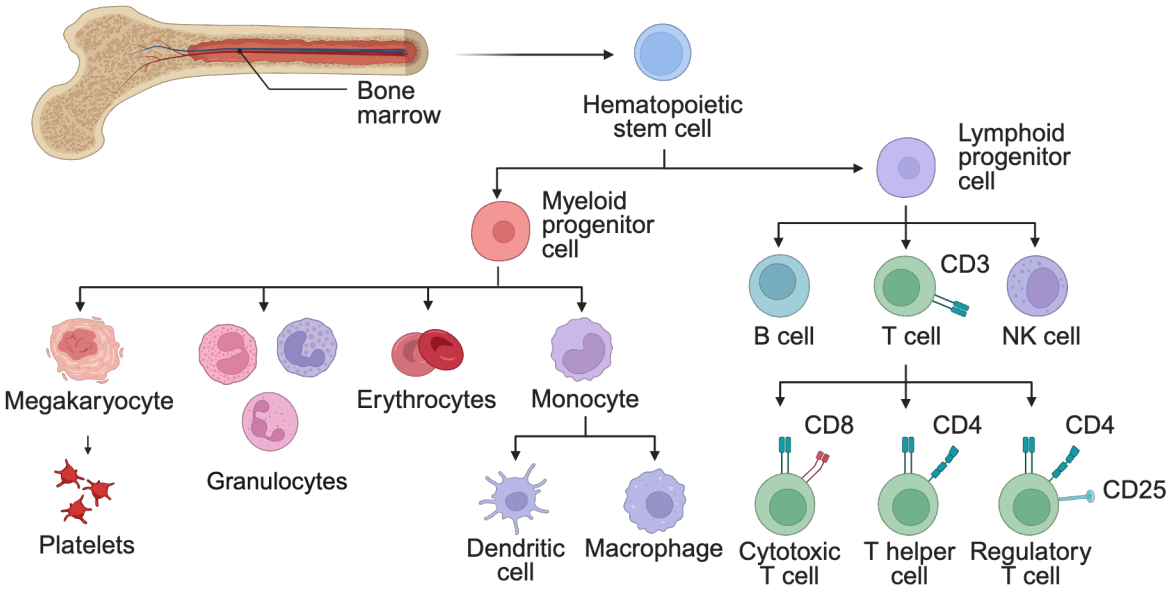



**Figure 1: Schematic representation of the haematopoietic system highlighting key T cell subtypes and their surface markers.**

In gene therapy, lymphocytes are highly attractive targets due to their pivotal role in immune regulation and disease intervention. Unlike terminally differentiated tissues, lymphocytes are highly dynamic, proliferative, and capable of trafficking throughout the body, making them an ideal vehicle for systemic therapeutic effects. Targeting lymphocytes directly *in vivo* offers a transformative alternative to current *ex vivo* gene modification approaches, such as CAR T-cell therapies, which require complex, labour-intensive, and expensive manufacturing processes [6].

The success of CAR T-cell therapy highlights the therapeutic relevance of genetically modified immune cells, particularly in the field of cancer immunotherapy. The approval of

six CAR T-cell therapies by the EMA underscores their clinical impact and growing acceptance as a mainstream treatment strategy [7].

Despite their efficacy, *ex vivo* CAR T-cell therapies require autologous cell harvesting, genetic modification, expansion, and reinfusion, making them logistically challenging, time-consuming, and costly. In contrast, *in vivo* gene therapy approaches could enable the direct genetic engineering of immune cells within the patient, eliminating the need for *ex vivo* manipulation while broadening accessibility and scalability [6].

Beyond cancer immunotherapy, genetically modified immune cells hold promise for autoimmune diseases [8], infectious diseases [9], and regenerative medicine [10]. For example, T-cell and NK-cell engineering could be leveraged to suppress autoreactive immune responses in autoimmunity [8], enhance pathogen clearance in chronic infections [11], or support tissue repair [10].

## 1.3 T Cells

T cells are a critical component of the adaptive immune system, playing a central role in immune surveillance, pathogen clearance, and tumour defence. They are characterized by their T-cell receptor (TCR), which recognizes antigenic peptides presented by major histocompatibility complex (MHC) molecules. T cells are broadly classified into $CD4^+$ helper T cells, $CD8^+$ cytotoxic T cells, and regulatory T cells (Tregs) based on their surface marker expression and function (Figure 1).

While $CD4^+$ T cells primarily coordinate immune responses by secreting cytokines and activating other immune cells, $CD8^+$ T cells act as direct effectors, recognizing and eliminating infected or malignant cells via cytotoxic mechanisms [12]. Tregs are a specialized subset of $CD4^+$ T cells, play a crucial role in immune tolerance by suppressing excessive immune responses, preventing autoimmunity, and maintaining immune homeostasis. Their immunosuppressive function makes them a promising target for therapeutic interventions in autoimmune diseases and transplantation [13].

All T cells express CD3, a co-receptor essential for TCR signalling [14]. Additional key markers include CD28, which provides co-stimulatory signals for activation [15], and lymphocyte function-associated antigen 1 (LFA-1), which facilitates adhesion to antigen-presenting cells [16]. $CD8^+$ T cells specifically express the CD8 co-receptor, which binds to MHC class I molecules on target cells, stabilizing the TCR-MHC interaction and enhancing antigen recognition [17]. Upon activation, $CD8^+$ T cells upregulate markers such as CD25 (IL-2 receptor) [18], CD69 [19], and programmed cell death protein 1 (PD-1) [20], reflecting their engagement in immune responses. Activated $CD8^+$ T cells differentiate into cytotoxic

T lymphocytes, which mediate target cell killing via perforin and granzyme release or Fas-FasL interactions [21].

### 1.3.1 CD8: Structure, Isoforms, and Role in T-Cell Activation

CD8 is a disulfide-linked dimeric glycoprotein found on the surface of cytotoxic T cells, dendritic cells [22], and some NK cell subsets [23,24]. It exists in two isoforms: CD8α and CD8β, which typically form CD8αβ heterodimers, although CD8αα homodimers can be found in some specialized T-cell subsets [17]. The extracellular domain of CD8 binds to the α3 domain of MHC class I, stabilizing the TCR-pMHC interaction and enhancing signal transduction. The intracellular tail of CD8 is associated with the tyrosine kinase Lck, which amplifies TCR signalling, promoting T-cell activation and cytotoxic function [17].

### 1.3.2 CD8 T Cells in Gene Therapy

$CD8^+$ T cells are particularly attractive targets for *in vivo* gene therapy due to their unique biological properties, therapeutic relevance, and widespread implications in immune regulation. First, their antigen specificity and cytotoxic function make them crucial for cancer immunotherapy, where engineered $CD8^+$ T cells can selectively target tumour cells while sparing healthy tissues. This has already been demonstrated in CAR T-cell therapies, which have revolutionized treatment for haematological malignancies [7].

Additionally, $CD8^+$ T cells play a key role in antiviral immunity, making them prime candidates for gene therapy approaches in chronic infections such as HIV and hepatitis B [11,25]. Modifying $CD8^+$ T cells *in vivo* would eliminate the need for *ex vivo* expansion and reinfusion, simplifying therapeutic workflows and making treatments more accessible [6]. Furthermore, CD8 is a relatively T-cell-specific marker, making it a suitable targeting receptor for gene delivery without significant off-target effects.

Another advantage $CD8^+$ T cells is their self-renewing and memory-forming capacity, allowing gene modifications to persist over long periods. Memory $CD8^+$ T cells can maintain protective immunity for years, which is highly beneficial for durable therapeutic effects in diseases requiring long-term immune modulation [26].

## 1.4 NK Cells

NK cells are a critical component of the innate immune system, specializing in immune surveillance, viral defence, and tumour cell elimination. Unlike T cells, NK cells do not rely on antigen-specific receptors but instead use a balance of activating and inhibitory receptors to recognize and eliminate stressed, infected, or malignant cells. They possess potent cytotoxic activity, mediated through perforin and granzyme-dependent mechanisms or death receptor pathways such as Fas-FasL and TRAIL-TRAILR interactions [27].

Additionally, NK cells secrete cytokines such as IFN-γ and TNF-α, influencing adaptive immune responses and shaping the inflammatory environment [28].

NK cells are traditionally classified by the absence of CD3 and the presence of CD56, which serves as their defining surface marker. CD56 expression defines functionally distinct NK cell subsets, with $CD56^{bright}$ NK cells being more responsive to soluble factors (such as cytokines) and $CD56^{dim}$ NK cells demonstrating superior cytotoxicity through direct cell contact. This phenotypic distinction is accompanied by differences in receptor expression, migratory behaviour, and activation response, contributing to their specialized roles in immune surveillance and response [29]:

Other key NK cell markers include CD16 (FcγRIIIa), which mediates antibody-dependent cellular cytotoxicity [30], NKG2D, an activating receptor that recognizes stress ligands [31], and NKp46 (NCR1), a natural cytotoxicity receptor that plays a crucial role in NK cell activation and tumour surveillance [32]. Additionally, inhibitory receptors such as killer-cell immunoglobulin-like receptors (KIRs) and NKG2A regulate self-tolerance by preventing NK cells from attacking healthy, MHC class I-expressing cells [33]. However, in some cases, NKG2A can also suppress NK cell-mediated tumour killing, as certain cancers exploit HLA-E expression to engage NKG2A and evade immune surveillance, effectively inhibiting NK cell cytotoxicity [34].

### 1.4.1 CD56: Structure, Function and Expression

The ectodomain of CD56, also known as neural cell adhesion molecule 1 (NCAM1), is composed of five immunoglobulin-like (Ig) domains and two fibronectin type III (FnIII) domains, which mediate cell adhesion, migration, and immune synapse formation [35]. Unlike other NK cell receptors directly involved in cytotoxic signalling, the CD56 ectodomain functions primarily as a cell adhesion molecule, facilitating NK-target cell interactions and modulating immune responses. Despite variations in the cytoplasmic domain across different CD56 isoforms, the extracellular domain is highly conserved across species, with near-identical homology between human, mouse, and other vertebrates [36]. This structural conservation suggests an evolutionarily preserved function in cell-cell adhesion and immune regulation, making it a robust target for cross-species studies and gene therapy applications.

CD56 represents a promising target for NK cell-directed gene therapy due to its consistent expression across NK cell subsets, including both $CD56^{bright}$ and $CD56^{dim}$ populations, and its relatively limited expression in non-haematopoietic tissues. However, its expression on neurons, glial cells, and skeletal muscle necessitates careful consideration for *in vivo* targeting strategies to avoid off-target effects [35,37]. Using tissue-specific promoters, such

as NCR1 (NKp46) for NK cells or granzyme B (GZMB) for cytotoxic NK cells, can ensure that transgene expression remains NK cell-restricted [38,39].

### 1.4.2 NK Cells in Gene Therapy

Like T cells, NK cells are highly promising candidates for *in vivo* gene therapy, particularly in cancer immunotherapy [40], infectious disease treatment [41], and autoimmune modulation [42]. Their innate ability to recognize and eliminate abnormal cells without prior antigen sensitization makes them faster-acting than T cells, providing broad and immediate immune surveillance [43]. Additionally, NK cells are equally valuable for *ex vivo* gene therapy. Unlike T-cell-based therapies, which require strict MHC matching, NK cells operate independently of MHC molecules, allowing for off-the-shelf allogeneic applications without the risk of graft-versus-host disease [44].

Another key advantage of NK cells in gene therapy is their innate ability to recognize and eliminate tumours without prior antigen priming, making them particularly effective in targeting solid tumours where CAR T cells have struggled [45]. NK cells have already been explored in CAR-NK therapies, which combine CAR-T-like antigen specificity with the natural cytotoxic properties of NK cells, and an *in vivo* approach could further enhance their therapeutic potential by eliminating the need for *ex vivo* expansion and reinfusion.

Furthermore, the $CD56^{+}$ NK-92 cell line is widely utilized in several pre-clinical and clinical trials as an off-the-shelf option for cancer therapy [46,47]. NK-92 cells are frequently subjected to extensive genetic modification, making the development of safe and efficient gene transfer methods in these cells particularly relevant for next-generation NK cell-based therapies.

## 1.5 Adeno-Associated Viral Vectors

Adeno-associated viral vectors (AAVs) have become one of the most widely used and versatile gene delivery tools in modern gene therapy. Originally derived from the adeno-associated virus, a small, non-enveloped, single-stranded DNA virus classified as a *Dependoparvovirus*, this virus is naturally replication-deficient and requires the presence of a helper virus, such as adenovirus or herpesvirus, for productive infection [48]. Importantly, numerous naturally occurring AAV serotypes have been identified across various species, each with distinct tissue tropisms and cellular entry pathways, allowing for tailored vector development [49]. Despite isolated reports linking wild-type adeno-associated virus integration to hepatocellular carcinoma in the context of insertional mutagenesis [50], it is not associated with any known human disease, which underscores its general safety and supports its continued use as a gene delivery vector [5,51].

Throughout this thesis, the abbreviation AAV refers exclusively to the recombinant gene therapy vector and not to the wild-type virus. Recombinant AAVs have been genetically modified by removing all viral coding sequences and replacing them with a therapeutic transgene and regulatory elements, making it a highly adaptable platform for a wide range of gene therapy applications, including monogenic disorders, neurological diseases, and cancer therapy [5].

The key advantages of AAVs include their ability to efficiently transduce both dividing and non-dividing cells, their low immunogenicity, and their potential for long-term gene expression in post-mitotic tissues such as the brain, liver, and muscle [5,51]. AAV entry into host cells is a multi-step process that begins with the virus binding to specific attachment receptors on the cell surface, such as heparan sulfate proteoglycans (HSPG), sialic acids, or other glycan structures depending on the AAV serotype [52]. These attachment receptors mediate the initial adhesion of the virus to the cell membrane, increasing local viral concentration and facilitating subsequent engagement with essential entry receptors.

Following attachment, AAV particles are typically internalized via clathrin-mediated endocytosis, although other uptake routes can also occur depending on the cell type and capsid composition [53,54]. Once internalized, AAV traffics through the endosomal system, where it undergoes escape from late endosomes or lysosomes to reach the cytoplasm [54]. A critical step in this process involves the interaction with the AAV receptor (AAVR; also named KIAA0319L), a transmembrane protein essential for productive infection across multiple serotypes. AAVR facilitates intracellular trafficking and likely plays a key role in mediating endosomal escape and delivery of the capsid to the perinuclear region [55–57]. After successful trafficking, the viral particle undergoes uncoating, releasing its single-stranded DNA genome, which must undergo second-strand synthesis or annealing of self-complementary sequences to enable transgene expression (Figure 2) [54]. Unlike integrating viral vectors, AAV transgenes primarily remain episomal, significantly reducing the risk of insertional mutagenesis [58].

However, AAV-based gene therapy does come with limitations. The most notable challenges include a limited packaging capacity (~4.7 kb), which restricts the size of the therapeutic payload [59], the presence of pre-existing neutralizing antibodies, which can reduce transduction efficiency in patients with prior AAV exposure and the transduction of non-target tissue [60,61]. Untargeted gene delivery poses a significant risk of off-target gene expression in non-intended tissues, potentially leading to adverse effects or toxicity [62,63].

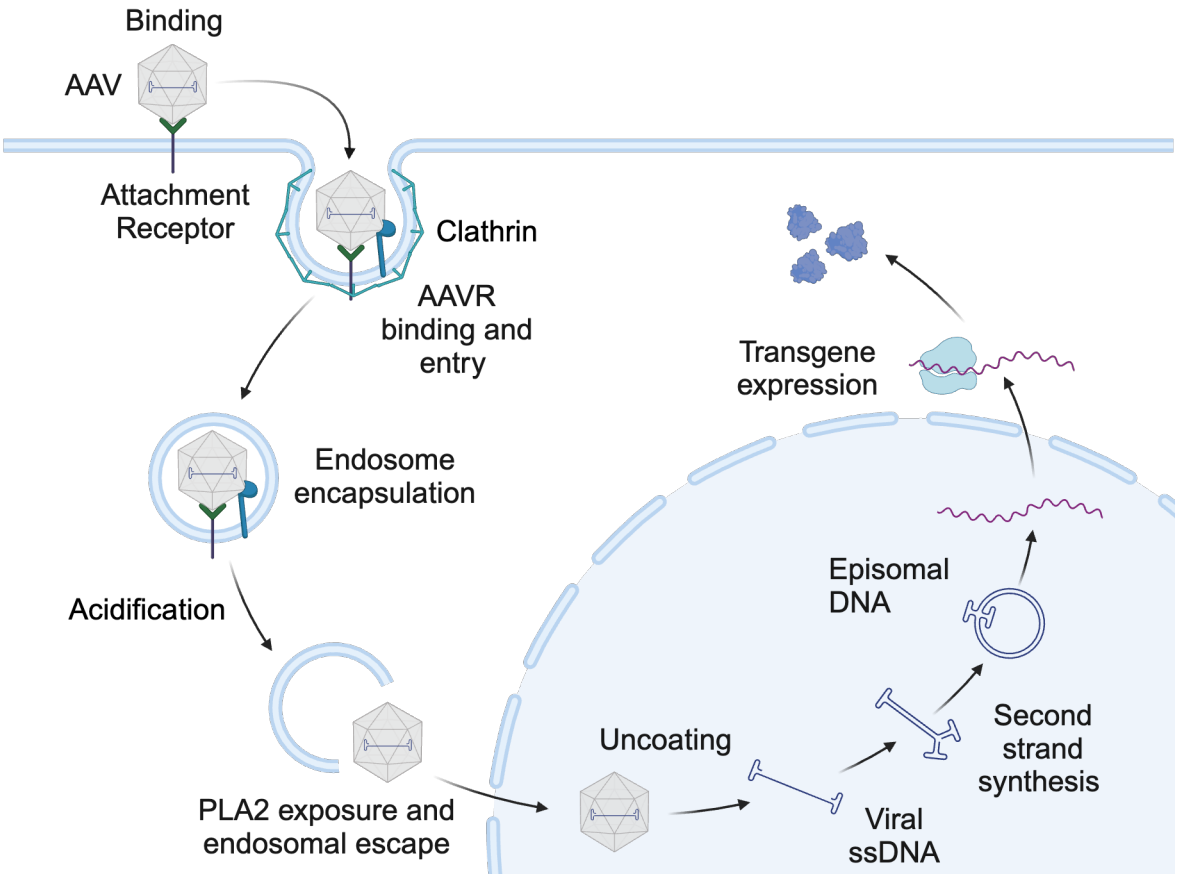



**Figure 2: Schematic representation of the AAV transduction process.**

Additionally, transgene expression in antigen-presenting cells can elicit immune responses against the therapeutic product, potentially reducing efficacy and compromising long-term treatment outcomes [64].

Importantly, many genetic and acquired diseases, including inherited disorders, autoimmune conditions, and cancer, affect only specific tissues or cell populations. Achieving highly selective transgene expression is therefore critical to improving therapeutic efficacy while minimizing systemic toxicity. Addressing these challenges through capsid engineering and improved vector designs will be crucial for expanding the clinical utility of AAV gene therapies.

AAV capsids exhibit icosahedral symmetry, consisting of 20 equilateral triangular faces, 12 vertices, and 30 edges, a structural feature that is essential for capsid stability and uniformity [65]. Modifications must be carefully designed to maintain capsid integrity.

The AAV capsid is composed of three structurally related viral proteins: VP1, VP2, and VP3, which assemble into a complete particle in an approximate ratio of 5:5:50 (VP1:VP2:VP3) [66]. All VP proteins share a common C-terminal core region, while VP1 possesses the largest N-terminal extension, containing phospholipase A2 (PLA2) activity,

which is critical for endosomal escape and intracellular trafficking [54]. VP3, the most abundant and smallest capsid protein, serves as the primary structural framework for capsid assembly. However, due to the stochastic nature of capsid formation, individual particles within an AAV stock often exhibit unique stochiospatial configurations, deviating from the theoretical ratio [67].

A crucial feature for capsid engineering and receptor interactions lies in the protrusions at the three-fold symmetry axis, which harbour hypervariable regions (HVRs). These highly divergent regions differ among AAV serotypes and are key determinants of tissue tropism, host-cell interactions, and immunogenicity [68]. Modifications within the HVRs have been extensively leveraged in capsid engineering, enabling enhanced targeting and transduction efficiency while preserving overall capsid integrity and stability [69–71].

### 1.5.1 AAV Serotypes and Capsid Engineering

AAVs encompass a variety of natural serotypes, each exhibiting distinct tissue tropisms that make them valuable for different gene therapy applications. Among these, AAV2, AAV5, AAV6, AAV8, and AAV9, as well as primate-derived AAVrh10 and AAVrh74, have been widely studied and applied in clinical research [49,72]. AAV9 is particularly efficient in muscle transduction and is one of the few serotypes capable of crossing the blood-brain barrier (BBB), making it highly suitable for neuromuscular and neurological disorders [73–75]. AAV8 is among the most effective vectors for liver-directed gene therapy, frequently used in treatments for metabolic disorders and haemophilia [76,77]. AAV2 has a well-characterized affinity for retinal cells, making it the preferred choice for ocular gene therapies [78]. Both AAV5 and AAV6 exhibit strong liver tropism, but AAV6 also demonstrates enhanced lung transduction and is widely regarded as the preferred serotype for lymphocyte transduction [79–82]. Meanwhile, AAVrh10 and AAVrh74, originally derived from non-human primates, have shown strong potential for central nervous system (CNS)-directed therapies, particularly in the treatment of neurodegenerative or neuromuscular diseases [83,84].

Thus far, only naturally occurring AAV serotypes have been utilized in EMA-approved gene therapies (Table 1). AAV2 is the vector used in Luxturna (voretigene neparvovec) for retinal dystrophy and Upstaza (eladocagene exuparvovec) for AADC deficiency [78,85]. AAV5 serves as the delivery vector in Roctavian (valoctocogene roxaparvovec) and Hemgenix (etranacogene dezaparvovec), both approved for the treatment of haemophilia A and B, respectively [82]. AAV9, known for its high neuronal tropism and ability to penetrate the BBB, is the vector of choice in Zolgensma (onasemnogene abeparvovec), the first approved gene therapy for spinal muscular atrophy (SMA) [86].

Despite the broad utility of natural serotypes, their limitations in immune evasion, tissue specificity, and transduction efficiency have driven the development of engineered AAV capsids. These next-generation vectors are designed to overcome pre-existing immunity, enhance cell-specific targeting, and optimize gene delivery efficiency, expanding the therapeutic potential of AAV-based gene therapies [51].

One approach to improving AAV vectors is directed evolution, where large libraries of AAV variants are generated and screened for superior transduction properties. This method has led to significant breakthroughs, such as AAV-PHP.B and AAV-PHP.eB, which exhibit dramatically improved blood-brain barrier penetration [87,88], and AAV-LK03, optimized for human hepatocyte transduction [89]. In addition to random mutagenesis, rational design strategies have been applied by modifying known structural elements of the capsid to enhance receptor binding, reduce immunogenicity, or improve intracellular trafficking. This method has resulted in AAV2.5, which balances improved muscle transduction with reduced binding to heparan sulfate proteoglycans, decreasing off-target effects [90].

Hybrid capsid design has further expanded the potential of AAV vectors by combining sequences from different serotypes to create chimeric variants with improved infectivity and tissue targeting. AAV-DJ, a synthetic capsid derived from AAV2, AAV8, and AAV9, exemplifies this approach, demonstrating enhanced transduction across multiple tissues compared to its wild-type counterparts [91].

Another powerful strategy in AAV capsid engineering is peptide display, where short targeting peptides are inserted into exposed capsid loops to enhance cell-specific tropism and receptor interactions. This approach has been successfully utilized to generate AAV variants with improved transduction efficiency and selectivity. A notable example is AAV2.GL, which enables widespread, high-efficiency retinal transduction [92].

Among rational capsid engineering strategies, one previously explored approach involves attaching targeting ligands to the N-terminus of the VP2 capsid protein [93,94]. However, a key limitation of this strategy lies in the natural orientation of VP2: in the native AAV capsid, the N-terminus of VP2 is located internally and only becomes surface-exposed through partial capsid rearrangement during cell entry. As a result, fusions at this site can adversely affect capsid assembly, ligand presentation, reduce particle stability, and significantly lower vector titers.

In contrast, recent studies have identified the HVR within the GH2/GH3 loop as a highly promising insertion site for engineered ligands, such as designed ankyrin repeat proteins (DARPins) and nanobodies [69,70]. Modifications at this site have been shown to significantly enhance targeted gene transfer efficiency while maintaining capsid integrity

and high production yields, addressing many of the limitations of the VP2-based approach [95]. Unlike traditional directed evolution approaches, which rely on random mutagenesis and selection, this rational design strategy allows for the precise incorporation of targeting ligands, ensuring that AAV vectors preferentially transduce specific cell populations.

Additionally, rationally designed AAV capsids can be engineered to disrupt or mask interactions with their natural receptors, effectively preventing transduction of non-target tissues and improving therapeutic specificity [96]. This approach is superior to other capsid engineering methods, as it allows for greater control over tropism and avoids the labour-intensive nature of large-scale library screening. By systematically modifying the GH2/GH3 loop, this strategy may enable the development of next-generation AAVs with enhanced precision, efficiency, and safety, advancing the potential of targeted gene therapy.

## 1.6 Designed Ankyrin Repeat Proteins for AAV Retargeting

DARPins are a class of synthetic binding proteins engineered from naturally occurring ankyrin repeat motifs, which are one of the most abundant structural domains in nature [97,98]. These repeat motifs enable DARPins to form rigid, elongated scaffolds with a high degree of structural stability, modularity, and specificity, making them an ideal tool for AAV retargeting [93]. Unlike antibodies or nanobodies, which often rely on disulfide bonds and complex folding, DARPins are composed of compact, rigid α-helical structures that ensure high stability [99], efficient expression, and optimal integration into AAV capsids. Their small size (~14-18 kDa) allows seamless insertion into AAV capsid loops, such as the GH2/GH3 region, without disrupting capsid assembly or viral infectivity [69,96].

One of the major advantages of DARPins over nanobodies is their structural compatibility with AAV capsids. While nanobodies retain β-sheet-rich antibody-like structures, which can lead to folding challenges and steric hindrance, DARPins are fully α-helical and easily integrate into capsid proteins, preserving capsid integrity while enhancing targeting specificity. Additionally, DARPins lack an Fc region, eliminating the risk of unwanted immune activation and making them safer for *in vivo* applications.

Beyond their structural advantages, DARPins enable multiplexed targeting strategies by allowing simultaneous engagement of multiple receptors, which can enhance cellular binding specificity and functional outcomes [100]. Their capacity to be engineered into bivalent or bispecific constructs expands their utility for precision-targeting applications in biotechnology and therapeutics.

In contrast to traditional directed evolution approaches, which rely on random mutagenesis, DARPins offer a rationally designed approach for retargeting AAVs to specific cell populations, increasing transduction efficiency while minimizing off-target effects. With their

unmatched stability, modularity, and compatibility with AAV capsids, DARPins will influence the future of AAV-based gene therapy by enabling precise receptor-mediated targeting, making them the optimal solution for next-generation *in vivo* gene delivery systems.

## 1.7 Aim

The aim of this dissertation was to develop immune cell-specific AAV-based gene therapy vectors, with a particular focus on $CD8^+$ T cells and NK cells. To achieve precise and efficient *in vivo* immune cell targeting and provide advanced tools for *ex vivo* immune cell editing, this study employed DARPins for receptor-specific AAV retargeting

A key objective was the generation of CD8-targeted AAVs, enabling selective gene delivery to cytotoxic T lymphocytes—a crucial cell population in cancer immunotherapy and the control of infectious diseases. In this context, the study also addressed fundamental mechanistic and biological questions, including whether the extracellular domain of CD8α could function as an effective entry receptor for retargeted AAVs, how T-cell activation impacts AAV transduction efficiency, and how CD8-DART-AAVs compare to other DART-AAV variants.

Additionally, this thesis focused on the selection and validation of novel DARPins targeting CD56 to facilitate the generation of CD56-specific AAVs for NK cell-directed gene therapies. Mechanistic investigations examined whether combining CD56-binding DARPins—either with themselves or with additional receptor-specific DARPins—could enhance vector binding and uptake and how these interactions influenced transduction efficiency in both immortalized cell lines and primary NK cells.

By integrating capsid engineering with DARPin-based receptor targeting, this work aimed to develop highly specific, efficient, and clinically relevant AAVs, laying the foundation for next-generation gene therapy approaches in immune modulation, cancer immunotherapy, and infectious disease treatment.

# 2 Results

## 2.1 CD56 Expression, Purification, and Preliminary Testing for Ribosome Display

The first essential step in developing receptor-targeted gene therapy vectors is the identification of a suitable cell surface receptor. In this study, CD56 was selected as the primary target for NK cell transduction due to its high expression on NK cells and its relevance for translational applications. This was part of a separate approach conducted in parallel with CD8-based T cell targeting. For the successful selection of high-affinity binders via ribosome display, it was critical to produce a natively folded, biotinylated protein of high purity. To this end—and to potentially broaden the applicability of selected DARPins across species—the ectodomain of murine CD56 (mCD56) (UniProt P13595; residues 20–711), which shares approximately 95 % sequence identity with human CD56 (huCD56), was synthesized *de novo* and fused to a human Fc (huFc)-Avi tag, enabling efficient purification and enzymatic biotinylation.

### 2.1.1 Successful Expression and Purification of CD56

The mCD56-huFc-Avi protein was expressed in 293F suspension cells. Biotinylation was achieved by co-transfecting the cells with plasmids encoding soluble and membrane-bound biotin ligases in biotin-supplemented medium, as described for CD8 before [101]. The expressed proteins were purified from the cell culture supernatant using Protein A Sepharose beads, followed by size exclusion chromatography (SEC). SEC chromatograms revealed minimal impurities, with a single distinct peak for both, the biotinylated and unbiotinylated mCD56 protein (Figure 3A). Further analysis by SDS-PAGE and Western blot confirmed these results, showing distinct bands migrating above 130 kDa (Figure 3B). The calculated molecular weight of the ectodomain of the mCD56 polypeptide (77 kDa), combined with the huFc-Avi tag (28 kDa), is approximately 105 kDa. Assuming 12 glycosylation sites (Asn222, Asn316, Asn348, Asn424, Asn450, Asn479, Thr876, Thr882, Ser911, Thr950, Thr952, Thr980 [102–104]), with an estimated glycosylation contribution of 2 kDa per site, an additional 24 kDa is added, resulting in a total molecular weight of approximately 129 kDa. However, the observed molecular weight by SDS-PAGE (Figure 3B) was above 130 kDa. This aligns with the migration pattern of commercially available recombinant mCD56 (ECD, hFc tag) protein expressed in HEK293 cells (SinoBiological, 56977-M02H), which migrates at approximately 140 kDa, suggesting glycosylation levels higher than initially estimated.

### 2.1.2 High Levels of Biotinylation

The degree of mCD56 biotinylation was evaluated using a plate-based colorimetric assay and compared against biotinylated and unbiotinylated standard proteins. The assay confirmed that the biotinylated mCD56 protein achieved a biotinylation efficiency of 90-100 %, comparable to the fully biotinylated standard, making it suitable for use in ribosome display. As anticipated, the unbiotinylated mCD56 protein showed no detectable biotinylation. (Figure 3C).

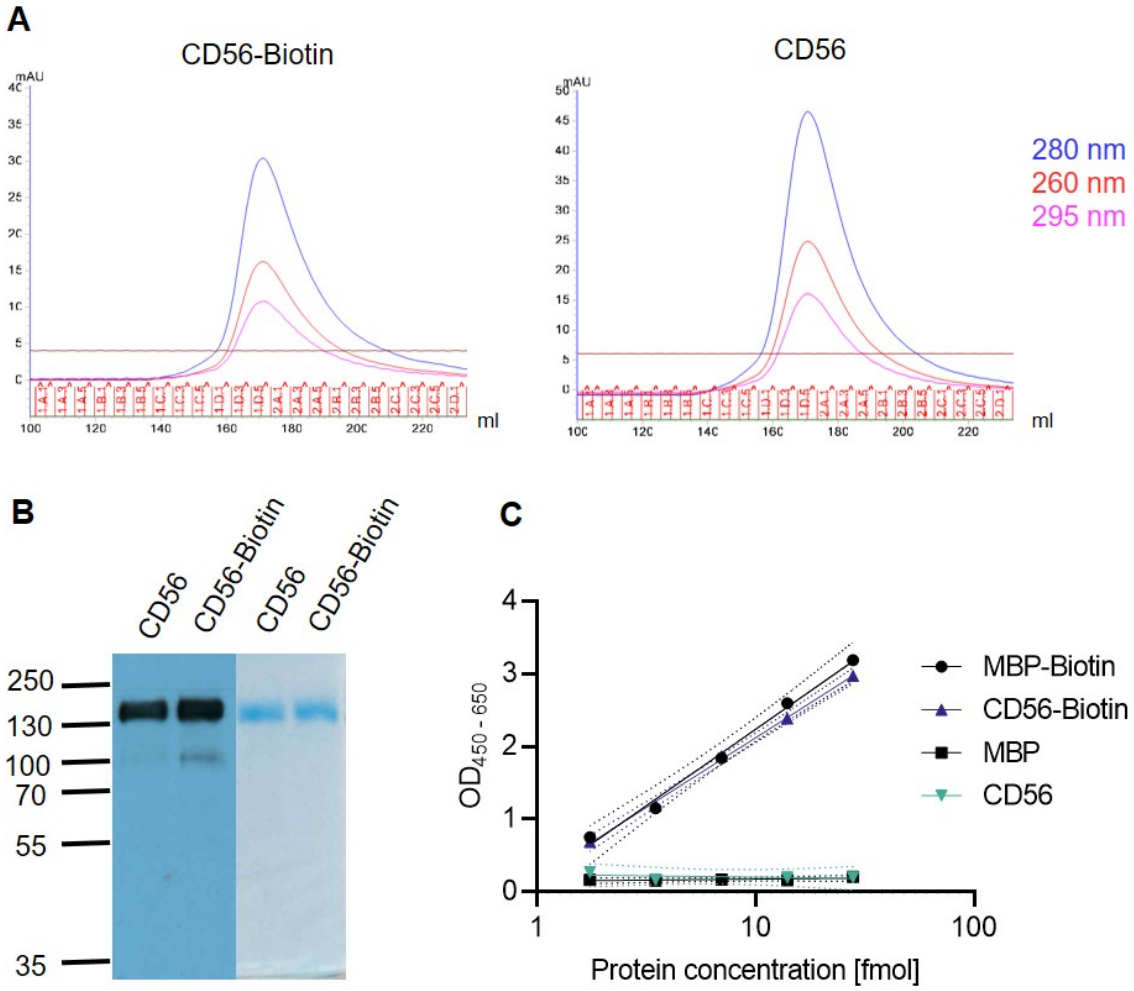



**Figure 3: CD56 purity and biotinylation. (A)** SEC of the Ni-NTA purified biotinylated and unbiotinylated mCD56 proteins with a HiLoad® 26/600 Superdex® 200 prep grade column. **(B)** SDS-PAGE of the purified mCD56 proteins. Both biotinylated und unbiotinylated mCD56 were loaded on a 10 % Tris-glycine gel. Direct protein detection with PageBlue™ Protein Staining Solution (right) and with an anti-huIgG-Fc-HRP antibody after blotting on a nitrocellulose membrane (left). **(C)** Determination of biotinylation of mCD56 compared to unbiotinylated standard protein (MBP) and a fully biotinylated standard protein (MBP-Biotin). Plate based colorimetric assay with streptavidin-HRP as detecting agent. Linear semilogarithmic interpolation with 95 % confidence interval (CI).

### 2.1.3 Finding the Optimal CD56 Concentration for Ribosome Display

To determine the optimal concentration of mCD56 for panning in ribosome display, an ELISA was performed. Biotinylated mCD56 as well as biotinylated and unbiotinylated huFc-Avi standard proteins were immobilized on neutravidin-coated 96-well plates in 1:1 serial dilutions ranging from 160 nM to 0.08 nM. Detection with anti-human IgG Fc HRP and its substrate showed that the $OD_{450-650}$ for biotinylated CD56 increased with concentration, reaching a plateau at 10 nM. Higher concentrations did not further enhance the signal. Controls with unbiotinylated mCD56 and both biotinylated and unbiotinylated standards showed only minor signals above background without concentration dependency. The biotinylated standard exhibited a concentration-dependent signal increase similar to biotinylated mCD56, confirming comparable immobilization and detection behaviour (Figure 4A). Based on these results, 10 nM was selected as the optimal concentration for panning in ribosome display

This represents a significantly lower protein requirement compared to previous DARPin selections, where 40 nM protein solutions were used for GluA4, NKp46, and CD105 [105], 425 nM for CD32a [106], and 50 nM for CD8αβ [101]. Considering the high degree of biotinylation (90–100 %) and the larger size of CD56, this lower concentration requirement is not surprising. For comparison, the GluA4 protein used in prior selections had a biotinylation efficiency of only 76 % and a molecular weight of 70–80 kDa [105], necessitating higher protein concentrations to saturate the plate.

### 2.1.4 Immobilized CD56 is Readily Detectable by Antibodies

To verify that immobilized mCD56 can be recognized by antibodies an additional ELISA was conducted. Both biotinylated and unbiotinylated mCD56 proteins were directly coated onto a plate and successfully detected using a mCD56-specific antibody. However, when immobilized via neutravidin, only biotinylated mCD56 was detectable, as expected (Figure 4B). This confirms that biotinylated mCD56 retains proper orientation and accessibility when immobilized via neutravidin.

Together, these findings validate the successful expression, purification, and biotinylation of mCD56-huFc-Avi, establishing a robust foundation for subsequent binder selection through ribosome display

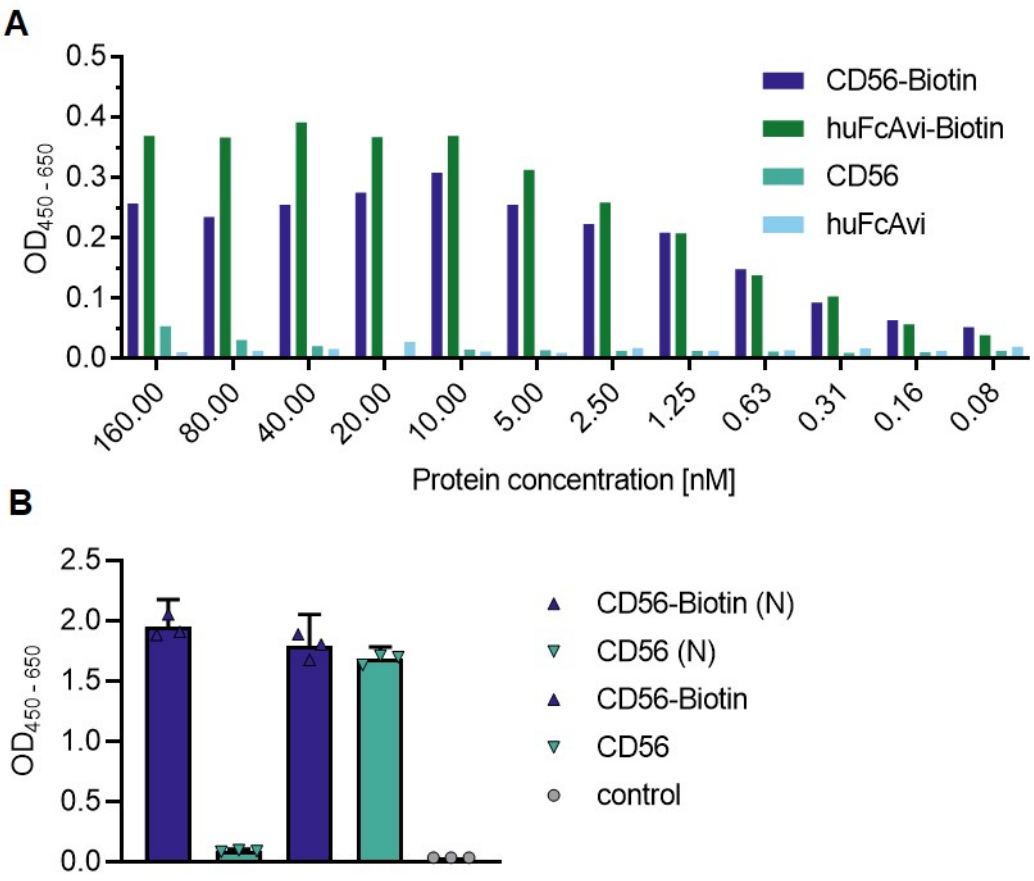
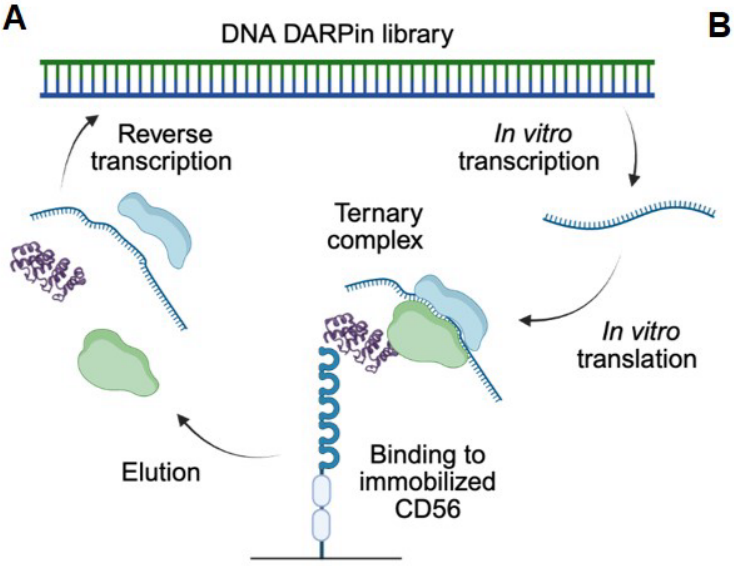



**Figure 4: Determination of CD56 protein amount needed for panning and antibody binding. (A)** Two-fold serial dilutions of biotinylated and unbiotinylated mCD56, along with a huFc-Avi standard protein, were immobilized on a neutravidin-coated plate. The dilution range covered concentrations from 160 nM to 0.08 nM. Detection was performed using an anti-human IgG Fc HRP-conjugated antibody, followed by colorimetric read-out with SureBlue™ TMB 1-Component Microwell Peroxidase Substrate. **(B)** mCD56 Antibody Recognition: Biotinylated or unbiotinylated mCD56 (10 nM) was either directly coated onto a plate or immobilized on neutravidin. Detection was performed using the anti-mCD56 antibody EPR21827, followed by an HRP-conjugated anti-rabbit IgG-Fc antibody. Colorimetric read-out was achieved using SureBlue™ TMB 1-Component Microwell Peroxidase Substrate. The experiment was conducted in technical triplicates, and error bars represent the 95 % CI.

## 2.2 Ribosome Display

The ribosome display was performed with the purified mCD56 proteins in five selection rounds with 2-3 different DARPin DNA libraries, as depicted in Figure 5A. Two DARPin libraries consisting of two (N2C) and three (N3C) diversified repeats in-between the constant N- and C-capping modules (kindly provided by Jessica Hartmann, PEI [105]) and one additional N3C library (kindly provided by Markus A. Seeger, University of Zürich [107]) referred to as Seeger library. cDNA recovery after each round was verified via agarose gel electrophoreses.

A




| Round | Selection steps |
|---|---|
| 1 | Pre-panning: Neutravidin<br>Target: 10 nM CD56 |
| 2 | Pre-panning: Streptavidin<br>Target: 10 nM CD56 |
| 3 | Pre-panning: Neutravidin<br>Target: 10 nM CD56 |
| 4 | Pre-panning: Streptavidin<br>Target: 1.95 pmol CD56<br>Off-Rate: 195 pmol CD56<br>5 min |
| 5 | Pre-panning: Neutravidin<br>Target: 1.95 pmol CD56<br>Off-Rate: 195 pmol CD56<br>1 h |

Cloning in pQE-HisHA bacterial expression backbone

B



**Figure 5: Ribosome display. (A)** Schematic overview of the DARPin screening process using ribosome display. **(B)** Agarose gel images illustrating cDNA recovery after each round of ribosome display, displayed sequentially from top to bottom. Lanes are labelled as follows: 1. reverse transcribed and amplified N2C library; 2. non-reverse transcribed N2C library control; 3. reverse transcribed and amplified N3C library; 4. non-reverse transcribed N3C library control; 5. reverse transcribed and amplified Seeger library; 6. non-reverse transcribed Seeger library control. White arrows highlight visible PCR products, indicating successful amplification.

cDNA was successfully recovered from the N2C and N3C starting libraries across all rounds of ribosome display, though band intensity in agarose gels varied slightly from round to round—particularly in early cycles when target-specific binders were still rare. As expected, enrichment of high-affinity binders over successive rounds led to stronger PCR signals and more intense bands. In contrast, the Seeger library yielded no detectable cDNA recovery after the first round (Figure 5B) and was therefore excluded from further analysis. Following the fourth and fifth rounds, output libraries from N2C and N3C were cloned into the pQE-HisHA bacterial expression plasmid to enable crude lysate screening for DARPin binder identification.

## 2.3 Crude Lysate Screening

For the production of DARPin crude lysates, the pQE-HisHA-N2C and pQE-HisHA-N3C libraries were transformed into *E. coli*. From the output library of the fifth round of ribosome display, 282 clones were selected, and DARPins were expressed in small culture volumes of 1,100–1,300 µL per clone. To verify DARPin expression, two-thirds of the clones from each library were analysed using a plate-based colorimetric assay alongside positive and negative controls. Detection of DARPins was performed using a His-tag-specific antibody as depicted in Figure 6A. Encouragingly, most tested crude lysates contained detectable His-tagged protein, with $OD_{450\text{-}600}$ values significantly above the background of the negative control (Figure 6B).

All 282 crude lysates were evaluated for their binding properties to the $CD56^{+}$ murine glioblastoma cell line GL261. Equal volumes of each lysate were incubated with the cells, followed by detection using a His-tag-specific antibody, as previously used for expression control. Most crude lysates failed to bind the cells reproducibly and efficiently. Only four lysates demonstrated binding to more than 5 % of the cells, and among these, only clone 533E12 caused a distinct shift of the entire cell population (Figure 6C). This outcome aligns with expectations, considering the uniform expression of the target receptor on the cells (Figure 6D), which should result in consistent binding across the population.

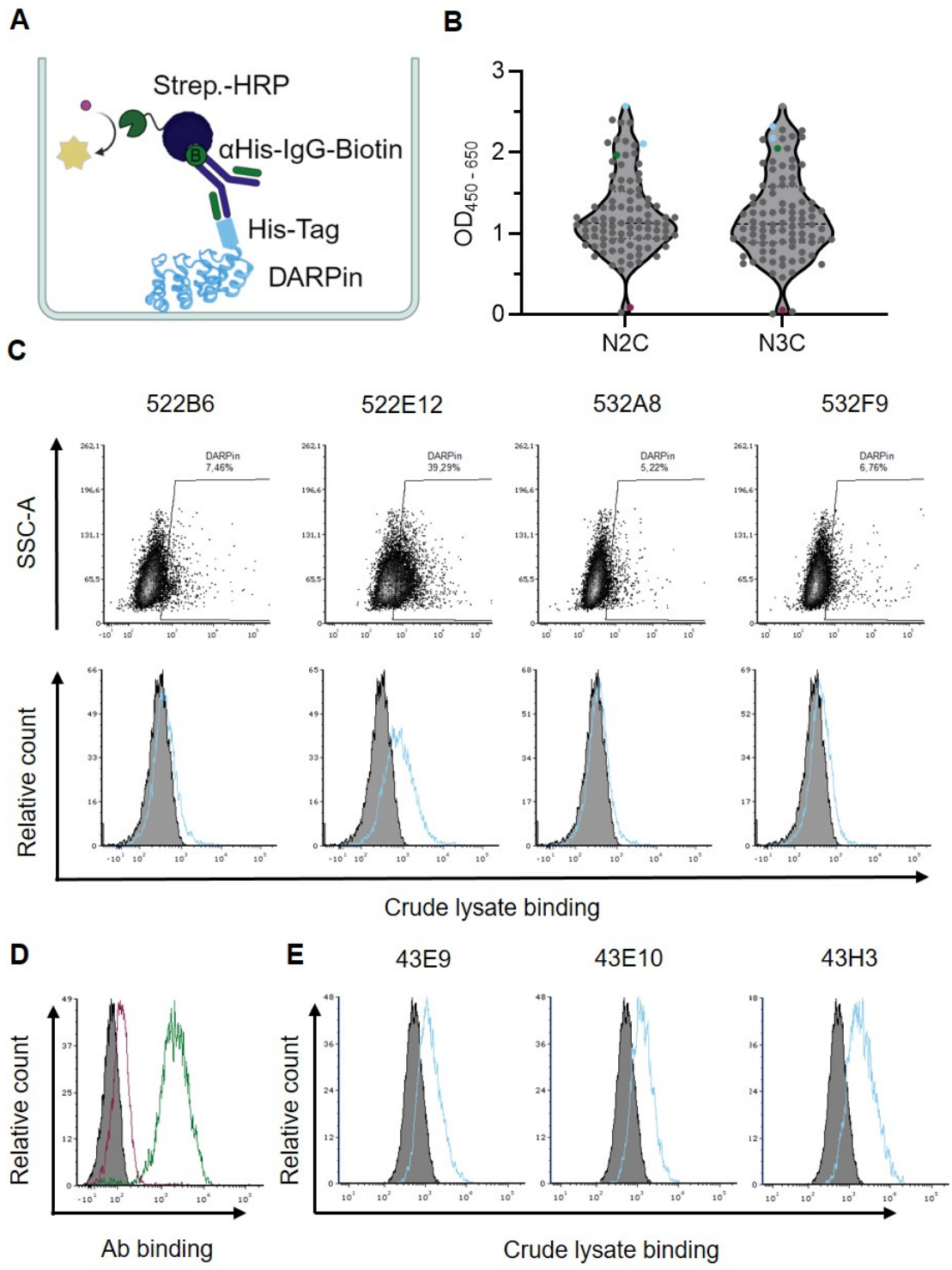



**Figure 6: Crude lysate screening after ribosome display. (A)** Schematic representation of the DARPin detection process using a plate-based colorimetric assay. Detection was achieved with an anti-Penta His biotin-conjugated antibody, followed by streptavidin-HRP. **(B)** Violin plot showing DARPin expression levels from a single batch of N2C and N3C crude lysates. Red dots represent the negative control (*E. coli* crude lysate without DARPin), green dots indicate the positive control (DARPin from prior screening, kindly provided by Vanessa Riechert, PEI), and blue dots correspond to crude lysates with binding efficiencies exceeding 5 % to target cells, as detailed in (C). **(C-E)** Binding analysis of crude lysates or an anti-CD56 antibody to viable GL261 single cells was performed via flow cytometry.

**(C)** DARPins from the fifth round ribosome display output library were detected using a PE-labeled anti-HA antibody. Density plots (top) illustrate fluorescence intensity distributions, while histograms (bottom) compare DARPin binding (blue) to background levels. Only the highest-binding candidates are shown. **(D)** Validation of CD56 expression on GL261 cells was performed using the anti-CD56 antibody EPR21827 (green), with comparison to an IgG isotype control (red) and background binding of the secondary antibody (grey). **(E)** Binding analysis of DARPins from the fourth round of the ribosome display output library, detected as described in (C). Data from (E) adapted from Lea Knapp.

Nevertheless, all four binding candidates were sent for Sanger sequencing (Eurofins Genomics). Sequencing revealed that the 532F9 clone's sequence was non-alignable with the DARPin template library, while 532A8 aligned only up to the N-Cap sequence, suggesting significant unwanted mutations in both clones. The remaining two clones, 522B6 and 522E12, had fully alignable sequences but contained single base pair deletions at different positions, leading to frame shifts. Interestingly, although the mutations were 98 base pairs apart, both resulted in similar truncated amino acid sequences. This was verified through an amino acid alignment with the N2C DARPin consensus sequence (Figure 7A).

Since it was not possible to identify a single functional DARPin clone from the 282 crude lysates screened from the output library after the fifth round of ribosome display, my bachelor student, Lea Knapp, conducted a second screening under my supervision. This screening included 190 crude lysates from the output library after the fourth round of ribosome display. This resulted in the identification of three DARPins (43E9, 43E10, and 43H3) with strong binding properties to murine GL261 cells (Figure 6E). Among these, two DARPins (43E10 and 43H3) shared 98.09 % sequence identity, with 43H3 exhibiting significantly stronger binding (Figure 7B-C). Consequently, 43H3 was selected for further experiments alongside the structurally less related DARPin, 43E9.

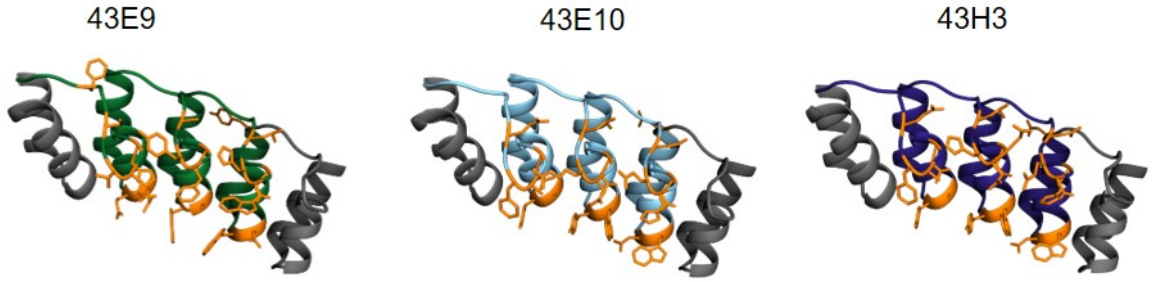
**A**

```
N2C:      DLGKKLLEAARAGQDDEVRILMANGADVNAXDXXGXTPLHLAAXXGHLEIVEVLLKXGAD  60
522B6     DLGKKLLEAARAGQDDEVRILMANGAD--------------------------------  27
522E12    DLGKKLLEAARAGQDDEVRILVANGADVNAVDDAGETPLHLAAQSGHLEIVEVLLKHGAD  60
          *****

N2C:      VNAXDXXGXTPLHLAAXXGHLEIVEVLLKXGADVNAQDKFGKTPFDLAIDNGNEDIAEVL  120
522B6     VNAHDPKVLHRCI----WQHLKVIWKLLKCC*---------------------------  54
522E12    VNAHDLLVLHRCI----WQQLRVIWKLLKCC*---------------------------  87


N2C:      QKAA* 124
522B6     ----- 54
522E12    ----- 87
```

**B**




**C**

```
N3C:      DLGKKLLEAARAGQDDEVRILMANGADVNAXDXXGXTPLHLAAXXGHLEIVEVLLKXGAD   60
43E9      DLGKKLLEAARAGQDDEVRILMANGADFNASDTEGHTPLHLAAQEGHLEIVEVLLKYGAD   60
43E10     DLGKKLLEAARAGQDDEVRILMANGADVNAQDFAGLTPLHLAAAYGHLEIVEVLLKYGAD   60
43H3      DLGKKLLEAARAGQNDEVRILMANGADVNAQDFAGLTPLHLAAAYGHLEIVEVLLKYGAD   60
          **************:************.** *  * *******  *********** ***

N3C:      VNAXDXXGXTPLHLAAXXGHLEIVEVLLKXGADVNAXDXXGXTPLHLAAXXGHLEIVEVL   120
43E9      VNAADYFGATPLHLAAAWCHLEIVEVLLKYGADVNALDHWGDTPLHLAADWGHLEIVEVL   120
43E10     VNALDRNGLTPLHLAAWYGHLEIVEVLLKYGADVNANDEHGDTPLHLAAWDGHLEIVEVL   120
43H3      VNALDHNDLTPLHLAAWYGHLEIVEVLLKYGADVNANDEHGDTPLHLAAWDGHLEIVEVL   120
          *** *   . *******   ********** ****** *  * *******  *********

N3C:      LKXGADVNAQDKFGKTPFDLAIDNGNEDIAEVLQKAA  157
43E9      LKYGADVNAQDKFGKTPFDLAIDNGNEDIAEVLQKAA  157
43E10     LKNGADVNAQDKFGKTPFDLAIDNGNEDIAEVLQKAA  157
43H3      LKNGADVNAQDKFGKTPFDLAIDNGNEDIAEVLQKAA  157
          ** ********************************
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**Figure 7: DARPin sequences and structures**. Amino acid alignment of cellular binders with the N2C or N3C DARPin consensus sequence using Clustal Omega [3]. Mutations are highlighted in red, variable regions in yellow, and cap sequences in grey. **(A)** Sequence comparison of clone 522B6 and 522E12 **(B)** Structure of the 43E9 DARPin (green), 43 E10 (light blue), and 43H3 (dark blue) with the variable amino acids highlighted in orange. **(C)** Sequence comparison of clone 43E9, 43E10 and 43H4.

## 2.4 Binding Properties of Purified DARPins

The sequences of 43H3 and 43E9 DARPins were cloned into the pET32a plasmid for efficient protein expression in a bacterial system. Following expression in *E. coli* BL21, the DARPins were purified via their His-tags using Ni-NTA agarose and SEC, as shown in Figure 8A. The purity of the proteins was confirmed through SDS-PAGE and Western blot analysis. Interestingly, although both DARPins have the same number of variable repeats and a calculated molecular weight of approximately 21 kDa, they migrated differently on SDS-PAGE. The 43E9 bands appeared at the expected ~21 kDa, while 43H3 migrated closer to ~15 kDa (Figure 8C-D). This discrepancy in migration behaviour was observed only under denaturing conditions; during SEC with natively folded proteins, both DARPins eluted at similar times under identical chromatography system settings (Figure 8B). Despite this anomaly in the SDS-PAGE, both DARPins were expressed in high yields of 400–700 µg per 100 mL of bacterial culture and high purity after SEC was achieved.

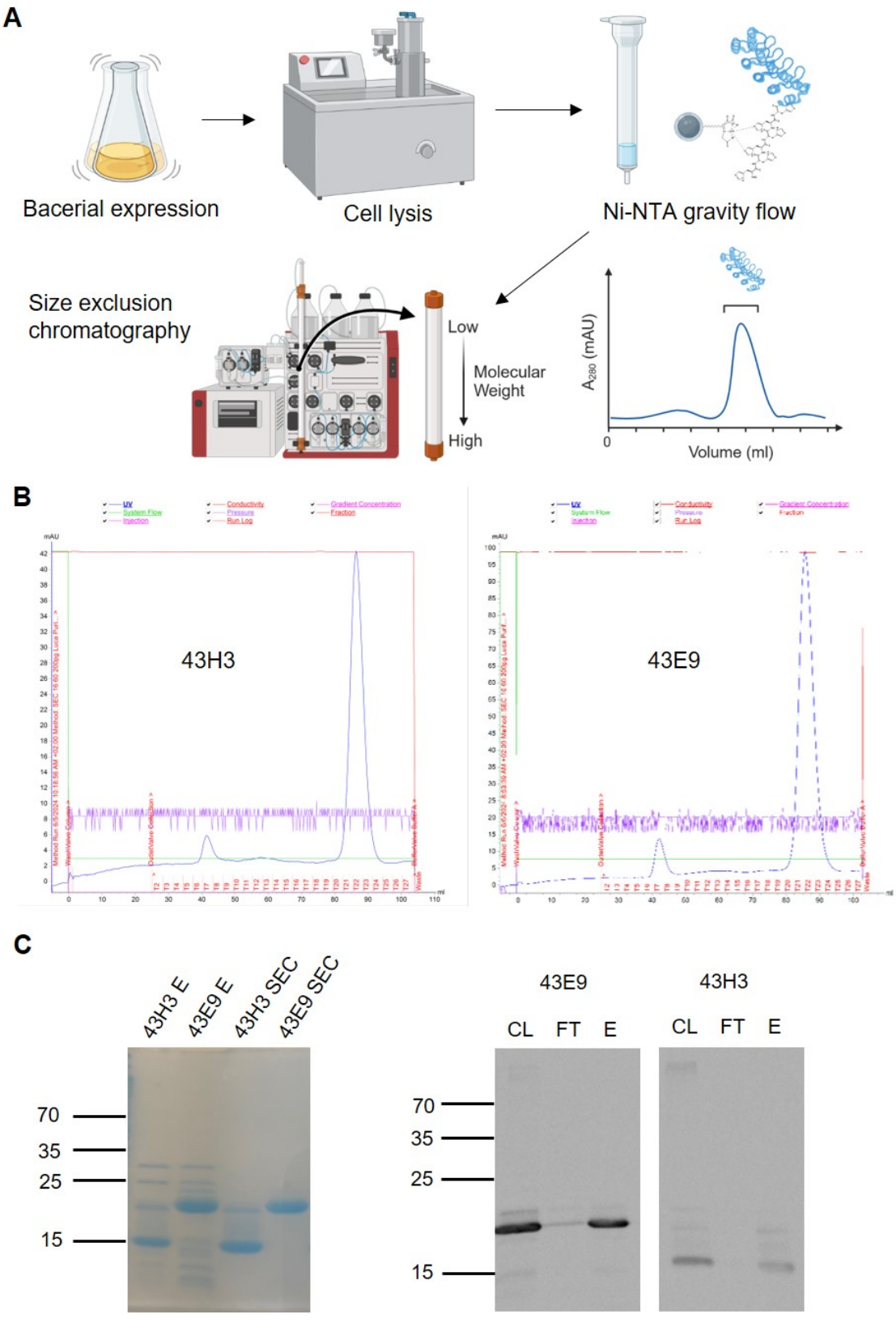



**Figure 8: DARPin purification. (A)** Schematic representation of the DARPin purification workflow. **(B)** SEC chromatograms of Ni-NTA purified DARPins 43H3 (left) and 43E9 (right) with a HiLoad™ 16/60 Superdex™ 200 prep grade column. **(C-D)** SDS-PAGE and Western blot analysis of purified 43E9 and 43H3 DARPins. Purified DARPins before (43E9 E, 43H3 E) and after size exclusion chromatography (43E9 SEC, 43H3 SEC) were analysed on a

12 % SDS-PAGE gel and stained with GelCode™ Blue Safe Protein Stain. **(C)** SDS-PAGE showing protein separation and purity, and **(D)** Western blot of samples transferred to a nitrocellulose membrane. Samples include crude lysate (CL), flow-through from Ni-NTA purification (FT), and eluted fractions from Ni-NTA purification (E). Detection was carried out using an anti-RGS-His antibody.

### 2.4.1 43H3 Exhibits Cross-Reactivity between Mouse and Human CD56

Since the DARPin crude lysates had been tested on $CD56^+$ mouse cells, the cross-reactivity of the purified DARPins with human CD56 was assessed next by testing them on primary human NK cells. Only 43H3 demonstrated reliable binding to human NK cells (Figure 9A). To further characterize its binding properties, an affinity analysis of 43H3 using surface plasmon resonance (SPR) was performed. All tested binding models consistently yielded a dissociation constant ($K_D$) of 11–15 nM for 43H3 binding to human CD56 (Figure 9B and Table 2). The 1:1 binding model describes a straightforward interaction where one ligand binds to one analyte, governed by a single set of rate constants ($k_a$ and $k_d$). In contrast, two-state binding involves an additional step, such as a conformational change or secondary interaction, requiring two sets of rate constants ($k_{a1}$, $k_{a2}$ and $k_{d1,}$ $k_{d2}$). Steady-state binding evaluates the equilibrium response at different analyte concentrations without analysing kinetic rates. These results confirm the successful selection of a high-affinity DARPin that is cross-reactive between mouse and human CD56.

**Table 2: Tested binding models and values of SPR measurement.**

| Model | | $K_D$ [nM] | $k_{on}$ [1/Ms] | $k_{off}$ [1/s] |
|---|---|---|---|---|
| 1:1 Binding | Mean | 13 | $8.7 \times 10^5$ | $1.1 \times 10^{-2}$ |
| | SD | 1.8 | $1.4 \times 10^5$ | $2.9 \times 10^{-4}$ |
| Two State | Mean | 11 | $9.8 \times 10^5$ | $1.9 \times 10^{-2}$ |
| | SD | 0.3 | $3.9 \times 10^{-3}$ | $3.9 \times 10^{-3}$ |
| Steady State | Mean | 15 | - | - |
| | SD | 1.0 | - | - |

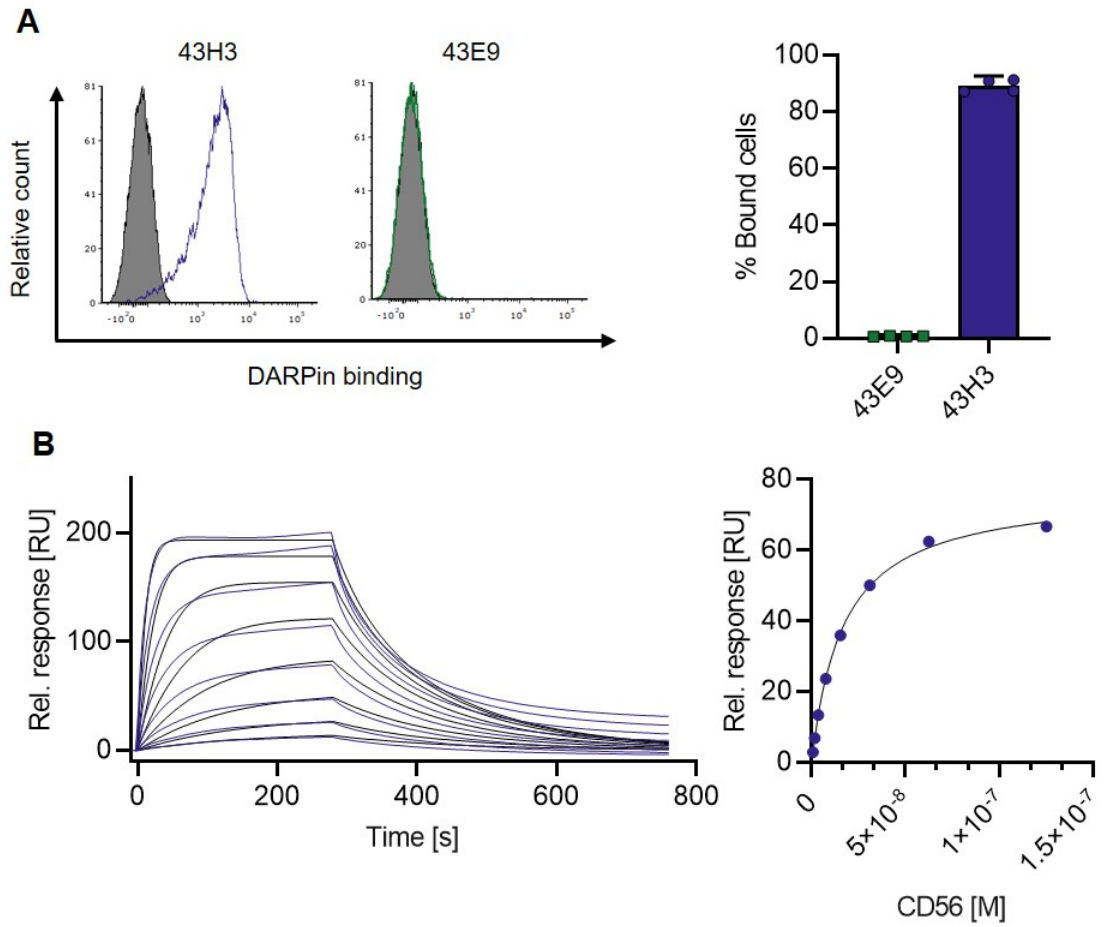



**Figure 9: Cellular binding of purified DARPins and surface plasmon resonance analysis. (A)** Representative histograms from flow cytometry immunostaining show the binding of purified 43E9 and 43H3 DARPins to primary human NK cells (left). Data normalized to the peak value. Bar graphs (right) summarize replicate data using NK cells from four different donors, with error bars representing the 95 % CI. **(B)** Binding affinity of the 43H3 DARPin to huCD56 was determined using surface plasmon resonance (SPR). DARPins were immobilized onto the sensor chip via amine coupling, and binding was assessed with huCD56 as analyte. Shown are response curves with a 280-second association phase and a 480-second dissociation phase (right), alongside the response versus concentration plot (left).

## 2.5 DART-AAV Generation and Biophysical Characterisation

After successfully selecting a suitable CD56 DARPin for viral vector retargeting, this DARPin and the previously described anti-human CD8 DARPin 63A4 were each independently integrated for display on the surface of AAV2 vector particles [101] (Figure 10A). For ideal display of 43H3 and 63A4 on the AAV particle surface, the coding sequences for both DARPins were genetically inserted into the GH2/3 loop of VP1 at amino acid positions 453–459, connected by a flexible $(G_4S)_5$ linker. The production of VP2 and VP3 with integrated DARPins was prevented by deleting the splice acceptor site downstream of the VP1 translation initiation site. To generate DART-AAVs, a combination of the VP1-DARPin plasmids and the complementary capsid plasmid pRC22-VP1-KO was

used [69]. The pRC22-VP1-KO plasmid encoded only VP2 and VP3, ensuring proper capsid assembly. Additionally, the natural affinity of the capsid for HSPG was ablated via point mutations, as demonstrated in previous studies [93], to enhance specificity and reduce off-target interactions (Figure 10B). Western blot analysis confirmed the successful incorporation of DARPins into both vector stocks. However, the DARPin-modified VP1 exhibited a lower signal intensity compared to the unmodified VP1 (Figure 10C).

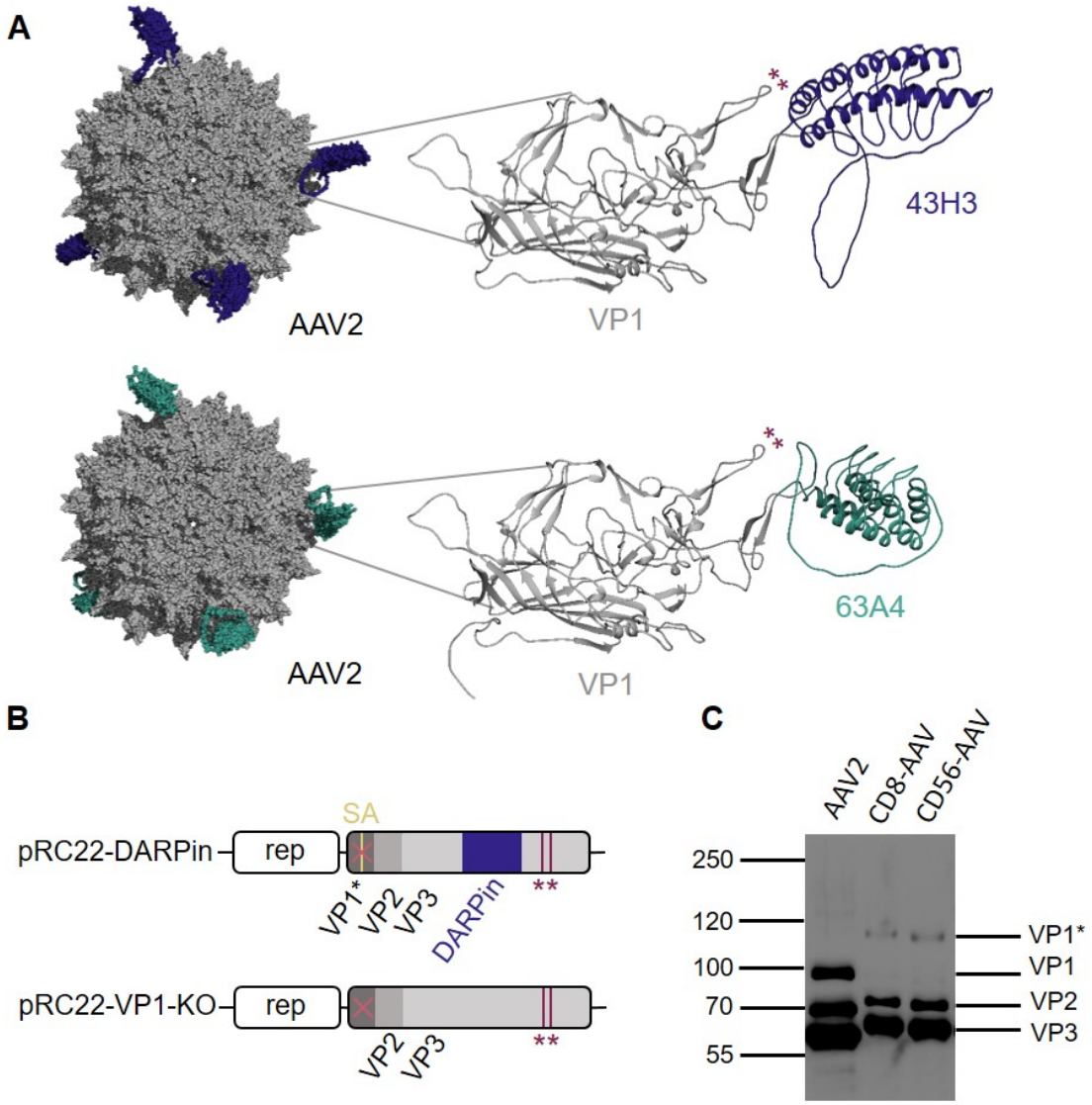



**Figure 10: Structural prediction of DART-AAVs and confirmation of DARPin integration. (A)** Predicted structural models of CD56-AAV2 (top) and CD8-AAV2 (bottom) were based on the AAV2 VP3 capsid structure (PDB 1LP3) fused to 43H3 and 63A4, respectively. Insets highlight the incorporated DARPins, with 43H3 in dark blue and 63A4 in turquoise, linked via a $(G_4S)_5$ linker inserted in the GH2/GH3 loop of VP1 (excluding the N-terminal tail). The AAV2 capsid is depicted in grey, while HSPG blinding mutations (R585A and R588A) are marked with purple asterisks. **(B)** Schematic representation of the DART-AAV2 capsid plasmids. The overlapping sequences for VP1, VP2, and VP3 are depicted in different shades of grey. The DARPin insertion site is marked in blue, while HSPG blinding mutations are shown as purple lines and asterisks. The splice acceptor site

(SA) downstream of the VP1 translation initiation site is highlighted in yellow, with its deletion or the deletion of VP1 itself indicated by red crosses. **(C)** Western blot analysis of unmodified AAV2, CD8-AAV and CD56-AAV. Vector stocks ($5 \times 10^{10}$ vg each) were heat-denatured in urea buffer, separated by 8 % SDS-PAGE, and transferred onto a PVDF membrane. AAV viral proteins were detected using the B1 antibody.

### 2.5.1 DART-AAVs Retain an Unaltered AAV Core Structure

For single-particle cryogenic electron microscopy (cryo-EM) analysis of CD8-AAV2, samples were sent to Petr Chlanda (BioQuant, Heidelberg University). The cryo-EM study confirmed that the core capsid structure remained unaltered and was identical to the atomic structure of unmodified AAV2 (PDB:1LP3 [108]) at a resolution of 3.3 Å (Figure 11A-C). Due to the stochastic nature of particle assembly and the low abundance of VP1, the DARPin protrusions were not visible in the cryo-EM map. These protrusions were likely averaged out because of their random distribution across the capsid surface.

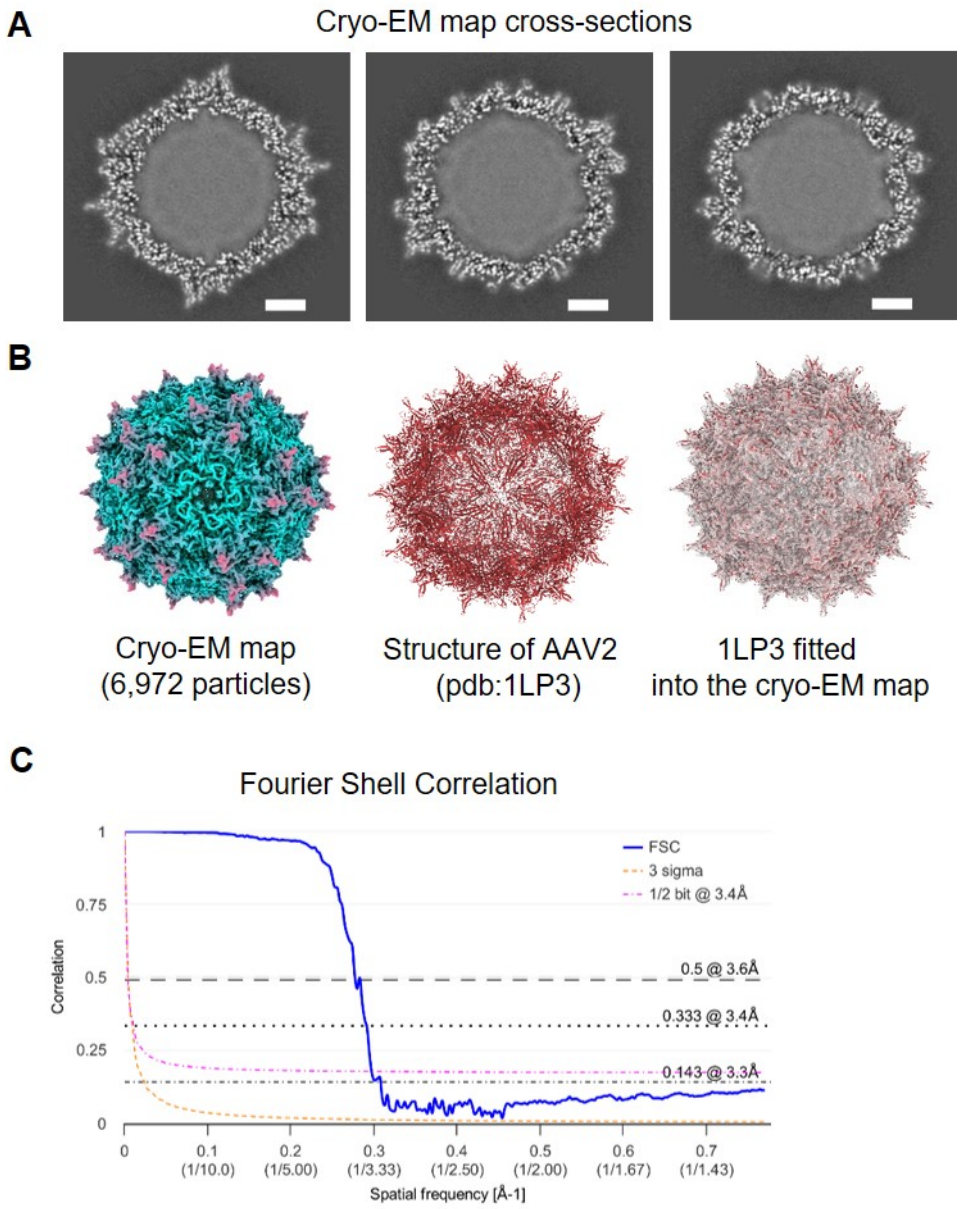



**Figure 11: Cryogenic electron microscopy of CD8-AAV2. (A)** Cryo-EM map cross sections. Scale bar: 50 Å. **(B)** Cryo-EM particle analysis map derived from 6,972 particles of CD8-AAV2 virions, compared to the unmodified AAV2 structure (pdb: 1LP3), which was fitted to the map. **(C)** Fourier shell correlation resolution estimation plot showing resolution cut-offs at different thresholds. Adapted from *Demircan and Zinser et al.* [96].

## 2.6 CD8-AAVs are Highly Efficient on Target Receptor Positive Cell Lines

The experiments with cell lines aimed to evaluate the targeting efficiency of CD8-AAV2 compared to unmodified AAV2 and the HSPG-binding-deficient AAV2 mutant ($AAV2_{mut}$), all packaging GFP. By comparing transduction efficiency and GFP expression across these vector variants, the study sought to confirm whether CD8-AAV2 successfully bypasses reliance on HSPG for cell entry and achieves selective targeting of $CD8^+$ cells.

When tested on $CD8^+$ cell lines such as MOLT4.8, no observable differences were detected between CD8-AAV2 and unmodified AAV2. Flow cytometry analysis of GFP expression revealed no significant differences in either the number of transduced cells or the mean fluorescence intensity across single cells. However, when compared to $AAV2_{mut}$, CD8-AAV2 demonstrated significantly higher transduction efficiency. Notably, the transduction efficiency of $AAV2_{mut}$ was comparable to the untransduced background signal, demonstrating successful retargeting of CD8-AAV2 due to the integration of the DARPin (Figure 12A-B).

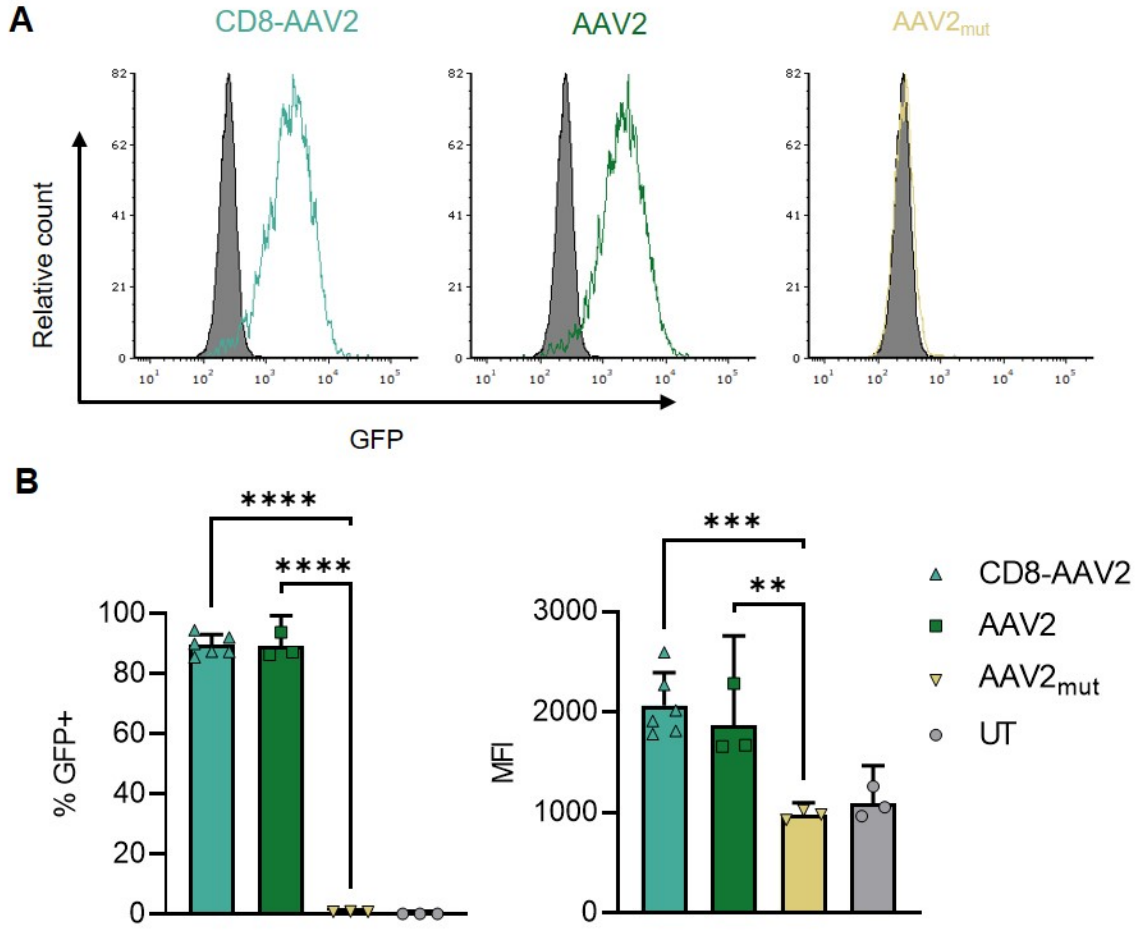



**Figure 12: Transduction efficiency of CD8-AAV2 compared to $AAV2_{mut}$ and AAV2.** Transduction of MOLT4.8 cells with same amounts of CD8-AAV2, AAV2 and $AAV2_{mut}$ packaging GFP as reporter gene analysed by flow cytometry five days post transduction. **(A)** Representative histograms show GFP expression levels of single cells for CD8-AAV2 (light blue), AAV2 (green), and $AAV2_{mut}$ (yellow) compared to untransduced cells (grey). Data normalized to the peak value. **(B)** Bar graphs summarize data from technical replicates, with error bars representing the 95 % CI. Left: Percentage of GFP+ single cells. Right: Arithmetic mean fluorescence intensity (MFI) of single cells (right). Statistical analyses were performed using ordinary one-way ANOVA and Tukey's multiple comparisons test, $**p < 0.01$, $***p < 0.001$, $****p < 0.0001$. Adapted from *Demircan and Zinser et al.* [96].

## 2.7 DART-AAVs Can Also Be Retargeted to Other T Cell Receptors

To rule out a cell line- or DARPin-specific effect, the same targeting strategy was applied to CD4-AAV2. In this case, the 63A4 DARPin was replaced with the previously described 55.2 DARPin, which specifically targets CD4, a marker for another T-cell population [94,109]. The experiment was adapted using CD4-AAV2 instead of CD8-AAV2 and $CD4^+$ A3.01 cells in place of $CD8^+$ MOLT4.8 cells. Interestingly, the transduction efficiency of AAV2 was not fully restored with the CD4 DARPin insertion. Although more than 60 % of A3.01 cells were transduced with CD4-AAV2, and $AAV2_{mut}$ transduction remained at background levels, CD4-AAV2 did not achieve the same efficiency as unmodified AAV2 (Figure 13A-B). Nevertheless, this experiment validated the general feasibility of the targeting concept and demonstrated that AAV2's ability to transduce cells is primarily determined by receptor binding, which can be restored through DARPin insertion into a receptor-blinded capsid.

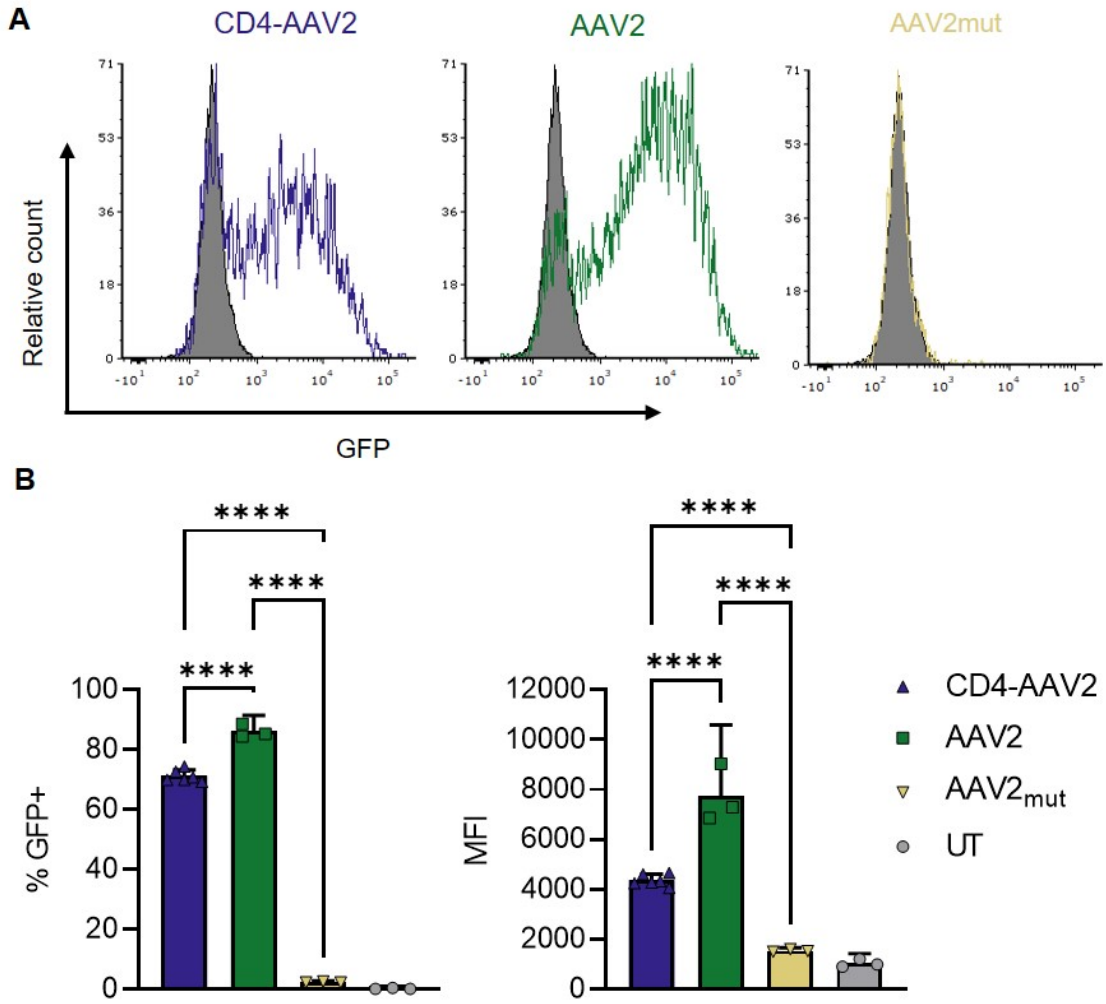



**Figure 13: Transduction efficiency of CD4-AAV2 compared to $AAV2_{mut}$ and AAV2.** Transduction of A3.01 cells with same amounts of CD4-AAV2, AAV2 and $AAV2_{mut}$ packaging GFP as reporter gene analysed by flow cytometry five days post transduction. **(A)** Representative histograms show GFP expression levels of single cells for CD4-AAV2 (blue), AAV2 (green), and $AAV2_{mut}$ (yellow) compared to the untransduced background

(grey). Data normalized to the peak value. **(B)** Bar graphs summarize data from technical replicates, with error bars representing the 95 % CI. Left: Percentage of GFP+ single cells. Right: Arithmetic mean fluorescence intensity (MFI) of single cells (right). Statistical analyses were performed using ordinary one-way ANOVA and Tukey's multiple comparisons test, ****$p < 0.0001$.

## 2.8 Expanding Targets Beyond T Cells: Establishing CD56-AAV

These experiments aimed to evaluate the selectivity and efficiency of CD56-AAV in transducing CD56-expressing cells. By comparing CD56-AAV with unmodified AAV2, the study sought to determine whether CD56-AAV selectively targets CD56-positive cells while avoiding non-specific transduction of CD56-negative cells.

HeLa cells, which are generally considered CD56-negative, were first transduced with a VSV-pseudotyped lentiviral vector encoding full-length CD56. These cells were then cultivated in puromycin-containing selection medium for at least two weeks to ensure stable expression. CD56 expression was confirmed via flow cytometry (Figure 14A). When exposed to CD56-AAV and unmodified AAV2, only CD56-expressing HeLa cells were transduced by CD56-AAV, achieving transduction levels comparable to unmodified AAV2. In contrast, unmodified AAV2 transduced both CD56-positive and CD56-negative HeLa cells indiscriminately (Figure 14B-E). Transduction rates for both vectors remained relatively low, with less than 50 % of cells transduced at a multiplicity of infection (MOI) of $1 \times 10^6$ vg/cell, indicating potential for further optimization. Nevertheless, the selective transduction of CD56-positive cells by CD56-AAV was encouraging, highlighting its promise for further experiments and targeted applications.

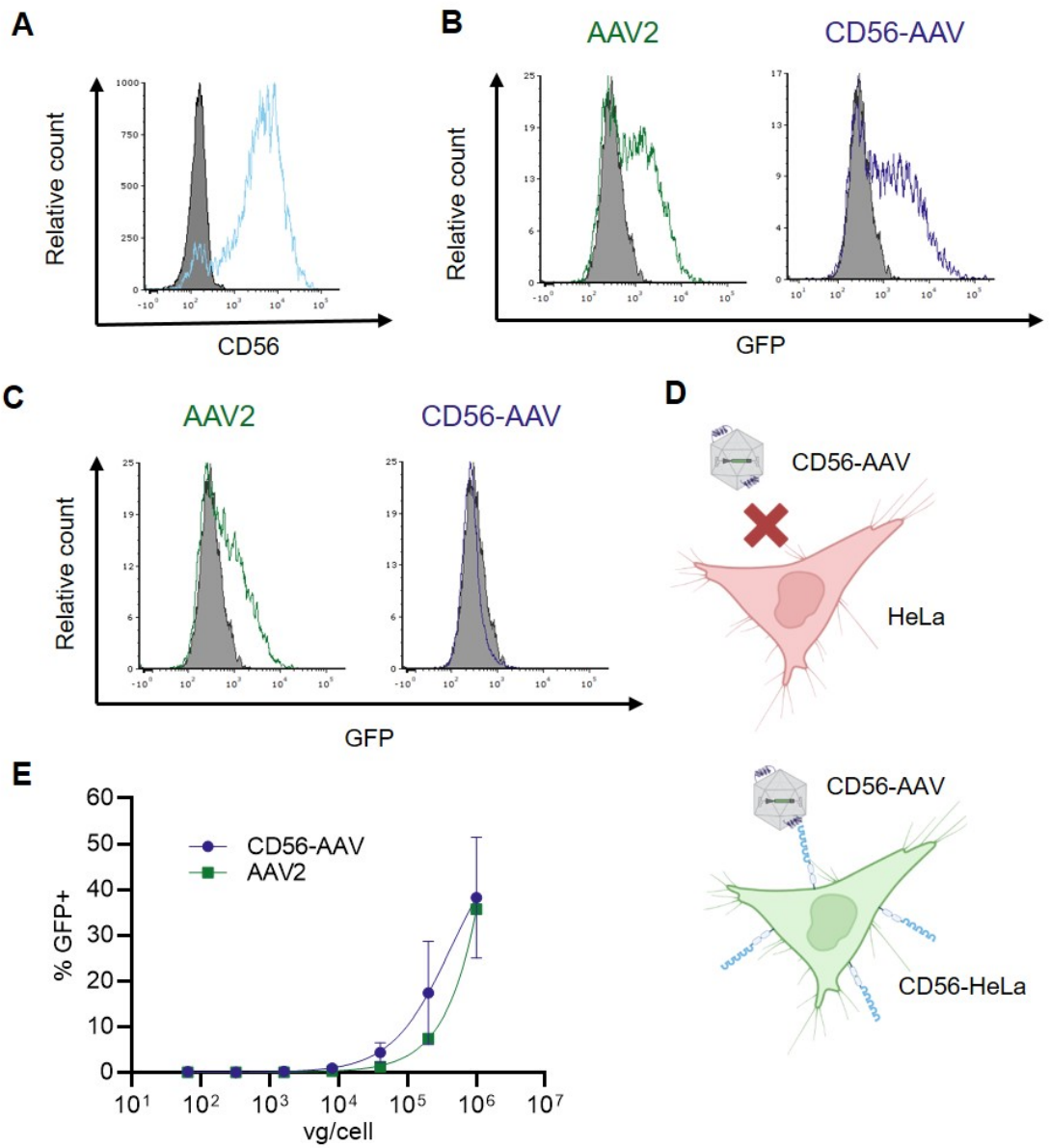



**Figure 14: CD56-AAVs selectively transduce CD56⁺ cells. (A)** CD56 expression in CD56-HeLa cells (light blue) was analysed by flow cytometry two weeks post-transduction with a VSV-pseudotyped lentiviral vector encoding full-length CD56. Cells were cultured in puromycin-containing selection medium. Histograms compare CD56-HeLa cells to untransduced HeLa cells (grey). **(B)** Transduction of CD56-HeLa cells with CD56-AAV (dark blue) and unmodified AAV2 (green) vectors packaging GFP as reporter gene. Representative histograms show GFP expression at 1 $10^6$ vg/cell. **(C)** Transduction of unmodified HeLa cells with CD56-AAV and unmodified AAV2 vectors packaging GFP as reporter gene. GFP expression was analysed by flow cytometry three days post-transduction. Representative histograms show GFP expression at 1 × $10^6$ vg/cell. All histograms are normalized to their peak values. **(D)** Schematic illustration summarizing experimental results: CD56-AAV transduces CD56-HeLa cells (green) but not regular HeLa cells (red). Full-length CD56 is depicted in blue. **(E)** GFP reporter gene expression mediated by the indicated vectors in CD56-HeLa cells. Titration curves were generated using five-fold serial dilutions of vector stocks, ranging from 1 × $10^6$ to 64 vg/cell. Data represent technical triplicates for CD56-AAV, with error bars indicating the 95 % CI. Nonlinear regression fitting was performed with the HillSlope fixed at 1 and the bottom constrained to a constant value of 0.

## 2.9 The DARPin-Retargeting Strategy is Adaptable to Other AAV Serotypes

To evaluate whether the targeting strategy could be applied to other serotypes and given AAV6's status as the preferred serotype for lymphocyte transduction, CD8-AAV6 was generated. Following the approach used for CD8-AAV2, the natural affinity of the AAV6 capsid for its primary attachment receptors was disrupted through point mutations. Besides HSPG [110], AAV6 primarily interacts with sialic acid [111]. Binding to sialic acid can be impeded by the V473D mutations [111], while the K531E mutation prevents HSPG contact [110]. The introduction of both mutations resulted in the modified capsid, $AAV6_{mut}$. Given the more than 95 % structural similarity between AAV2 and AAV6 [65], the amino acids 453–459 of the GH2/GH3 loop of $AAV6_{mut}$ VP1 were replaced with the 63A4 DARPin (Figure 15A-B). Transmission electron microscopy (TEM) imaging confirmed that the structural integrity of CD8-AAV6 particles remained unaltered and appeared identical to unmodified AAV6 particles (Figure 15C).

When tested for functionality, $AAV6_{mut}$ showed significantly reduced transduction rates on the lymphocyte cell lines SUP-T1 and NALM-6, confirming the disruption of natural receptor interactions. Remarkably, the transduction efficiency of CD8-AAV6 was not only restored but also enhanced 37-fold compared to unmodified AAV6 on CD8-positive SUP-T1 cells. Furthermore, signals on $CD8^-$ NALM-6 cells were reduced to levels two-fold below those of $AAV6_{mut}$, demonstrating the successful and specific retargeting of AAV6 to $CD8^+$ cells (Figure 16A-B).

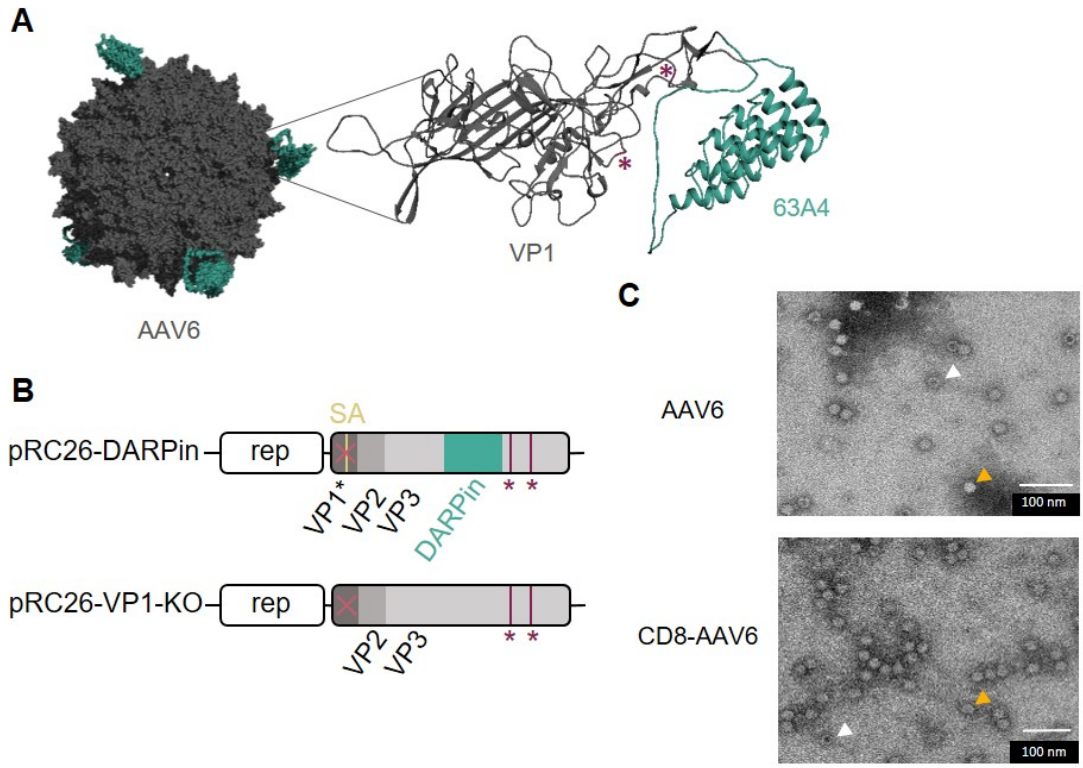



**Figure 15: CD8-AAV6 structure and assembly. (A)** Structure prediction of CD8-AAV6 based on AAV6 VP3 (PDB: 3OAH) fused to 63A4 via a $(G_4S)_2$ linker. The inlay shows 63A4 and the linker in the GH2/GH3 loop in turquoise, the blinding mutations (V473D, K531E) indicated as asterisks, and VP1 (without the N-terminal tail) of AAV6 in grey. **(B)** Schematic representation of the DART-AAV6 capsid plasmids. Overlapping sequences for VP1, VP2, and VP3 are shown in different shades of grey. The DARPin insertion site is marked in turquoise, while the blinding mutations are shown as purple lines and asterisks. The splice acceptor site (SA) downstream of the VP1 translation initiation site is highlighted in yellow, with its deletion or the deletion of VP1 itself indicated by red crosses. **(C)** TEM analysis of CD8-AAV6 and unmodified AAV6. Orange arrows point to full, white arrows to empty particles. Images were taken with 50,000-fold magnification and the scale bar represents 100 nm. Adapted from *Demircan and Zinser et al.* [96].

## 2.10 CD8-AAVs Selectively Transduce CD8$^+$ T-Cells *Ex Vivo.*

Until now, all DART-AAVs have been tested exclusively on distinct cell lines, without simultaneously evaluating target and off-target cells in the same transduction. To advance beyond isolated cell line testing, it was essential to examine the performance of DART-AAVs in more complex and biologically relevant settings. Therefore, the selectivity of CD8-AAV2 was next evaluated on primary cells.

PBMCs were isolated from anonymous human donor blood and, three days after T-cell activation with anti-CD3 and anti-CD28 antibodies as well as IL-7 and IL-15, the PBMCs were transduced with non-purified vector supernatant. Compared to unmodified AAV2, CD8-AAV2 showed substantially increased transduction rates on target cells, achieving an 11-fold increase, while $AAV2_{mut}$ was inactive in gene transfer. Notably, CD8-AAV2

demonstrated almost absolute selectivity for $CD8^+$ cells. However, with only 15 % of $CD8^+$ cells transduced, its efficiency left room for improvement (Figure 17A-C).

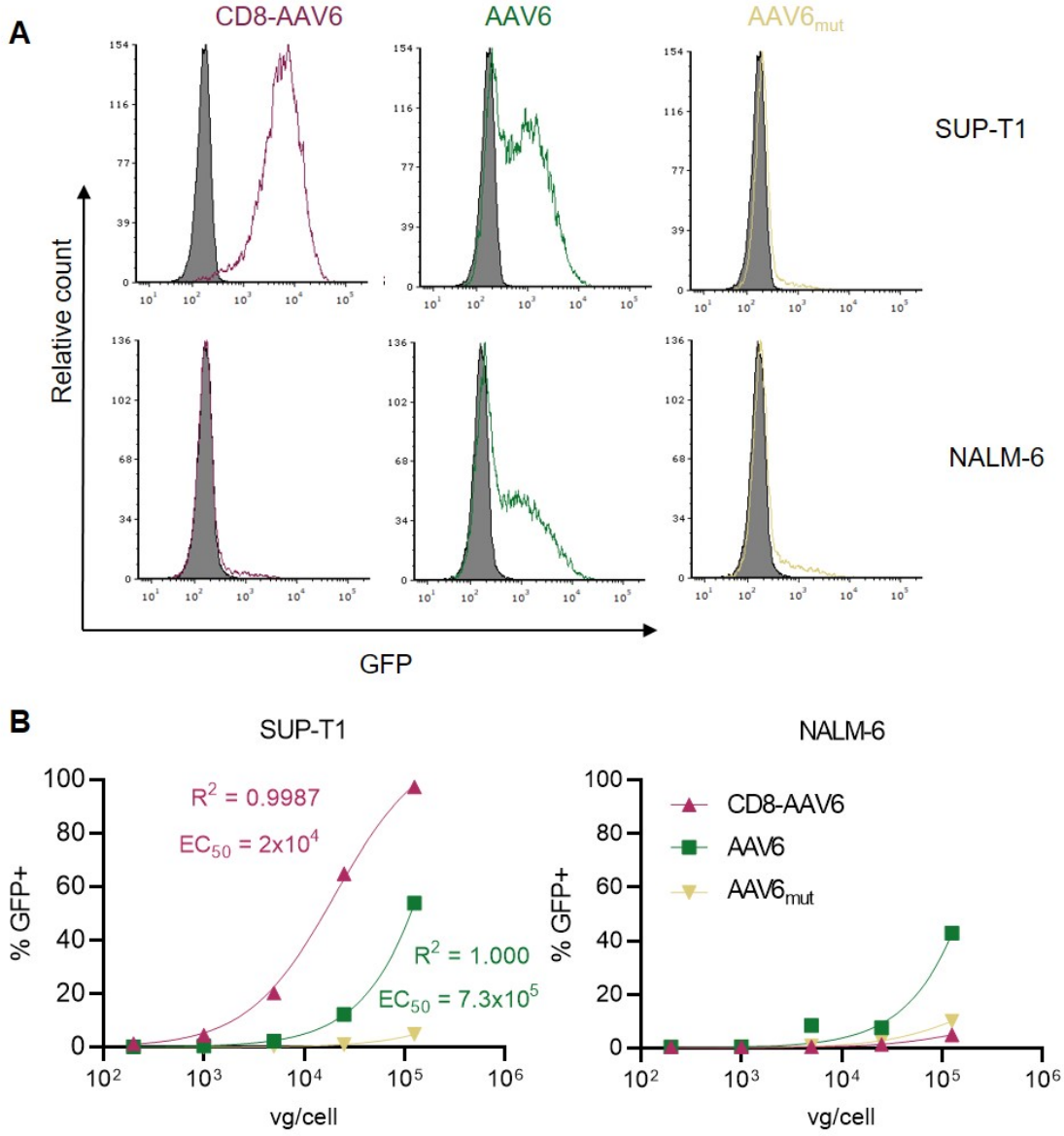



**Figure 16: Gene transfer activity of CD8-AAV6 on cell lines**. Titration of CD8-AAV6 (red), AAV6 (green) and $AAV6_{mut}$ (yellow) delivering GFP as reporter on $CD8^+$ SUP-T1 and $CD8^-$ NALM-6 cells. Vector stocks were purified and concentrated from cell culture supernatant by sucrose cushion centrifugation. GFP positivity was determined three days post-transduction via flow cytometry. **(A)** Representative histograms at 1.25 x $10^5$ vg/cell show GFP expression compared to untransduced cells (grey). Data normalized to the peak value. **(B)** Titration curves with five-fold serial dilutions of the vector stocks ranging from 2 x $10^2$ to 1.25 x $10^5$ vg/cell on SUP-T1 (left) and NALM-6 (right). Nonlinear regression for $EC_{50}$ fitted with HillSlope = 1 and Bottom constrained to a constant value of 0. Adapted from *Demircan and Zinser et al.* [96].

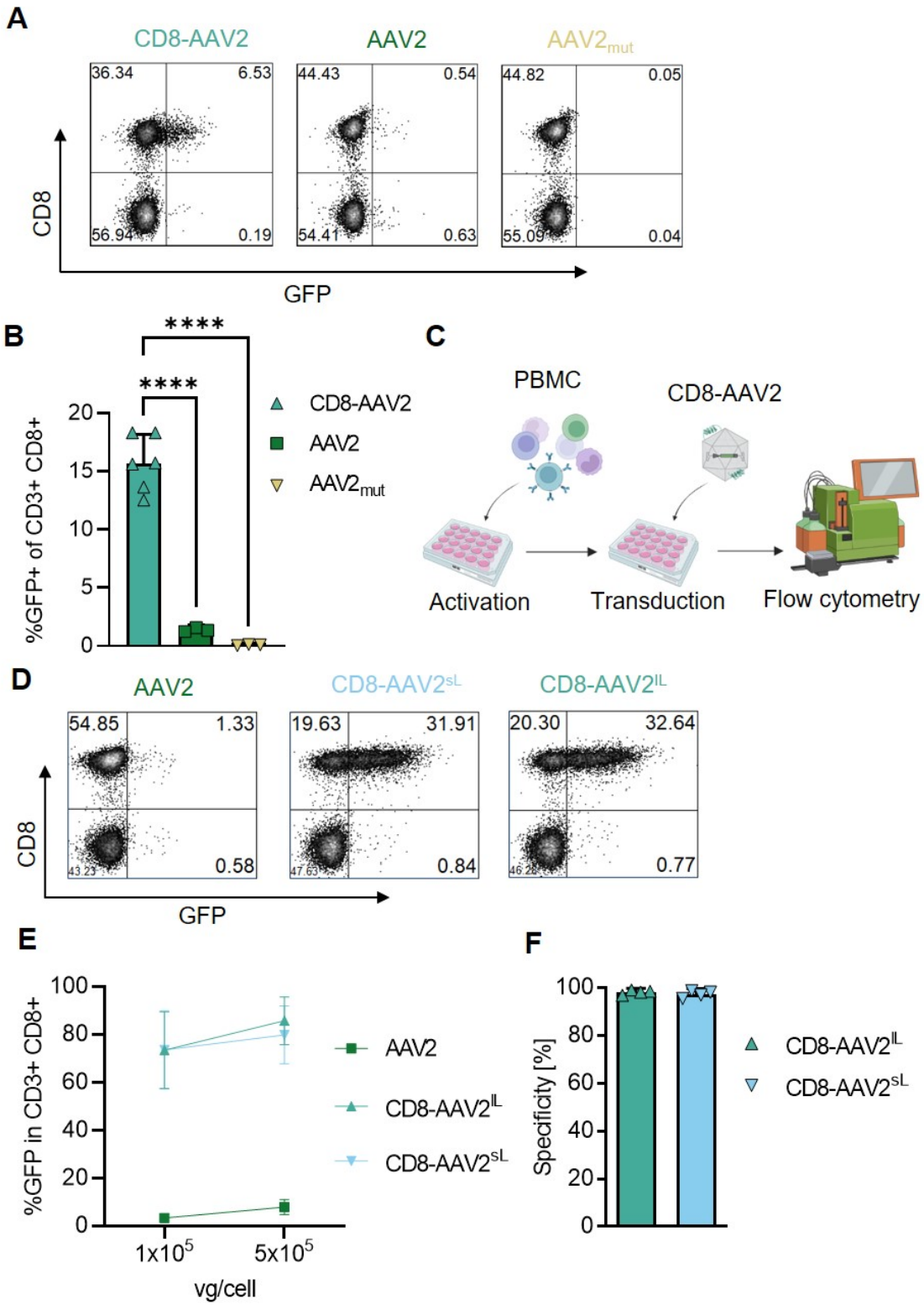



**Figure 17: Selective gene transfer of CD8-AAV2 into human T cells. (A-B)** Transduction of PBMCs. Activated PBMCs were transduced with 50 µL of non-purified vector supernatant containing CD8-AAV2, unmodified AAV2, or AAV2mut packaging GFP. **(A)** Representative flow cytometry plots show the percentage of $GFP^+$ cells within the $CD8^+$ population. **(B)** Bar graphs summarize data from technical replicates, with error bars indicating the 95 % CI. Transduction efficiency was calculated as the percentage of $GFP^+$ cells within the $CD3^+/CD8^+$ subset. **(C)** Schematic representation of the PBMC transduction process. **(D-F)**

Transduction across donors and purified AAV vectors. Activated PBMCs from four different donors were transduced with $1 \times 10^5$ and $5 \times 10^5$ vg/cell of CD8-AAV2 vectors featuring either the longer $(G_4S)_5$ linker (CD8-AAV2$^{lL}$) or the shorter $(G_4S)^2$ linker (CD8-AAV2$^{sL}$), and compared to unmodified AAV2 vectors all delivering GFP **(D)** Representative flow cytometry plots from one donor at $5 \times 10^5$ vg/cell illustrate GFP expression. **(E)** Transduction efficiency, calculated as the percentage of GFP$^+$ cells within the CD3$^+$/CD8$^+$ population, is summarized. **(F)** Transduction specificity, expressed as the percentage of CD8$^+$ cells among all GFP$^+$ cells, is shown. Bars represent mean values, with error bars indicating 95 % CI. Statistical significance is denoted as ****p < 0.0001. Adapted from *Demircan and Zinser et al.* [96].

To address this, vector stocks were purified using iodixanol density gradient centrifugation to remove empty, non-functional AAV capsids. Additionally, in Fabian John's master thesis, which I supervised, the impact of linker length and flexibility on DARPin attachment to the VP1 capsid protein was investigated. A shorter linker improved transduction efficiency in SUP-T1 cells, reducing the $(G_4S)_5$ linker of CD8-AAV2 to a shorter $(G_4S)_2$ linker [96]. Both linker variants—CD8-AAV2 with the longer $(G_4S)_5$ linker (CD8-AAV2$^{lL}$) and the shorter $(G_4S)_5$ linker (CD8-AAV2$^{sL}$)—were subsequently tested on primary T-cells.

For primary T-cells, no significant difference was observed between the two linker variants in either selectivity or gene transfer efficiency, with both reaching transduction rates of around 80 % in CD8$^+$ cells (Figure 17D-E). Remarkably, the on-target selectivity, determined as the percentage of CD8$^+$ cells among all GFP$^+$ cells, exceeded 98 % (Figure 17F) Compared to unmodified AAV2, both CD8-AAV variants achieved substantially higher transduction rates on target cells, reaching up to a 22-fold increase.

### 2.10.1 CD4-AAV2 Demonstrates Lower Efficiency Compared to CD8-AAV2

To ensure that a target cell- or DARPin-specific effect, already tested on cell lines, was not influencing the results, the same experiment was performed using CD4-AAV2 vectors. Both linker variants—CD4-AAV2 with the longer $(G_4S)_5$ linker (CD4-AAV2$^{lL}$) and the shorter $(G_4S)_2$ linker (CD4-AAV2$^{sL}$)—were tested on primary T-cells. Consistent with observations on the A3.01 cell line, both CD4-AAV2 variants exhibited lower transduction efficiency compared to CD8-AAV2. With only up to 50 % of CD4$^+$ target cells transduced, the on-target efficacy was even lower than on A3.01 cells (Figure 18A-B).

Interestingly, the on-target selectivity, defined as the percentage of CD4$^+$ cells among all GFP$^+$ cells, was also reduced compared to CD8-AAV2. While 88 % of all transduced cells were CD4$^+$, demonstrating a preference for target receptor-positive cells, this selectivity was notably lower than the near-absolute specificity achieved with CD8-AAV2 (Figure 18C). However, in contrast to the A3.01 cell line, the transduction efficiency of CD4-AAV2 on primary T-cells was 10-fold higher than that of unmodified AAV2. These findings confirm the previously observed impact of receptor usage and cell type on transduction efficiency,

underscoring the distinct entry mechanisms of DART-AAVs. As no significant differences were detected between the two linker variants in either transduction efficiency or specificity, all subsequent experiments were performed using the short linker variants CD8-AAV.

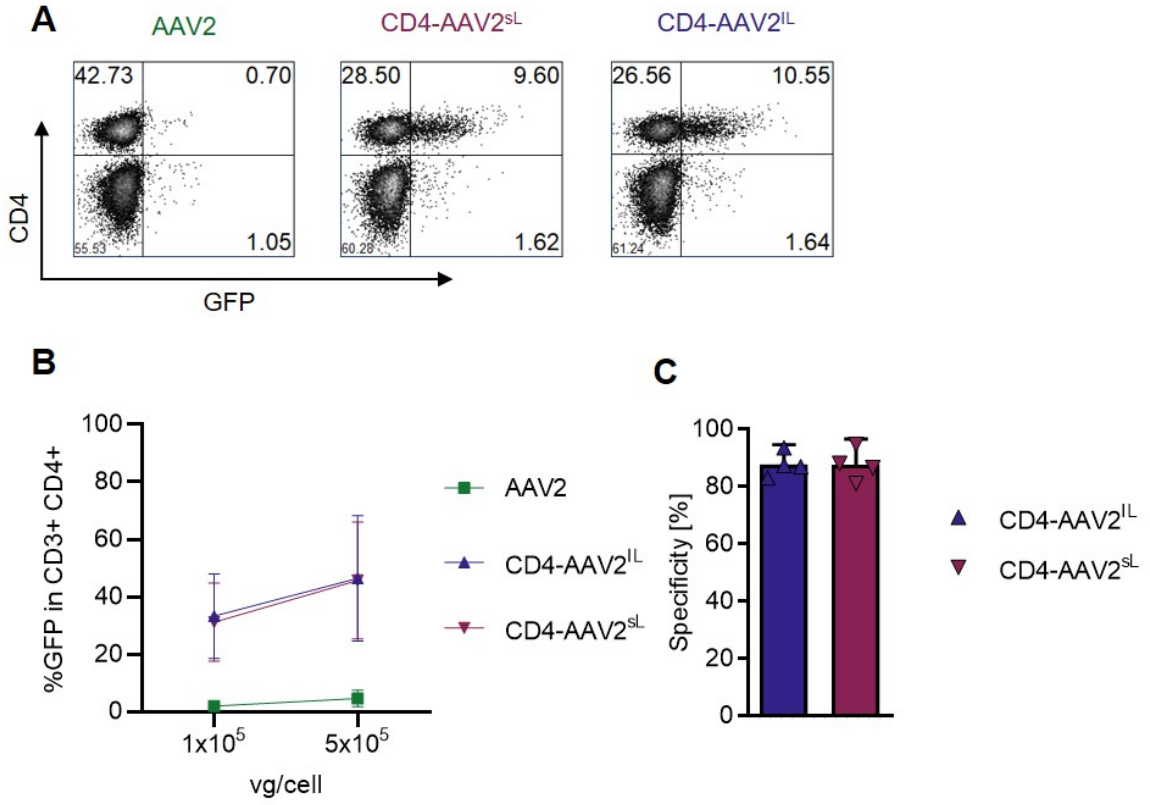



**Figure 18: Selective gene transfer by CD4-AAV2 into human T cells.** Activated PBMCs from four different donors were transduced with 1 × $10^5$ and 5 × $10^5$ vg/cell of purified CD4-AAV2 vectors featuring either the longer $(G_4S)_5$ linker (CD4-AAV2$^{IL}$) or the shorter $(G_4S)_2$ linker (CD4-AAV2$^{sL}$), delivering GFP alongside unmodified AAV2 as a control. **(A)** Representative flow cytometry plots from one donor at 5 × $10^5$ vg/cell illustrate GFP expression. **(B)** Transduction efficiency, calculated as the percentage of GFP$^+$ cells within the CD3$^+$/CD4$^+$ population, is summarized. **(C)** Transduction specificity, expressed as the percentage of CD4$^+$ cells among all GFP$^+$ cells, is shown. Bars represent mean values, with error bars indicating 95 % CI. Adapted from *Demircan and Zinser et al.* [96].

### 2.10.2 CD8-AAV6 Achieves Potent Transduction While Preserving Cell Viability

Given that CD8-AAV6 demonstrated comparable performance to CD8-AAV2 on cell lines, and considering AAV6's established reputation as the serotype of choice for efficient lymphocyte transduction, its potential for targeting primary activated T cells warranted further investigation. Like CD8-AAV2, CD8-AAV6 exhibited high selectivity for CD8-positive cells. While CD8-AAV6 achieved near-absolute selectivity, the unmodified AAV6 vector was completely non-selective (Figure 19A-B)

Notably, AAV6 demonstrated significantly greater potency in transducing $CD8^+$ T cells, particularly at lower doses. The number of transduced cells at a dose of $1 \times 10^5$ vg/cell and the MFI of $GFP^+$ cells within the $CD3^+$ $CD8^+$ population were highest for unmodified AAV6 at both tested doses (Figure 19C). However, this increased potency came with a trade-off: unmodified AAV6 caused a significant reduction in cell viability at a dose of $1 \times 10^5$ vg/cell, whereas CD8-AAV6 maintained cell viability across both tested doses (Figure 19D). This highlights CD8-AAV6's ability to achieve targeted transduction without compromising cellular health.

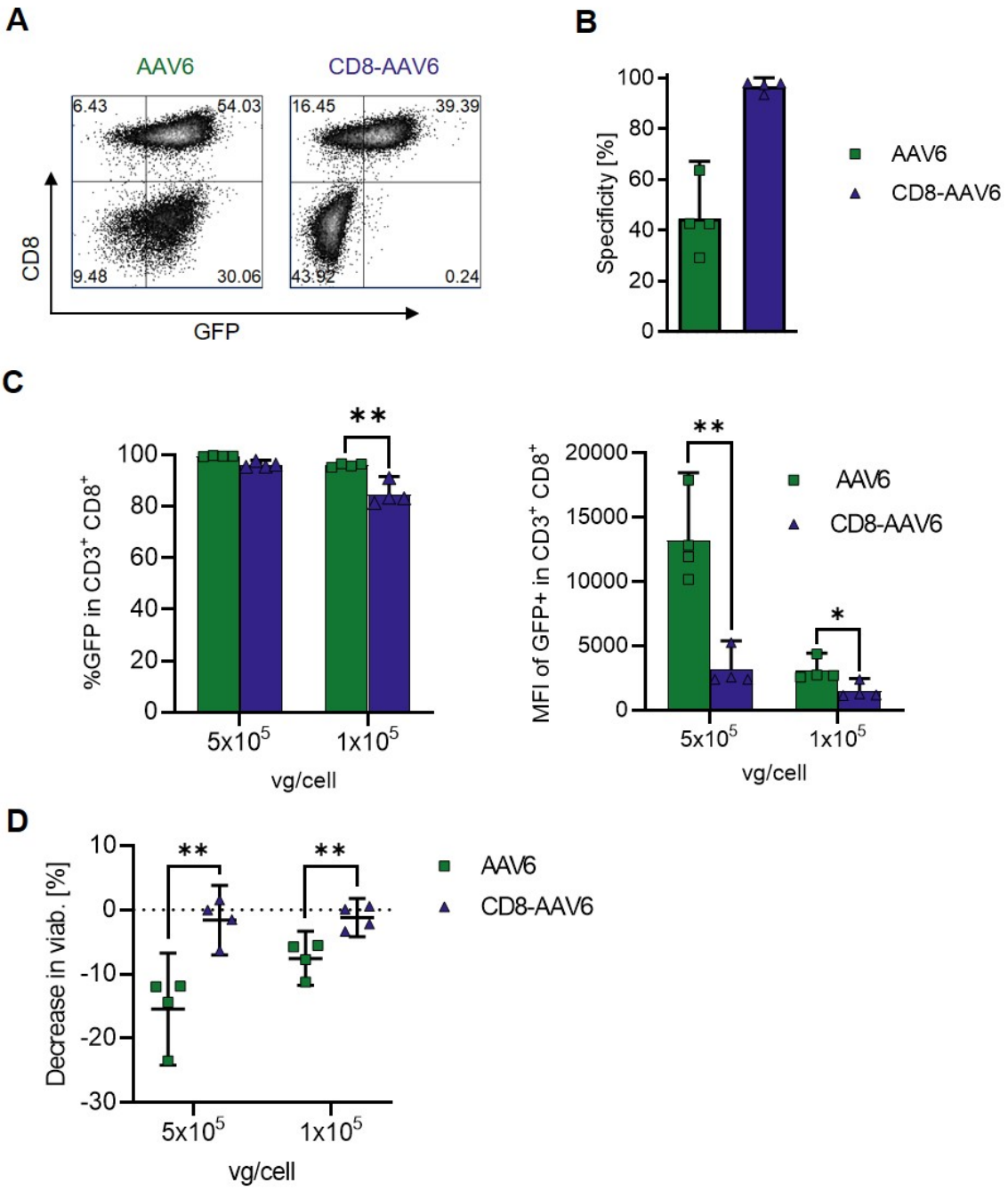



**Figure 19: Selective gene transfer of CD8-AAV6 into human T cells.** Transduction of activated PBMCs from four different donors with CD8-AAV6 compared with unmodified AAV6 delivering GFP. GFP expression was quantified three days post-transduction via flow cytometry. **(A)** Representative flow cytometry plots from one donor at $5 \times 10^5$ vg/cell. **(B)** Bar diagrams display transduction specificity, quantified as percentage of $CD8^+$ cells within the $GFP^+$ population. **(C)** Transduction efficiency is shown as both the percentage of $GFP^+$ cells within $CD3^+/CD8^+$ subset (left) and the MFI of $GFP^+$ cells within $CD3^+/CD8^+$ (right). **(D)** Relative viability decrease in PBMCs exposed to CD8-AAV6 or AAV6 compared to untransduced cells. Viability was assessed using eFluor780 staining and flow cytometry. Statistical analyses were performed with an unpaired t test. Bars represent means, error bars indicate the 95 % CI, with significance denoted as $*p < 0.05$ and $**p < 0.01$. Adapted from *Demircan and Zinser et al.* [96].

## 2.11 Optimization of CD56-AAV by Combining with NKp46-Specific DARPins

Due to the inadequate efficiency of CD56-AAV on HeLa cells, improvements were necessary before testing the vector on more difficult-to-transduce NK cell lines. Additionally, since CD56 is not exclusive to NK cells and is expressed on various non-immune tissues as well, combining it with a more specific NK cell marker was a logical step. As previously demonstrated with the successful combination of the CD4-specific 55.2 DARPin and the CD32a-specific F11 DARPin [112], I aimed to insert another DARPin alongside CD56.43H3 into the GH2/GH3 loop of the VP1 AAV2 capsid protein.

Among others, NKp46 is considered an NK cell-specific receptor because it is uniquely and consistently expressed on NK cells across species, playing a critical role in their activation and cytotoxic functions. Fortunately, a new DARPin selection via ribosome display was unnecessary, as NKp46-specific DARPins had already been identified for use in lentiviral vectors and AAVs [105]. The five most promising candidates (46.3C7, 46.3F9, 46.3A8, 46.3E7, and 46.3C9), previously used for lentiviral vector retargeting, were inserted into the GH2/GH3 loop of the VP1 AAV2 capsid protein to generate NKp46-AAV2 vector stocks packaging GFP as a transgene, following the established approach for other DART-AAVs.

To determine the most suitable NKp46 DARPin for combination with CD56.43H3, an artificial NKp46-positive HT-1080 cell line, previously used for lentiviral transduction [105], was transduced with the NKp46-AAV2 vector stocks, and transduction efficiency was assessed via flow cytometry. Titration curves revealed similar transduction efficiencies for most vector stocks on the NKp46-HT1080 cells, with the exception of 46.3C7-AAV, which transduced noticeably fewer cells (Figure 20A-B). As no significant differences were observed among the remaining NKp46-AAVs, the 46.3A8 DARPin was selected for the generation of bispecific CD56-NKp46-AAVs.

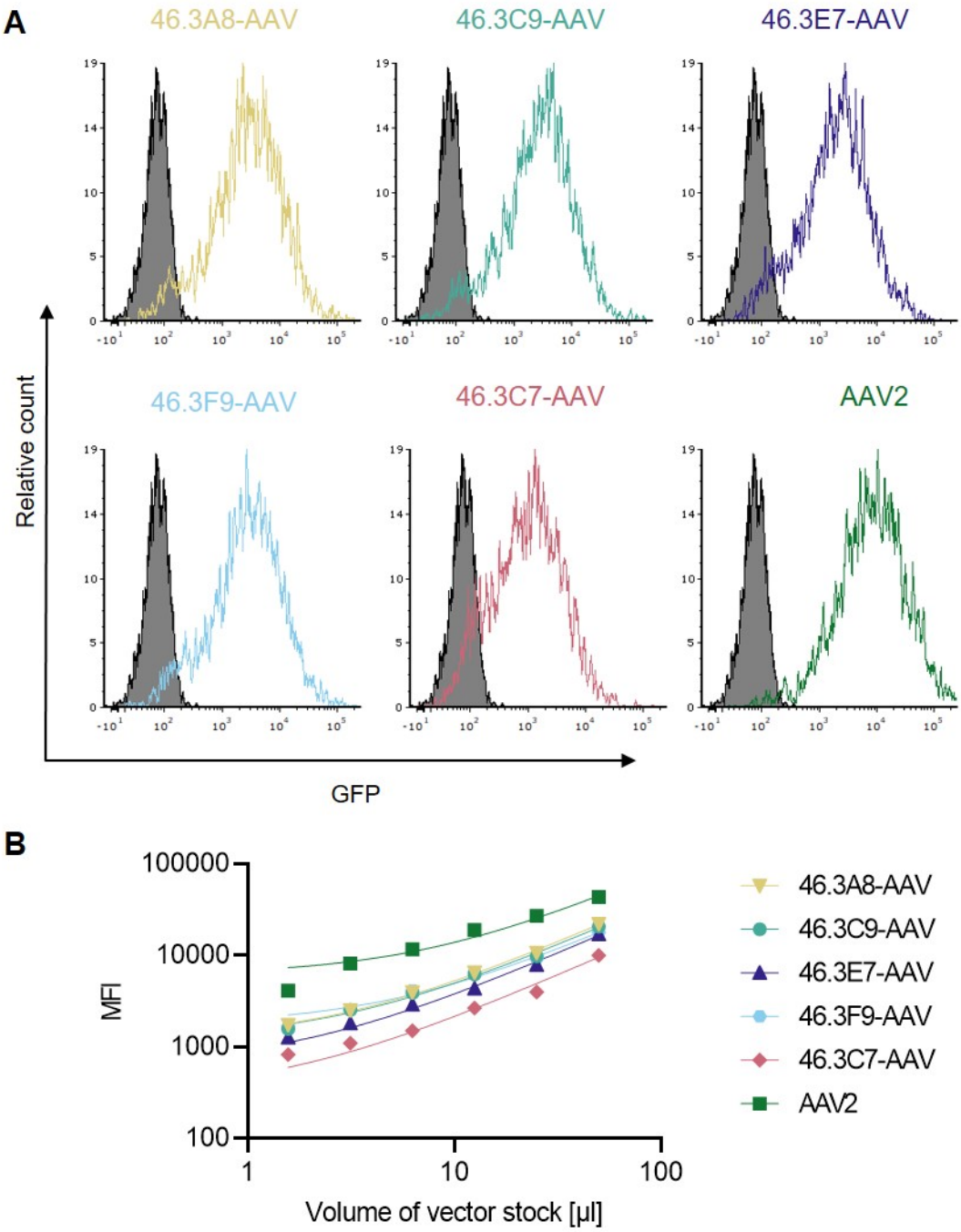



**Figure 20: NKp46-AAV screening.** 8 × 10³ NKp46-HT-1080 cells were transduced with 1:2 serial dilutions of non-purified NKp46 vectors with various DARPins and compared to unmodified AAV2. The dilution range spanned from 50 µl to 1.56 µl non-purified vector **(A)** Histograms depict GFP expression at 12.5 µL for all tested vectors in comparison to untransduced cells. **(B)** Serial dilution curves of all vector stocks with interpolated values.

The 46.3A8 DARPin was inserted alongside CD56.43H3 in both orientations (NKp46-CD56 and CD56-NKp46) within the GH2/GH3 loop of the VP1 AAV2 capsid protein, and proper particle assembly was evaluated via Western blot. Consistent with previous results from mono-DART-AAVs, the VP1 band of the dual-DART-AAVs showed a distinct shift to the expected molecular weight of approximately 120 kDa, reflecting the insertion of two

DARPins and confirming successful integration of the DARPin-modified VP1 protein into the AAV2 capsid (Figure 21A).

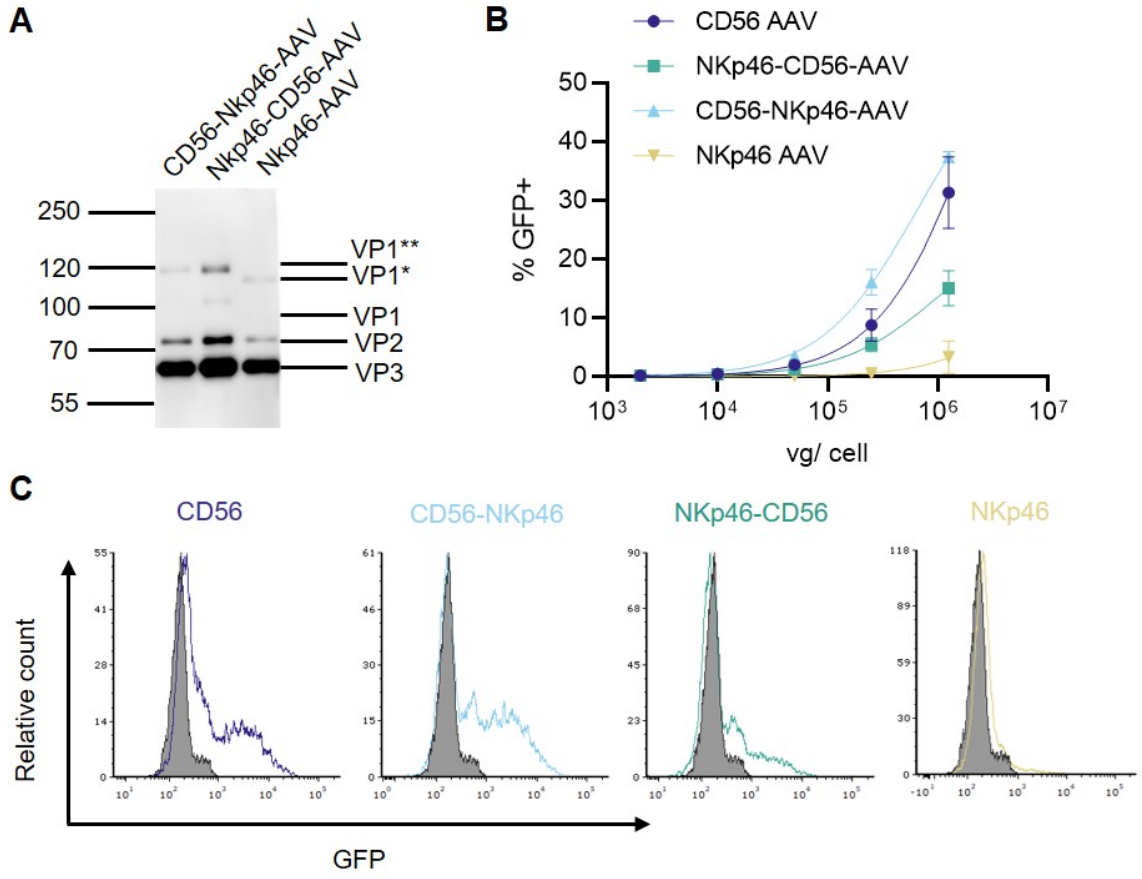



**Figure 21: Characterization of CD56-NKp46-AAVs. (A)** Western blot analysis of CD56-NKp46-AAV, NKp46-CD56-AAV and NKp46-AAV vectors. Each vector stock ($2 \times 10^{10}$ vg) was heat-denatured in urea buffer, separated on an 8 % SDS-PAGE gel, and transferred to a PVDF membrane. AAV viral proteins were detected using the B1 antibody. VP1*: VP1 with a single DARPin insertion; VP1**: VP1 with dual DARPin insertions. **(B)** Transduction of KHYG-1 cells with CD56-AAV, CD56-NKp46-AAV, NKp46-CD56-AAV, and NKp46-AAV vectors packaging GFP as a reporter gene. Titration curves display results from five-fold serial dilutions of vector stocks, ranging from $1.25 \times 10^6$ to 2000 vg/cell. Data points represent the means of technical triplicates, with error bars indicating the 95 % CI. Nonlinear regression was performed with the HillSlope set to 1 and the Bottom fixed at a constant value of 0. **(C)** Representative histograms show GFP expression normalized to peak value at $1.25 \times 10^6$ vg/cell, compared to untransduced cells.

The transduction efficiencies of both dual-DART-AAVs were tested against their mono-DART-AAV counterparts (CD56-AAV and NKp46-AAV) on CD56 and NKp46 double-positive KHYG-1 cells. Against the expectation, the dual-DART-AAVs exhibited minimal to no improvement in transduction efficiency compared to the mono-DART-AAVs. Notably, the construct with the 46.3A8 DARPin in the N-terminal position (NKp46-CD56-AAV) displayed even lower transduction of KHYG-1 cells than CD56-AAV (Figure 21B-C). These findings

underscored the need for further optimization or alternative strategies to enhance the efficacy of DART-AAVs.

## 2.12 Enhanced Transduction Efficiency with 2xCD56-AAV

To test the hypothesis that increasing receptor engagement through bivalent binding could enhance transduction efficiency of the CD56-AAVs, a bivalent display of the CD56-specific DARPin was engineered. This was achieved by tandem insertion of two identical copies of the 43H3 DARPin into the GH2/GH3 loop of the VP1 AAV2 capsid protein, creating 2xCD56-AAV. The correct folding and surface presentation of the DARPins on the AAV particle were assessed *in silico* using AlphaFold3 (Figure 22A). Proper incorporation of 2x43H3 was confirmed via Western blot analysis (Figure 22B).

Functionally, 2xCD56-AAV showed markedly improved performance compared to its monospecific counterpart. On KHYG-1 cells, transduction rates increased substantially with the bivalent vector (Figure 22C) and titration experiments revealed an approximately 100-fold enhancement in transduction efficiency relative to the mono-DARPin CD56-AAV (Figure 22F). Furthermore, the binding of 2xCD56-AAV to KHYG-1 cells was readily detectable by flow cytometry, whereas the binding signal of CD56-AAV remained at background levels (Figure 22D), supporting the enhanced avidity and cell surface engagement achieved through the bivalent design.

Encouraged by the observed improvements in binding and transduction, the 2xCD56-AAV was evaluated on the clinically relevant NK-92 cell line, a well-established model for studying NK cell behaviour and therapies. Remarkably, the 2xCD56-AAV achieved significantly higher transduction rates compared to the standard AAV6 vector (Figure 22E-F). At the highest tested vector dose, the transduction efficiency of the 2xCD56-AAV was approximately threefold higher than that of unmodified AAV6, underscoring its enhanced efficiency for NK cells. These results highlight the potential of 2xCD56-AAV as a powerful tool for gene delivery in NK cell-based research and therapeutic applications.

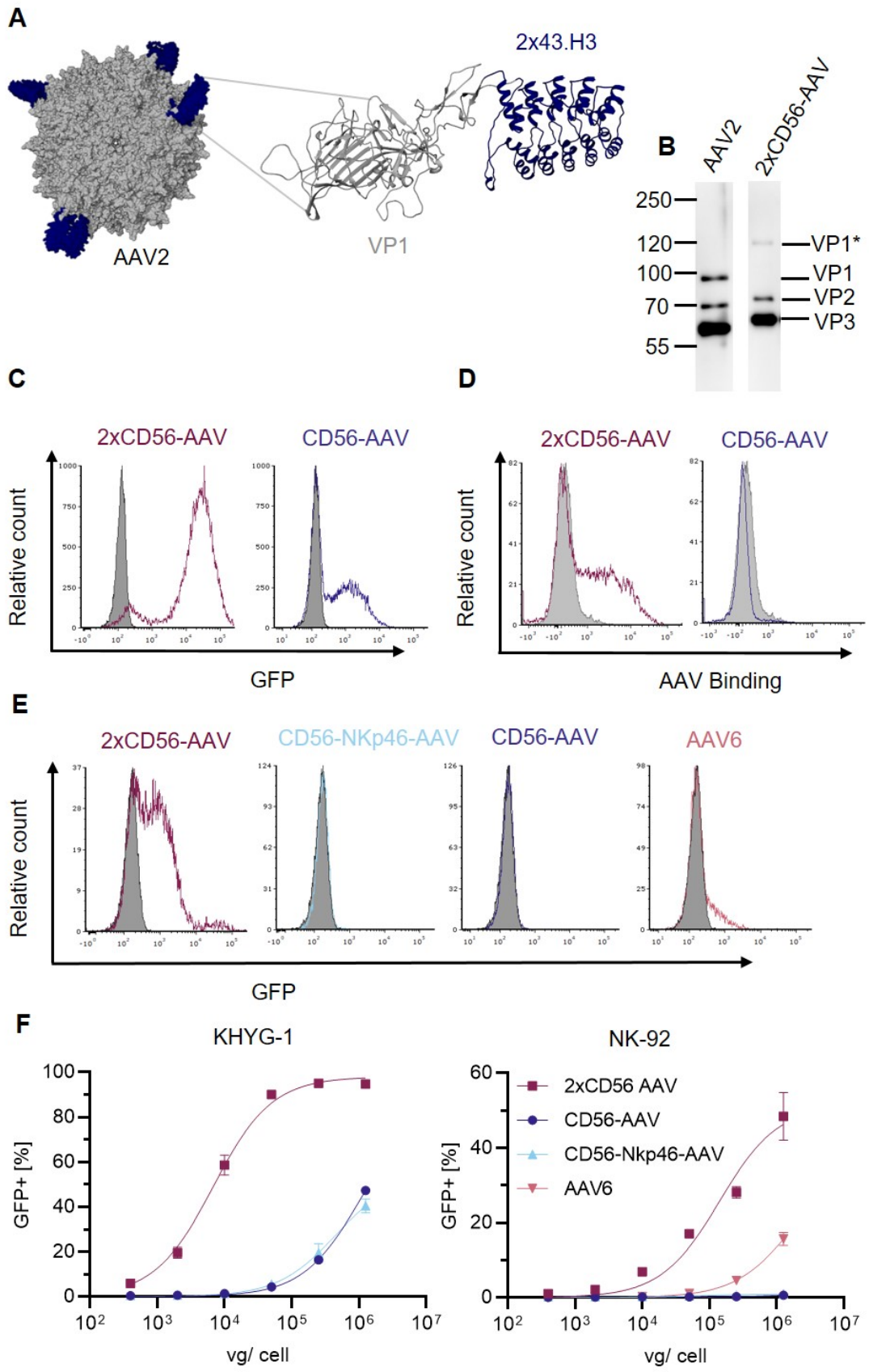



**Figure 22: Increased transduction of CD56$^+$ cells with 2xCD56-AAVs. (A)** Structure prediction of 2xCD56-AAV based on AAV2 VP3 (PDB 1LP3) fused to two 43H3 DARPins in tandem configuration. The inlay shows 43H3 and the linkers in the GH2/GH3 loop in blue

and VP1 (without the N-terminal tail) of AAV2 in grey. **(B)** Western blot analysis of 2xCD56-AAV and unmodified AAV2. 2 x $10^{10}$ vg of each vector stock were heat-denatured in urea buffer before separation with 8 % SDS-PAGE and blotting onto a PVDF membrane. AAV viral proteins were detected with the B1 antibody **(C)** Transduction of KHYG-1 cells with 2xCD56-AAV, CD56-NKp46-AAV and CD56-AAV vectors having GFP packed as reporter gene. Representative histograms at 1.25 x $10^6$ vg/cell **(D)** AAV cell binding assay with 5 x $10^5$ vg/cell and KHYG-1 **(E)** Transduction of NK-92 cells with 2xCD56 AAV, CD56-NKp46-AAV and CD56-AAV and unmodified AAV6 with GFP as reporter gene. Representative histograms at 1.25 x $10^6$ vg/cell. Histograms normalized to peak value. **(F)** GFP reporter gene transfer mediated by the indicated vectors on KHYG-1 (left) and NK-92 cells (right). Titration curves generated with five-fold serial dilutions of vector stocks, ranging from 1.25 × $10^6$ to 400 vg/cell, are presented. Data points represent the means of technical triplicates (duplicates for CD56-NKp46-AAV), with error bars indicating the SD. Nonlinear regression was performed with the HillSlope fixed at 1 and the Bottom constrained to a constant value of 0.

## 2.13 Translational Limitations of CD56-AAVs in Primary NK Cells

For the DART-AAV vector platform to have potential *in vivo* applications, achieving high transduction efficiency in primary NK cells is crucial and required evaluation. To this end, primary human NK cells were isolated from buffy coats of anonymous donors and expanded for three days in medium supplemented with IL-2 and IL-5. Successful isolation and expansion of NK cells were confirmed by flow cytometry, with inclusion criteria set at >90 % cell viability and CD56 surface expression. Given that primary NK cells are generally more challenging to transduce than the NK-92 cell line, the TBK1/IKKε inhibitor BX795, which has previously been shown to enhance lentiviral vector transduction in NK cells [113,114], was tested alongside with 2xCD56-AAV and AAV6 (Figure 23A).

As anticipated, the transduction efficiency of 2xCD56-AAV was substantially lower in primary NK cells compared to the NK-92 cell line (Figure 23B-C). Surprisingly, in contrast to the results observed with NK-92 cells, unmodified AAV6 significantly outperformed 2xCD56-AAV, with transduction of primary NK cells by 2xCD56-AAV being barely detectable. The addition of the TBK1/IKKε inhibitor BX795 did not have a detrimental effect in the viability of the cells. However, BX795 not only failed to enhance transduction for both 2xCD56-AAV and unmodified AAV6 but also reduced transduction rates by more than two-fold. Unfortunately, a direct comparison with lentiviral vectors could not be included as an activity control.

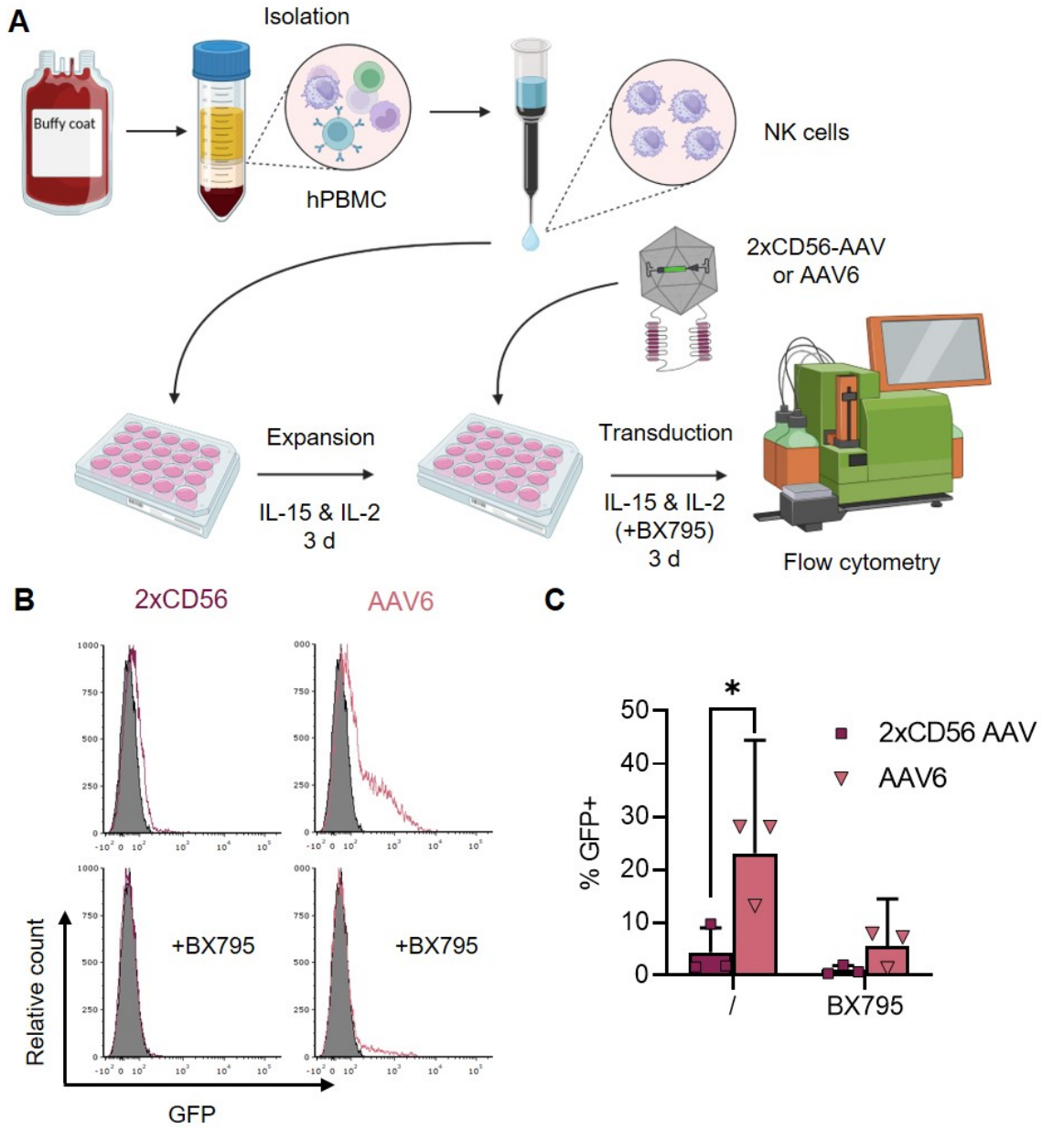



**Figure 23: Isolation and transduction of human NK cells. (A)** Experimental setup. **(B)** Flow cytometry histograms showing GFP expression in NK cells from a single donor transduced with $5 \times 10^5$ vg/cell of either 2xCD56-AAV or unmodified AAV6, both transferring GFP, with and without 2 µM BX795 added to the culture medium during transduction. **(C)** Bar diagrams summarizing transduction data from NK cells derived from three donors. Statistical analyses were performed with an unpaired t test. Bars represent means, error bars indicate the 95 % CI, with significance denoted as $*p < 0.05$.

These findings underscore the importance of evaluating gene delivery vectors in primary cells to assess their translatability beyond cell line models like NK-92. Promising results in immortalized cell lines do not necessarily predict performance in primary cells, where additional biological barriers may exist. Moreover, transduction enhancers such as BX795 must be carefully adapted to the specific vector system, as approaches effective for lentiviral vectors may not translate to AAV-based platforms. This highlights the need for system-

specific optimization to achieve reliable and effective gene transfer across experimental models.

## 2.14 The 43H3 DARPin Binds to Distal CD56 Epitope

To further investigate the 43H3 DARPin and its corresponding CD56-AAVs, particularly the 2xCD56-AAV, it was crucial to identify and evaluate the binding epitope of the DARPin. Understanding the precise interaction between the DARPin and CD56 is key to interpreting the performance of the (2x)CD56-AAV. To narrow down potential binding mechanisms and target sites, multiple CD56 and DARPin constructs were modelled *in silico* using the protein structure prediction capabilities of Alphafold3 AI. This computational approach provided high-confidence structural models for both the DARPin and the CD56 ectodomain. Following the structural predictions, docking simulations were conducted to identify the most likely interaction sites and assess the binding geometry of the DARPin-CD56 complex.

The most reliable binding predictions indicated that the DARPin interacts with the distal Ig-like C2-type-2 domain of CD56, specifically in the region encompassing amino acids 154-165 (Figure 24A). Notably, this region is highly conserved across vertebrate species and is 100 % identical in CD56 from humans, mice, rats, and other vertebrates, suggesting evolutionary significance and functional importance.

**Table 3: Predicted interactions between 43H3 DARPin and CD56.**

| Diversified residues in 43H3 DARPin | Position in VP1-2x43H3 Fusion Protein | Interacting residues in CD56 |
|---|---|---|
| F33 | F698 | L160 |
| A34 | A699 | L160 |
| N67 | N732 | K162 |
| E99 | E591, E764 | K162 |
| H100 | H592<br>H765 | R156, K162<br>D163 |
| D102 | D594 | R156 |

In the docking simulations, the variable amino acids F33, A34, N67, E99, H100, and D102 of the DARPin were found to form interactions with CD56 (Table 3), with distances ranging from 2 to 4 Å, indicating a strong and specific binding interface (Figure 24B). Interestingly, these amino acids are conserved between the 43E10 and 43H3 DARPins (Figure 7C), both of which bind to human CD56. In contrast, the 43E9 DARPin, which does not bind to human CD56, differs in these key residues (Figure 7C). This suggests that these conserved amino acids play a critical role in the binding specificity and affinity of the DARPin-CD56 interaction.

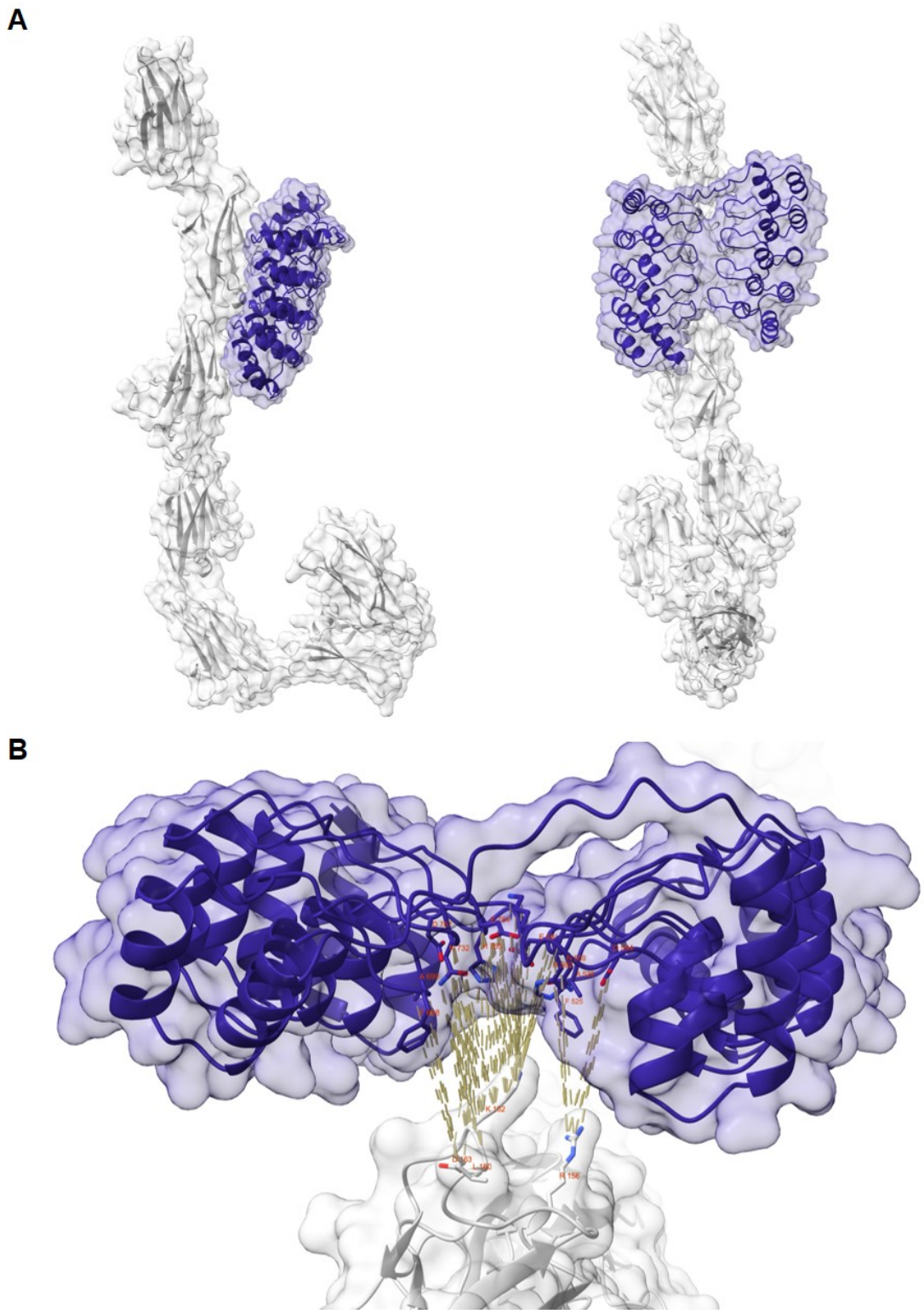



**Figure 24: AlphaFold3-based binding prediction of the 2x43H3 DARPin interaction with the ectodomain of human CD56. (A)** Predicted binding of the tandem-arranged 43H3 DARPins (blue), as displayed on AAV2 VP1 (not shown), to the ectodomain of human CD56 (grey). **(B)** Close-up view of the tandem 43H3 DARPin binding to the distal Ig-like C2-type-2 domain of CD56, with predicted contact sites (yellow pseudobonds) representing interatomic distances of less than 4 Å between distinct amino acids of the DARPins and CD56.

To experimentally confirm that the Ig-like C2-type-2 domain is essential for binding and transduction by 43H3 and 2xCD56-AAV, HeLa cells were engineered using a lentiviral vector to express CD56 constructs fused with CD4. Two constructs were generated: one included the CD56 N-terminus, encompassing both the Ig-like C2-type-1 and C2-type-2 domains (amino acids 1–212), fused to full-length CD4 (1-CD56-212-CD4), and the other attached the same CD56 fragment to the CD4 transmembrane domain only (1-CD56-212-CD4.TM) (Figure 25A).

CD4 was chosen as the fusion partner due to its previously demonstrated compatibility with DART-AAV-mediated transduction, its structural similarity as an Ig-like protein, and its predominantly monomeric expression on the cell surface, which reduces potential confounding effects from dimerization. Proper folding of both constructs was confirmed *in silico* using Alphafold3, which predicted stable and biologically relevant structures for the chimeric proteins. This ensured that the CD56 domains retained their structural integrity, allowing precise functional assessment of their roles in DARPin binding and AAV-mediated transduction.

Two weeks after lentiviral transduction and cultivation in puromycin-containing selection medium, the expression of CD56 domains was confirmed using the AF12-7H3 anti-CD56 antibody and flow cytometry. Previously generated CD56-HeLa cells served as a positive control, while unmodified HeLa cells were used as a negative control. Although expression levels varied across the CD56 constructs, with full-length CD56 showing the highest expression, the (partial) CD56 domains were clearly detectable on all transduced cells (Figure 25B).

More importantly, when these cells were transduced with 2xCD56-AAVs, all CD56-expressing HeLa cells showed robust GFP expression, while no GFP signal was detected in the unmodified HeLa negative control (Figure 25C). Interestingly, when the binding of purified 43H3 and 2x43H3 DARPins to the 1-CD56-212 construct was assessed, the 2x43H3 DARPin demonstrated significantly stronger binding, producing a superior signal intensity even at half the concentration of the single 43H3 DARPin (Figure 25D). This finding underscores the enhanced avidity of the 2x43H3 DARPin, further establishing its potential as an efficient tool for targeting CD56-positive cells. Moreover, it confirms that the distal region of CD56, comprising the Ig-like C2-type-1 and C2-type-2 domains, is both sufficient and necessary for effective binding and transduction via the 43H3 DARPin.

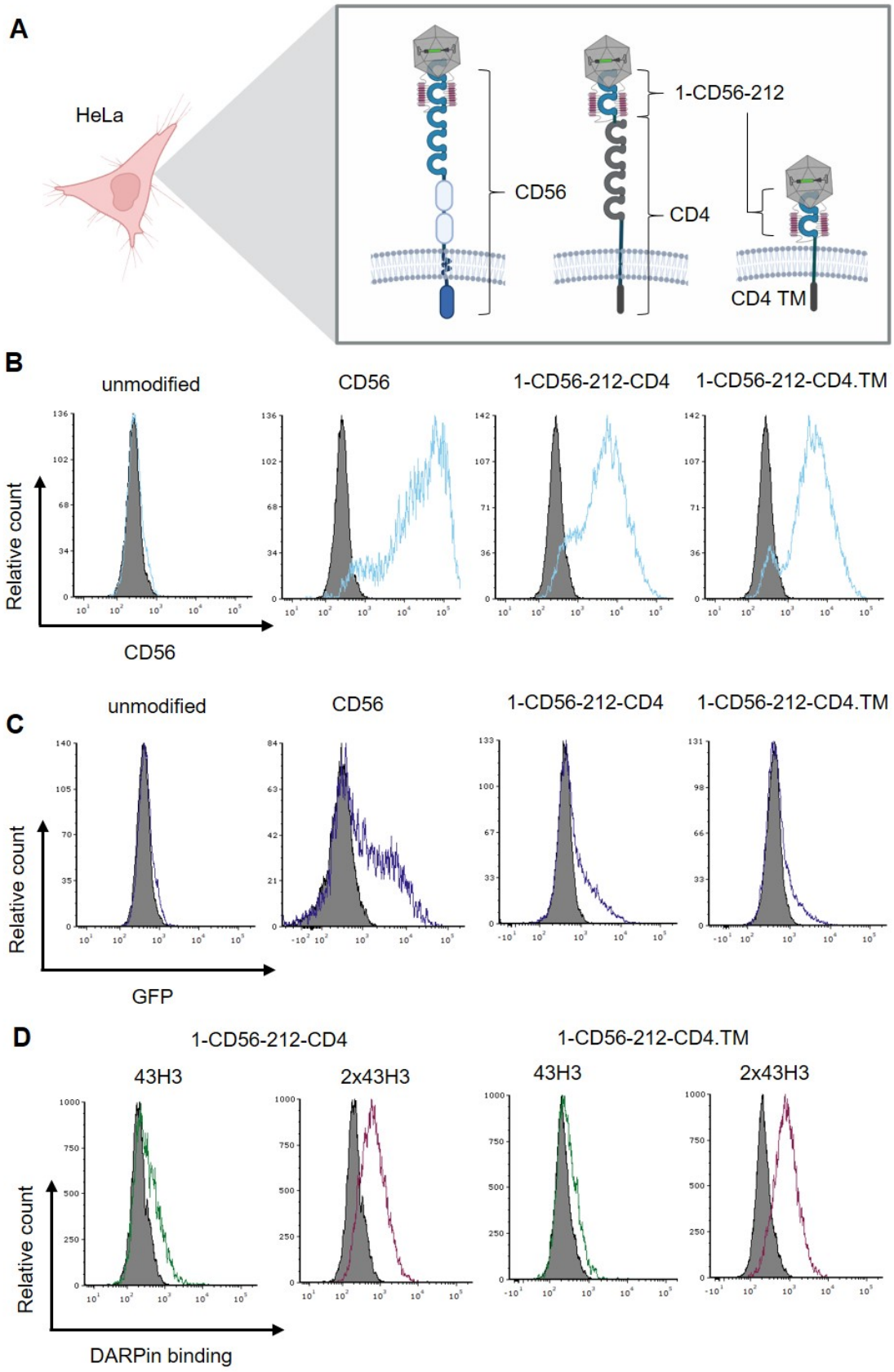



**Figure 25: Determination of the binding domain of the 43H3 DARPin. (A)** Schematic representation of CD56 and CD56/CD4 chimeric constructs expressed in HeLa cells, featuring amino acids 1–212 of CD56 fused with either the full-length CD4 protein or the

CD4 transmembrane domain (TM) and cytoplasmic tail alone. **(B)** Flow cytometry histograms showing CD56 expression in unmodified HeLa cells (unmodified) and HeLa cells expressing the constructs depicted in (A), using the AF12-7H3 anti-CD56 antibody. **(C)** GFP expression in HeLa cells transduced with $1 \times 10^5$ vg/cell of 2xCD56-AAV packaging GFP, analysed by flow cytometry three days post-transduction. **(D)** Binding of 5 µg 43H3 DARPin or 2.5 µg 2x43H3 DARPin to HeLa cells expressing the CD56 constructs (amino acids 1-212), determined via flow cytometry using an anti-HA antibody.

However, the transduction data on primary cells revealed notable limitations, underscoring the need for improvements in targeting efficiency. Consequently, all subsequent experiments focused exclusively on the more promising CD8-AAVs, which had demonstrated superior potential in earlier studies. This strategic shift aimed to refine and optimize the CD8-AAV platform for *in vivo* applications, where achieving robust and selective transduction is essential for success.

## 2.15 CD8-AAVs Outperform Unmodified AAV6 in Human Blood

To evaluate vector performance under conditions that closely approximate the clinical scenario, a whole-blood transduction assay was conducted. CD8-AAV2 and CD8-AAV6 were compared with unmodified AAV6 in a human whole-blood transduction assay (Figure 26A). This experimental design incorporated human serum and a diverse range of blood cell types, including monocytes, erythrocytes, and thrombocytes, to better simulate physiological conditions and assess the potential advantages of DART-AAVs over unmodified AAV6.

The results revealed a stark contrast in transduction efficiency: unmodified AAV6 achieved only about 3 % transduction of $CD8^+$ cells, approximately 25-fold lower than CD8-AAV2 (60–83 %) and CD8-AAV6 (67–85 %) (Figure 26B-C). Additionally, CD8-AAV2 and CD8-AAV6 exhibited a high specificity for $CD8^+$ cells, with selectivity rates of 93–97 % for CD8-AAV2 and 92–94 % for CD8-AAV6. In contrast, unmodified AAV6 showed no ability to discriminate between cell populations (Figure 26D).

These findings demonstrate that CD8-AAV2 and CD8-AAV6 outperform unmodified AAV6 in both transduction efficiency and target cell specificity under physiologically relevant conditions.

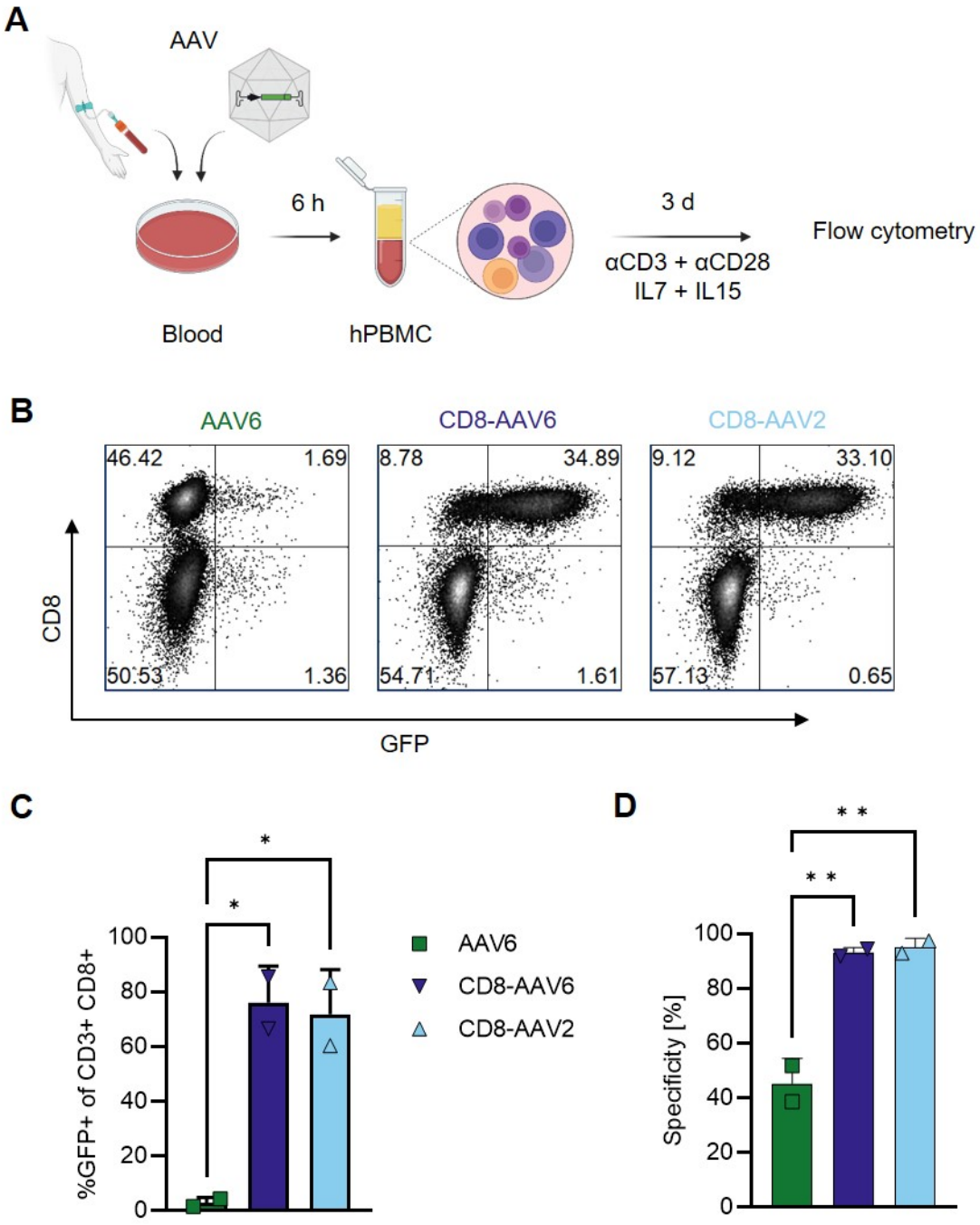



**Figure 26: Transduction of CD8⁺ T cells in whole blood. (A)** Experimental setup. **(B–D)** Flow cytometry results of two donors 3 days after addition of CD8-AAV or AAV6 particles ($2.5 \times 10^{11}$ vg/mL blood), PBMC purification, and activation. Representative flow cytometry plots depicting percentages of $GFP^+$ cells within $CD3^+/CD8^+$ cells of one donor **(B),** transduction efficiencies determined as percentage of GFP+ cells within $CD3^+/CD8^+$ cells **(C),** and transduction specificities determined as percentage of $CD8^+$ cells within all $GFP^+$ cells **(D)** are shown. Statistical analyses were performed using ordinary one-way ANOVA and Tukey's multiple comparisons test. Bars represent means, error bars represent SD, ∗p = 0.01–0.05, ∗∗p < 0.01. Adapted from *Demircan and Zinser et al.* [96].

### 2.15.1 No Significant Difference in Resistance to Serum Inhibition

To determine whether immunoglobulins or other neutralizing factors in serum contributed to the reduced transduction efficiency of unmodified AAV6, the impact of pooled human serum on vector performance was assessed. CD8-AAV6 and unmodified AAV6 were exposed to a range of serum concentrations to evaluate their sensitivity to neutralization. While both vectors showed reduced transduction in the presence of serum, CD8-AAV6 exhibited a slight tendency for higher resistance to serum inhibition compared to unmodified AAV6, though this difference was not statistically significant (Figure 27).

These findings highlight the influence of serum components, such as antibodies and complement proteins, on AAV vector efficiency and emphasize the need to account for such inhibitory factors in the development and optimization of AAV-based therapies for clinical applications.

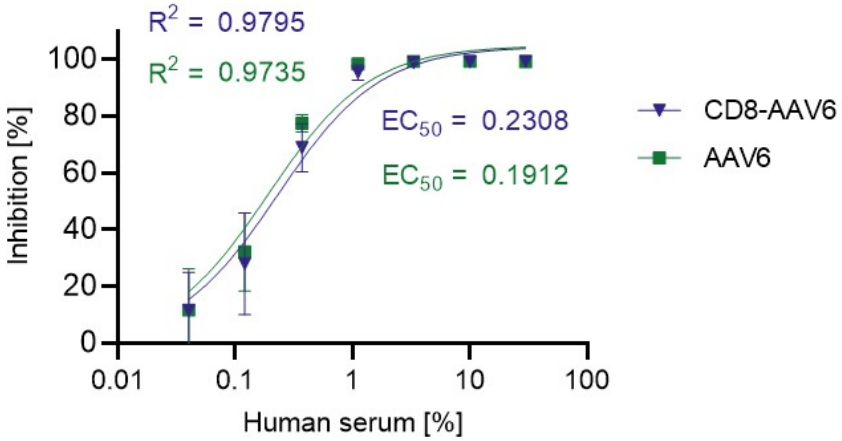



**Figure 27: Influence of human serum on gene transfer of CD8-AAV6 and unmodified AAV6.** Transduction of SUP-T1 cells with human serum pre-incubated CD8-AAV6 or AAV6. AAV vectors ($4 \times 10^4$ vg/cell) were pre-incubated with threefold serial dilutions of a commercially available pool of human off-the-clot serum before being added to SUP-T1 cells. GFP expression was analysed by flow cytometry three days post-transduction. Percent inhibition was calculated by normalizing GFP expression levels of CD8-AAV6 or AAV6 to their respective controls without serum. Nonlinear regression for $EC_{50}$ was fitted with the HillSlope set to 1 and the Bottom constrained to a constant value of 0. Data points represent the means of technical triplicates, with error bars indicating the 95 % CI. Adapted from *Demircan and Zinser et al.* [96].

## 2.16 CD8-AAVs Achieve Exceptional Selectivity in *In Vivo*

To evaluate the transduction efficiency and specificity of DART-AAVs in a physiologically relevant *in vivo* model, the performance of CD8-AAVs was compared to unmodified AAV6 in immunodeficient NOD.Cg-Prkdc$^{scid}$ Il2rg$^{tm1Wjl}$ (NSG) mice. These mice were transplanted with activated human PBMCs prior to systemic vector injection. The use of activated PBMCs ensured high levels of gene expression in T cells, providing a robust platform to assess the transduction capabilities of the AAVs through reporter gene expression.

Three days after the intravenous administration of AAV vectors, the mice were euthanized, and cells were isolated from blood and spleen for analysis. Flow cytometry was performed to evaluate the efficiency and specificity of transduction, with a focus on T cell populations, including $CD8^+$ cells, to determine the targeting accuracy and overall effectiveness of the AAV vectors (Figure 28A).

Although human PBMC recovery rates were low in some mice, making quantitative analysis infeasible for every individual, sufficient data were obtained from at least three animals per treatment group to identify clear trends (Figure 28B). *In vivo* transduction rates of $CD8^+$ cells with unmodified AAV6 were consistently below 1 % in blood and spleen. In contrast, CD8-AAV6 achieved approximately 25-fold higher transduction rates in blood, while CD8-AAV2 demonstrated an even 70-fold increase. Notably, CD8-AAV2 outperformed CD8-AAV6, achieving the highest transduction efficiencies in both blood and spleen (Figure 28C-D).

As expected, the superior performance of targeted vectors in this *in vivo* setting was driven by their high selectivity. In blood, specific transduction of $CD8^+$ cells reached up to 99 % with CD8-AAV2 and 98 % with CD8-AAV6. This specificity was further confirmed in PBMCs isolated from spleen, where in three out of five mice treated with CD8-AAV6, all transduced cells were CD8-positive (Figure 28E). Surprisingly, CD8-AAV2 demonstrated greater potency than CD8-AAV6 in this experimental setup, achieving significantly higher transduction rates on target-receptor-positive cells, with roughly a twofold difference in the percentage of $GFP^+$ $CD3^+$ $CD8^+$ cells (Figure 28D).

These results highlight the critical advantage of receptor-targeted AAVs in achieving efficient and selective transduction *in vivo*. The exceptional specificity and superior transduction rates particularly of CD8-AAV2, compared to unmodified AAV6, underscore its potential for targeted gene delivery in therapeutic applications.

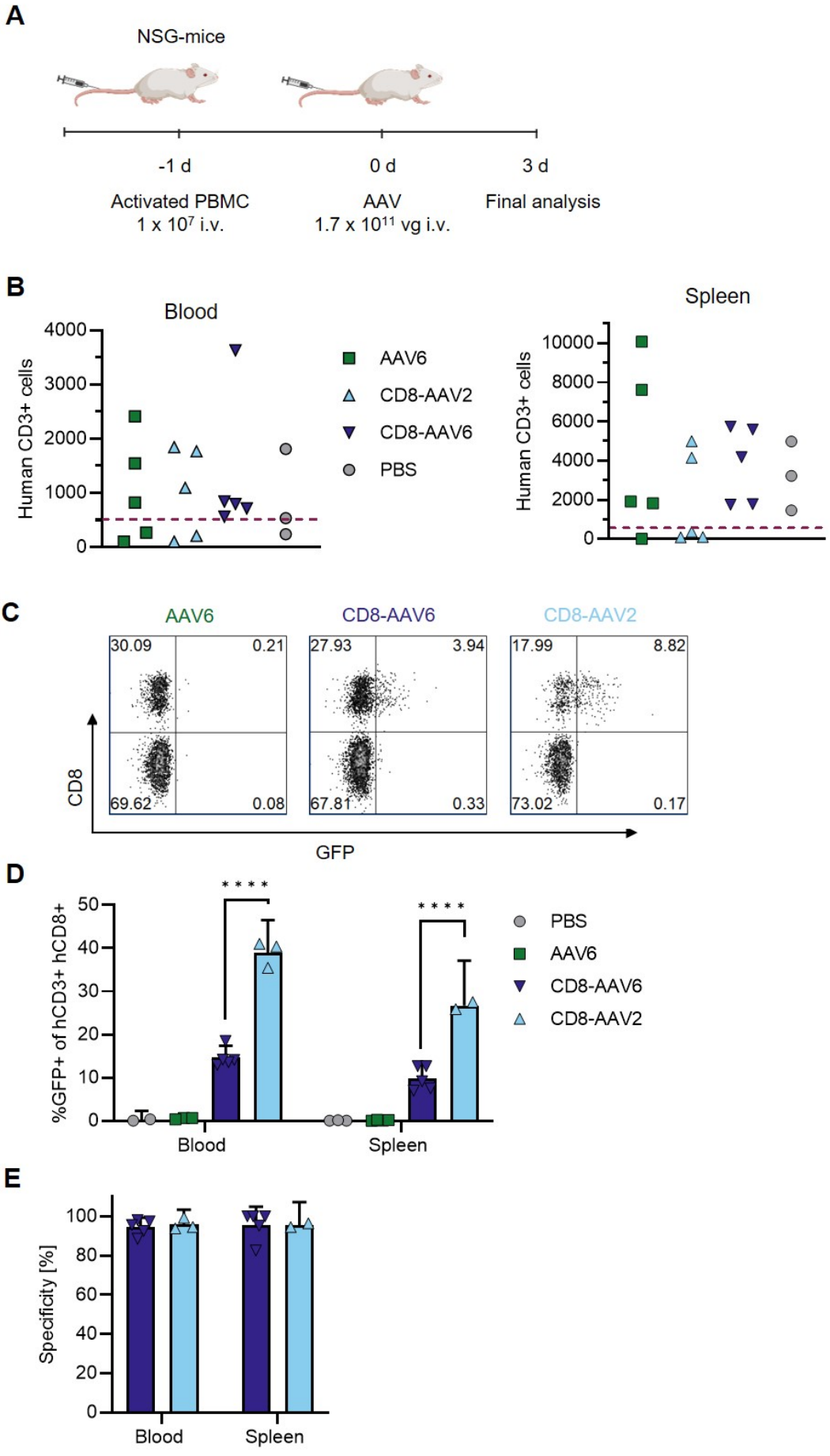

A
NSG-mice
-1 d
0 d
3 d
Activated PBMC
1 x 10^7 i.v.
AAV
1.7 x 10^11 vg i.v.
Final analysis
B
Blood
Spleen
Human CD3+ cells
AAV6
CD8-AAV2
CD8-AAV6
PBS
C
AAV6
CD8-AAV6
CD8-AAV2
30.09
0.21
69.62
0.08
27.93
3.94
67.81
0.33
17.99
8.82
73.02
0.17
CD8
GFP
D
%GFP+ of hCD3+ hCD8+
Blood
Spleen
E
Specificity [%]
Blood
Spleen

**Figure 28: *In vivo* gene transfer into activated human CD8⁺ T cells. (A)** Experimental setup. **(B)** Number of viable human $CD3^+$ cells retrieved from the blood (left) and spleen (right) of PBMC-transplanted NSG mice. A total of $5 \times 10^4$ blood cells were analysed by flow cytometry. Mice with fewer than 500 viable human $CD3^+$ cells were excluded from the analysis (red dotted line). **(C–E)** Flow cytometry results for human PBMCs isolated from mouse blood and spleen three days post-administration of AAV6 or CD8-AAV vectors. **(C)** Representative flow cytometry plots show percentages of $GFP^+$ cells among human $CD3^+/CD8^+$ cells for all three vector types in blood. **(D)** Transduction efficiencies, expressed as the percentage of $GFP^+$ cells within human $CD3^+/CD8^+$ cells in blood and spleen. **(E)** Transduction specificities, calculated as the percentage of human $CD8^+$ cells among all $GFP^+$ cells in blood or spleen. Statistical analyses were performed using one-way ANOVA followed by Tukey's multiple comparisons test. Bars represent mean values, error bars indicate the 95 % CI, and statistical significance is denoted as $****p < 0.0001$. Adapted from *Demircan and Zinser et al*. [96].

### 2.16.1 T-Cell Activation is Crucial for CD8-AAV Transduction Efficiency

To evaluate the impact of T-cell activation status on vector performance in a physiologically relevant *in vivo* model, non-activated human PBMCs were transplanted into NSG-MHC I/II double knock-out (DKO) mice to minimize T-cell activation caused by xenoreactivity. Three days after intravenous vector administration, the mice were euthanized, and isolated cells were analysed by flow cytometry (Figure 29A). In an additional experimental group, CD8-AAV2 administration was combined with *in vivo* T-cell conditioning using huIL-2 and a low dose of anti-huCD3 antibody (OKT3). Across all groups, no significant changes in body weight were observed compared to controls (Figure 29B). Human cell engraftment in the spleen reached approximately 5 %, with a CD8:CD4 ratio of roughly 1:2 (Figure 29C).

Although the number of $GFP^+$ cells was considerably lower in this model compared to the model using activated T cells, transduced cells were still clearly detectable within the human $CD3^+CD8^+$ compartment, particularly in the spleen. CD8-AAV2 achieved an average transduction rate of 2.2 % of human $CD8^+$ cells, while CD8-AAV6 showed even lower efficiency. Remarkably, the transduction efficiency of CD8-AAV2 dramatically increased when *in vivo* T-cell conditioning was applied, with transduction rates reaching up to 80 % of human $CD8^+$ cells in the spleen (Figure 29D-E). This remarkable improvement emphasizes the critical role of T-cell activation in enhancing vector performance and further underscores the exceptional specificity and efficiency of CD8-AAV2 in a conditioned *in vivo* environment.

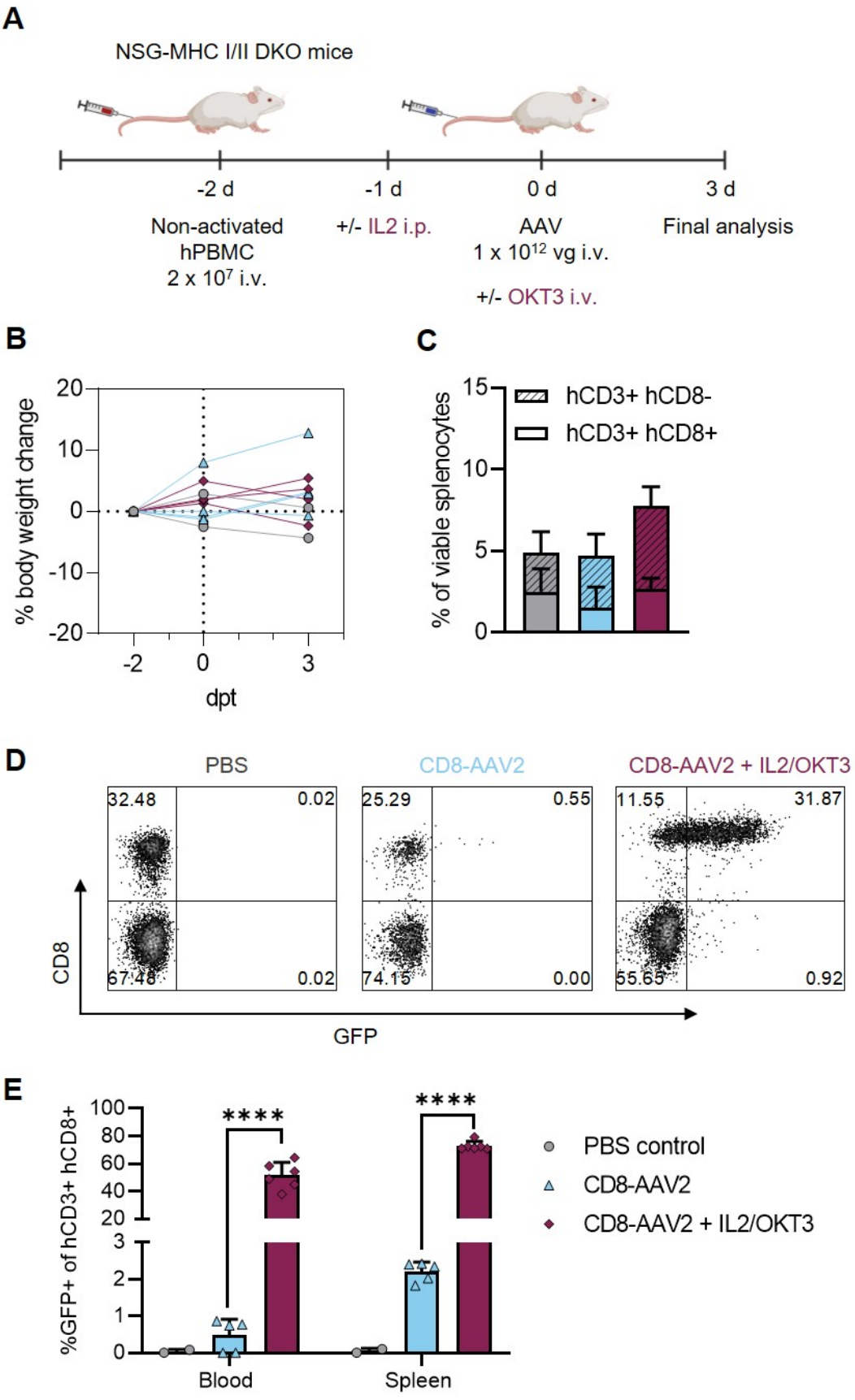



**Figure 29: *In vivo* gene transfer into non-activated human CD8⁺ T cells.** **(A)** Experimental setup. **(B)** Mouse body weight changes after treatment. **(C)** Bar diagrams

show cumulative data with the percentages of human $CD3^+/CD8^-$ cells and the percentage of human $CD3^+/CD8^+$ cells for all groups in the spleen, as determined by flow cytometry. **(D-E)** Flow cytometry analysis of human PBMCs isolated from mouse blood and spleen three days after administration of PBS (n = 2), CD8-AAV2 (n = 5), or CD8-AAV2 combined with IL-2/OKT3 (n = 6). **(D)** Representative flow cytometry plots with percentages of GFP+ cells among human $CD3^+/CD8^+$ cells for different groups in spleen. **(E)** Transduction efficiencies determined as percentages of GFP+ cells within human $CD3^+/CD8^+$ cells in blood or spleen. Statistical analyses were performed using one-way ANOVA followed by Tukey's multiple comparisons test. Bars represent mean values, error bars indicate the 95 % CI, and statistical significance is denoted as $****p < 0.0001$. Adapted from *Demircan and Zinser et al.* [96].

# 3 Discussion

## 3.1 Successful *De Novo* Selection of CD56-Specific DARPins

Identifying high-affinity binding proteins from large combinatorial libraries is a central objective in protein engineering and represents a key focus of this thesis. To enable this, several *in vitro* display technologies have been developed—most notably phage display [115], yeast surface display [116], and ribosome display—each offering distinct advantages depending on the scaffold, application, and desired throughput. While all these techniques enable the screening of large libraries to identify binders with desired specificities, ribosome display is particularly well-suited for DARPins due to its fully cell free nature, allowing the construction and screening of extremely large libraries (more than $10^{12}$ variants) without the transformation limitations of cell-based systems [105]. Furthermore, ribosome display enables direct linkage between the genotype (mRNA) and phenotype (DARPin), facilitating efficient selection and recovery of high-affinity binders. Its high-throughput potential and adaptability make it the method of choice for DARPin selection, especially when rapid and iterative affinity maturation steps are required. However, the linkage of the genotype and the phenotype is reliably maintained only under non-denaturing buffer conditions, in the absence of RNases and at low temperature [115].

A critical prerequisite for successful ribosome display-based selection of high-affinity binders is the availability of a natively folded, high-purity target protein. In this study, the recombinant mCD56 ectodomain met all relevant quality criteria, confirming its suitability for efficient binder selection. To further assess the bioactivity and accessibility of the target under ribosome display conditions, a simple binding assay using a CD56-specific antibody was performed. The native protein was successfully detected, suggesting the presence of accessible, properly folded epitopes. However, the assay has interpretational limitations, as the antibody used is reported to recognize both native and denatured CD56 in Western blot applications. Therefore, while the results support target accessibility, they do not unequivocally confirm correct native folding.

Unfortunately, further refinement of this experiment was not straightforward. The only conformation-dependent CD56 antibody, NCAM16.2, specifically recognizes human CD56, and no evidence suggests that it cross-reacts with mouse CD56. Additionally, more complex bioactivity assays, such as neurite growth experiments, were deemed impractical given the increased complexity of such methods and intrinsic stability of soluble CD56 variants. [117,118]. As a result, the current bioactivity assay, while limited, was the most feasible option under these constraints.

From the fifth round of ribosome display, 282 clones were selected and expressed in *E. coli*. While most crude lysates showed clear DARPin expression, only four exhibited binding to more than 5 % of $CD56^+$ target cells, and just one produced a distinct population shift. Subsequent analysis revealed that none of the binders retained an intact full-length DARPin sequence. This outcome is likely attributable to the stringent off-rate selection conditions applied in the fifth round. Specifically, this round employed a 100-fold molar excess of non-biotinylated competitor protein and a dissociation phase lasting one hour.

Such conditions tend to favour binders with exceptionally slow off-rates, allowing more time for weaker or moderately affine binders—such as full-length DARPins—to dissociate from the target, thereby eliminating them from the selection pool. This hypothesis is supported by SPR measurements presented in section 2.4, which demonstrated that a DARPin isolated from the fourth round, prior to the prolonged off-rate selection, maintained detectable binding despite a moderate dissociation rate.

Notably, *Dreier and Plückthun* advocate for a more balanced approach to off-rate selection to preserve binder diversity and functionality [119]. They recommend using a 10- to 100-fold excess of competitor and dissociation times of up to two hours [119]. While the total incubation time may be longer, the lower competitor concentration imposes gentler selective pressure, allowing useful full-length binders with moderate off-rates to remain in the selection pool. In contrast, the shorter one-hour dissociation combined with a high competitor excess, as used in this study, creates a bottleneck that may enrich only atypical or truncated binders with artificially low off-rates—at the expense of functional, structurally intact DARPins.

Truncated sequences or smaller polypeptides often arise during ribosome display due to PCR errors, (reverse-)transcription errors, or degradation. These fragments may bind to the target protein differently, such as through non-specific hydrophobic interactions, and can exhibit artificially low off-rates. This enables them to survive stringent selection conditions, outcompeting full-length DARPins. Non-specific binders with slow off-rates can also become enriched under these conditions [120]. Additionally, their simpler structure often results in higher expression and solubility in *E. coli,* contributing to their enrichment. This was consistent with experimental results, as all four truncated binders displayed even higher expression levels than the DARPin positive control (Figure 6B).

Importantly, three DARPins with strong binding properties to murine GL261 cells were successfully selected from the output library of the fourth round of ribosome display. This suggests that the failure to select functional DARPins after the fifth round was most likely due to the overly stringent off-rate conditions applied during the final round, as explained.

above. These findings emphasize the need for balanced selection stringency to retain full-length, high-affinity binders during ribosome display.

### 3.1.1 Binding Properties of Purified DARPins

The 43H3 and 43E9 DARPins were expressed and purified via their His-tags using Ni-NTA agarose and SEC. The purity of the proteins was confirmed through SDS-PAGE and Western blot analysis. Interestingly, despite both DARPins having the same number of variable repeats and a calculated molecular weight of approximately 21 kDa, they exhibited different migration patterns on SDS-PAGE. While the 43E9 DARPin bands appeared at the expected ~21 kDa, 43H3 migrated closer to ~15 kDa.

This discrepancy in migration behaviour was observed only under denaturing conditions; during SEC with natively folded proteins, both DARPins eluted at similar times under identical chromatography settings. DARPins are known for their extraordinary stability and resistance to boiling or denaturing agents like guanidine hydrochloride [99]. Additionally, a similar migration pattern to 43H3 has been previously reported for another DARPin from the same library [106].

Thus, it is reasonable to conclude that 43H3 did not fully denature under the chosen SDS-PAGE conditions, resulting in an altered migration pattern despite having a molecular weight similar to that of 43E9. This behaviour aligns with the unique structural properties of DARPins, which can retain partial conformational stability even under denaturing conditions. This hypothesis can be proven using differential scanning fluorimetry or native PAGE. Most importantly, both DARPins were of high purity following SEC, ensuring their suitability for reliable SPR measurements.

When the cross-reactivity of the purified DARPins with human CD56 was assessed, only 43H3 demonstrated reliable binding to human NK cells. Finding a cross-reactive DARPin is not surprising, as the ectodomain and full-length murine and human CD56 share approximately 95 % sequence identity. Moreover, the transmembrane and ectodomain regions of CD56 are generally highly conserved across vertebrate species, reflecting their critical functional roles in cell adhesion and signalling [36]. Notably, the variable regions of 43H3 and 43E9 differ substantially. Of the 20 variable amino acid positions in the N3C library, only three are identical between the two DARPins. This strongly suggests that they recognize distinct epitopes, with 43E9 likely targeting an epitope specific to murine CD56, explaining its lack of cross-reactivity with the human protein. This hypothesis is supported by *in silico* binding predictions for the 43H3 DARPin, which are discussed in detail in section 3.6.

The performed affinity analysis of 43H3 with SPR showed an $K_D$ of approximately 11-15 nM to human CD56, which is well in the expected range for DARPins [119]. Additionally, the off-rate was relatively high at 1.1 x $10^{-2}$ $s^{-1}$, especially when compared to most therapeutic antibodies and the anti-CD56 antibody Promiximab ($k_{off}$ < 1.0 x $10^{-7}$ $s^{-1}$) [121,122]. This outcome aligns with expectations, given the unsuccessful off-rate selection during the 5th round of ribosome display, as discussed above.

Despite this, the measured off-rate translates to a half-life of approximately 63 seconds for the DARPin-CD56 complex, calculated as:

$$t_{1/2} = \frac{\ln(2)}{k_{off}} \approx 63\ s$$

This residence time should be sufficient for the DARPin to mediate attachment of AAV particles to target cells. Moreover, a moderate off-rate may even offer specific advantages in tissue penetration of the AAV vector. Ligands with extremely high affinity (and correspondingly low $k_{off}$) may bind so tightly to cell surface receptors that their distribution within a tissue is limited. By contrast, ligands with a higher $k_{off}$ can dissociate and rebind to receptors, enabling more effective tissue penetration [123].

Additionally, AAV vectors typically rely on receptor-mediated binding and endocytosis for cell entry [54]. While high-affinity ligands are efficiently internalized, their prolonged binding can saturate receptors and impede subsequent rounds of uptake. Ligands with a higher $k_{off}$ enable receptor recycling, promoting multiple rounds of endocytosis and enhancing overall internalization efficiency [124].

In summary, while high-affinity ligands with very low off-rates ensure strong receptor binding, moderate off-rates can enhance tissue penetration and internalization efficiency, key features for effective gene delivery. Based on these considerations, 43H3 was selected for subsequent vector retargeting experiments.

## 3.2 Structural Implications of DARPin Incorporation into AAV Capsids

As explained in Section 4.5, the incorporation of the CD56-specific 43H3 and the CD8-specific DARPin into AAV particles was performed following the method described by *Michels et al.* [69]. Structurally, peptide incorporations into AAV particles have mainly been studied using immunoelectron microscopy, a technique previously applied to identify the GH2/GH3 loop as a potential insertion site for mCherry [125]. Biophysical characterization of the CD8-AAV revealed that inserting DARPins into the GH2/GH3 loop of VP1 does not disrupt the capsid core structure. Notably, the CD8-AAV2 capsid core retains high structural

integrity, consistent with the native AAV2 structure, as determined by cryo-EM single-particle analysis at 3.3 Å resolution.

However, due to the limited incorporation of utmost five randomly positioned DARPin molecules per particle when inserted only into VP1, detecting the DARPins was not feasible using cryo-EM. This limitation was addressed in a study by colleagues from the host group for this thesis, where DARPin detection became possible through cryo-ET when the DARPins were additionally inserted into VP2 and VP3, increasing their abundance theoretically to an average of one DARPin per two capsid proteins [96].

Cryo-EM analysis clearly demonstrates that the capsid core structure tolerates DARPin insertions but also reveals certain limitations. For example, particles carrying DARPins in all VP proteins failed to assemble, and hybrid capsids containing a 1:1 ratio of unmodified and DARPin-containing VP proteins exhibited reduced gene transfer activity [96].

This observation suggests that high-affinity binders targeting attachment receptors—such as CD56 or CD8—should be displayed at a controlled density on the AAV capsid to ensure efficient cell entry. If the density of these targeting ligands exceeds a certain threshold, it may impair downstream steps of the transduction process, including receptor-mediated endocytosis, intracellular trafficking, or interaction with essential co-receptors. In the case of AAVs, cell entry is not solely mediated by the initial attachment receptor, but also requires engagement with the AAVR, a key component of the AAV entry pathway that facilitates endosomal escape and nuclear trafficking [57,95,126]. Excessive or overly strong binding to surface receptors could hinder capsid mobility or prevent proper engagement with AAVR, thereby reducing overall transduction efficiency.

Supporting this hypothesis, a recent study demonstrated that gene transfer activity peaked when AAV particles were decorated with approximately 12 high-affinity ligands per capsid, introduced via chemical coupling [127]. This finding suggests the existence of an optimal ligand density that balances stable target cell binding with the flexibility needed for productive entry, underlining the importance of precise control over ligand valency in capsid engineering for targeted gene delivery.

## 3.3 Specific Transduction and Mechanistic Insights into DART-AAVs

This section explores the transduction specificity and mechanistic properties of DART-AAVs, with a particular focus on how receptor, identity, interactions, density, and endocytic trafficking influence vector performance. While previous studies have demonstrated the utility of ligand-modified AAVs—including nanobody- [128], peptide- [92], and DARPin-targeted capsids—for cell-type-specific gene delivery [93], this work extends these concepts

by employing rationally designed DARPins inserted into a hypervariable capsid region, further optimizing targeting specificity and transduction efficiency.

As discussed above, the low-level insertion of DARPins exclusively into the VP1 subunit of AAV particles was sufficient for receptor-targeted gene delivery by CD8-, CD4-, and CD56-AAVs. While CD8- and CD56-AAV particles achieved transduction rates comparable or even higher to unmodified AAV2 and AAV6 in receptor-positive cell lines, the transduction efficiency of CD4-AAV2 was significantly lower than that of unmodified AAV2, suggesting an influence of receptor-specific mechanisms on transduction.

A potential explanation for the reduced transduction of CD4-AAV2 lies in the number of available attachment sites per cell. Although precise quantification of CD4 and HSPG molecules on A3.01 cells is unavailable, human CD4+ T cells are estimated to express between 50,000 and 100,000 CD4 molecules per cell [129], whereas HSPG expression on cell surfaces typically ranges from 100,000 to 1,000,000 molecules per cell [130]. The higher receptor density for HSPG likely facilitates more efficient capsid binding and uptake, resulting in enhanced transduction rates for HSPG-mediated AAV2 compared to CD4-AAV2. Interestingly, this explanation does not account for observations in MOLT4.8 cells, where CD8-AAV and unmodified AAV2 exhibited comparable transduction efficiencies despite expected differences in receptor density. CD8 expression on T lymphocytes typically ranges between 10,000 and 30,000 molecules per cell, indicating that factors beyond receptor density may influence transduction efficiency. However, for precise quantification, experimental measurements such as flow cytometry or mass spectrometry would be necessary.

As already discussed in section 3.1.1, binding affinity to the attachment receptor is another important factor. The CD4-specific DARPin (55.2) used in this study exhibits an affinity of 0.84 nM and a $k_{off} = 1.2 \times 10^{-3}\ s^{-1}$ [109], while the unmodified AAV2 capsid binds HSPG with a similar affinity of 0.1 to 3.7 nM and a $k_{off} = 1.2 \times 10^{-3}\ s^{-1}$ [131]. Despite these similar binding properties, the interaction between AAV2 and HSPG is cooperative, with a single glycosaminoglycan chain capable of interacting with multiple capsid sites. This cooperative binding could amplify AAV2 attachment and uptake via HSPG, giving it an advantage over CD4-mediated uptake. At the time of writing this thesis, precise affinity data for the CD8-specific DARPin (63A4) were not yet available. Preliminary data, however, suggest a binding affinity of approximately 7 nM, indicating a lower affinity compared to the CD4-specific DARPin. Nevertheless, direct comparisons of binding kinetics between CD8- and CD4-specific DARPins remain limited due to incomplete characterization.

Receptor-specific endocytosis rates may also contribute to the differences in transduction. CD4 receptors undergo constitutive endocytosis at approximately 4 % of the cell surface pool per minute, leading to complete internalization within 25 min [132]. In contrast, HSPGs exhibit slower endocytosis rates, with half-lives ranging from 3-20 h [133]. CD8 itself is internalized at a slower basal rate but can be co-internalized alongside the TCR upon receptor activation [134,135]. While TCR internalization occurs constitutively at a relatively slow rate (~0.6 % per minute), this process accelerates significantly following ligand engagement [136]. Constitutive and activation-induced TCR internalization utilize distinct mechanisms [134,136]; however, the precise impact on CD8 internalization and the subsequent effect of these receptor-specific endocytosis rates on DART-AAV-mediated transduction require further investigation.

After endocytosis, successful transduction depends on the vector's ability to escape the endosomal network and avoid lysosomal or proteasomal degradation. Unmodified AAV2 is internalized predominantly through clathrin-mediated endocytosis in most cell types, although other pathways like the clathrin-independent carrier (CLIC)/GPI-anchored protein-enriched early endosomal compartment (GEEC) pathway may also play a role. Notably, the CLIC/GEEC pathway has been associated with productive transduction in certain cell types, such as HeLa and HEK293T cells [53]. However, some AAV particles are directed to lysosomes, leading to degradation and preventing successful gene delivery. A fraction of internalized AAVs may undergo ubiquitination and subsequent degradation by the proteasome. This non-productive processing can expose viral genomes to cytosolic DNA sensors, potentially triggering immune responses [137]. The type of attachment receptor plays a pivotal role in determining whether AAV endocytosis leads to productive infection or non-productive degradation. However, based on the current data, it is unclear whether CD8, CD4, or HSPG favours more productive AAV trafficking in this experimental context.

The significantly lower transduction rate of CD4-AAVs compared to unmodified AAV2 likely results from a combination of factors, including lower CD4 receptor density, differences in receptor-specific endocytosis, and the cooperative binding mechanism of AAV2 to HSPG. In contrast, CD8-AAV demonstrated superior performance, possibly benefiting from more favourable receptor engagement, efficient internalization kinetics, or improved intracellular trafficking. Further experiments are necessary to elucidate the precise receptor-specific trafficking and intracellular processing mechanisms contributing to these differences in transduction efficiency.

## 3.4 DART-AAV Retargeting Strategy is Compatible with AAV6

In contrast to AAV2, AAV6 is considered more potent for the transduction of haematopoietic stem cells and T lymphocytes of both human and murine origin [79,80]. As demonstrated in section 2.9 the DARPin retargeting strategy successfully established with AAV2 can also be applied to AAV6. This serotype switch was anticipated to provide advantages in modulating intracellular trafficking, degradation, and disassembly of AAV particles following endocytosis, thereby enhancing productive transduction in specific cell types [138,139].

Although AAV6 utilizes both HSPG and sialic acid as primary binding receptors—with sialic acid being the main determinant of transduction—capsid "blinding" was achieved through the V473D and K531E mutations, as shown in Figure 16, consistent with published literature [110,111]. Notably, when retargeted to CD8, the CD8-AAV6 significantly outperformed unmodified AAV6, exhibiting a more than 36-fold increase in transduction efficiency on $CD8^+$ SUP-T1 cells. This significant enhancement underscores the benefit of rational capsid engineering and receptor-specific targeting. Moreover, it parallels earlier findings with targeted AAV2, reinforcing the general principle that receptor-specific retargeting can substantially alter and often improve vector performance compared to wild-type vectors, depending on the properties of the chosen receptor and ligand pair.

Serum effects were also taken into consideration, as they can significantly influence AAV transduction efficiency. *In vitro* transduction experiments were performed with heat-inactivated FBS in the culture medium, which is noteworthy given AAV6's known sensitivity to FBS [79]. Interestingly, this inhibitory effect appears to be specific to unmodified AAV6 and FBS and complement proteins—typically inactivated by heat—are unlikely to be the primary inhibitory factor. Instead, other serum components like the soluble scavenger receptor galectin 3 binding protein (G3BP) or the C-reactive protein (CRP) may be responsible for reducing AAV6 activity [140]. In contrast, some serum components have been reported to enhance transduction. For instance, human serum albumin and low concentrations of human serum can promote AAV uptake by stabilizing capsids or facilitating receptor interactions for several serotypes [141,142]. However, at higher concentrations, human serum inhibits AAV transduction in a dose-dependent manner across most serotypes [142]. This inhibitory effect is primarily attributed to pre-existing neutralizing antibodies that can block capsid-receptor interactions or prevent endosomal escape [143]. These complex and often serotype-specific interactions between serum components and AAV vectors highlight the importance of evaluating serum compatibility in both *in vitro* and *in vivo* transduction models, particularly when optimizing vectors for therapeutic applications.

Since DARPins were inserted exclusively into the VP1 subunit of the AAV6 capsid, constituting approximately five of the 60 capsid proteins [67], the DARPin insertion itself is unlikely to impact serum effects. However, the blinding mutations V473D and K531E, present in every capsid protein of DART-AAV6, may influence interactions with serum or neutralizing antibodies. K531 was previously identified as a determinant for recognition by the mouse monoclonal antibody ADK6 [144]. While specific studies on the direct impact of V473D on antibody binding are limited, mutations in the capsid region encompassing this residue could potentially alter the capsid's surface topology. Both mutations are in the protrusion of Variable Region VIII near the threefold symmetry axis, a region prominently displayed on the capsid surface. As such, these mutations may influence the binding affinity of neutralizing antibodies [68]. However, a subsequent titration experiment evaluating the effects of human serum on SUP-T1 transduction with unmodified and CD8-AAV6 showed no significant impact of the V473D or K531E mutations, or the DARPin insertion, on serum neutralization (Figure 27).

The application of DARPin retargeting to AAV6 enhances its utility for transducing specific cell types, as demonstrated by the significant increase in CD8-AAV6 transduction efficiency. While the mutations V473D and K531E were effective for blinding AAV6 from primary receptors and potentially altering antibody binding, their functional impact on serum neutralization appears minimal. This highlights the versatility of AAV6 as a platform for further engineering and underscores the need for additional studies to fully elucidate its role in a broader therapeutic context.

## 3.5 Enhanced Performance of DART-AAVs in Primary Cells

The superiority of DART-AAVs became most evident when tested on primary cells in the presence of off-target cells. All DART-AAVs evaluated on activated human PBMCs (CD8-AAV2, CD4-AAV2, and CD8-AAV6) demonstrated high selectivity for target receptor-positive cells. Consistent with observations on cell lines, CD8-AAV2 achieved superior transduction efficiency and on-target selectivity (98 %) compared to CD4-AAV2, which not only showed reduced efficiency but also a lower selectivity of 88 %.

Even with the high affinity of the CD4 DARPin [109] , there may be low-level off-target interactions with non-CD4 molecules, resulting in some non-specific binding and internalization through clathrin-independent or macropinocytosis pathways [145]. This may reflect either a target-specific or DARPin-specific effect. Supporting this, when CD4-AAV was modified with capsid tyrosine mutations to enhance intracellular stability and prevent proteasomal degradation [71,146], the increase in on-target transduction was accompanied by a proportional increase in off-target transduction [96]. Residual tropism of the AAV

capsid, due to incomplete blinding for other receptors besides HSPG, appears less likely as it would similarly affect CD8-AAVs and $AAV_{mut}$ particles.

As discussed above for cell lines in section 3.3, the difference in transduction efficiency between these AAVs is likely multifactorial. However, it is notable that on primary T cells, CD4-AAV2 significantly outperformed unmodified AAV2, whereas on cell lines the reverse was observed. Cell lines can differ from primary cells in several aspects that influence the transduction of (DART-)AAVs, including target receptor density, function, localization, and intracellular trafficking [147]. This shift in relative performance is consistent with findings by *Hamann et al.,* who employed a CD4-specific nanobody to retarget AAV2 to $CD4^+$ T cells [128]. Importantly, both studies highlight that primary human T cells exhibit markedly different transduction profiles compared to immortalized cell lines, underscoring the necessity of validating targeted AAV vectors in physiologically relevant primary cell systems.

Unmodified AAV6 demonstrated significantly higher efficiency in transducing primary T lymphocytes compared to AAV2, consistent with previously published findings [80,148]. This enhanced performance is partially attributed to the superior nuclear import efficiency of AAV6 [148]. After endocytosis and endosomal escape, AAV6 capsids appear to engage more effectively with the host cell's nuclear import machinery, facilitating a greater proportion of vector genomes reaching the nucleus. The exact mechanisms remain under investigation, but differences in capsid structure and receptor interactions likely contribute to these serotype-specific trafficking efficiencies. Surprisingly, CD8-targeting rendered AAV2 as effective as AAV6 in mediating gene transfer, despite AAV6's known superiority in transducing hematopoietic cells. This finding suggests that in T cells, the key determinant of transduction efficiency may be the effective concentration of vector particles at the cell surface, mediated by the DARPin. This surface accumulation likely compensates for less favourable post-entry processing, indicating that intracellular barriers can be overcome by enhanced receptor-mediated uptake.

The cytotoxicity exhibited by AAV6 at a dose as low as $1 \times 10^5$ vg/cell supports this hypothesis. In contrast, CD8-AAV6 allowed T cells to tolerate over 25 times this dose without any adverse effects on cell viability [96]. This highlights that modifying the attachment site on target cells can mitigate not only transduction efficiency but also cytotoxicity. The adverse effects of AAV6 are likely attributable to its natural attachment receptors—primarily sialic acid and, to a lesser extent, HSPGs—which are ubiquitously expressed across many cell types, including T cells. As discussed in section 3.3, high receptor density may lead to excessive capsid binding and uptake, potentially triggering

intracellular stress or innate immune activation. In contrast, targeting CD8 may offer a more selective and physiologically relevant entry route, limiting off-target interactions and excessive intracellular burden. These differences in receptor engagement may also explain the lower transduction rates observed for CD8-AAV6 under high-titer *in vitro* conditions, where unmodified AAV6 benefits from abundant and nonspecific receptor availability.

Further research is required to elucidate whether the observed cytotoxicity is primarily due to viral attachment, transgene toxicity, or more complex downstream processes.

### 3.6 Optimization of CD56-AAV

This thesis presents the first generation of CD56-targeted AAVs. Due to the moderate transduction efficiency observed on CD56-expressing HeLa cells, the vector was subsequently optimized to improve performance on more challenging NK cell lines. Inspired by the successful combination of a CD4 DARPin and a CD32a DARPin [112], an additional DARPin was inserted alongside CD56.43H3 into the GH2/GH3 loop of the VP1 AAV2 capsid protein, creating 2xCD56-AAV. This optimisation increased the transduction rates more than 100-fold compared to the mono-DARPin equivalent, CD56-AAV. Binding studies demonstrated significantly enhanced association of 2xCD56-AAV with target cells, while the binding signal of CD56-AAV remained at background levels and was undetectable in the performed assays. This dramatic increase in detectable binding strongly suggests an avidity effect. Interestingly, the combination of CD56.43H3 with an NKp46 DARPin did not improve transduction efficiency in CD56/NKp46 double-positive cell lines.

AAV particles are small, comprising only 60 capsid subunits per particle [149]. Recent mass spectrometry data revealed variability in their polypeptide composition, with VP1—used for DARPin insertion—constituting, on average, five molecules per AAV2 particle [67]. Thus, a CD56-NKp46-AAV would present a maximum of ten binding sites each for CD56 and NKp46, while the 2xCD56-AAV could display up to twenty CD56-binding domains, randomly distributed across the capsid surface. Owing to this stochastic distribution, it is sterically unlikely that all binding sites can engage their respective receptors simultaneously upon contact with the cell membrane.

Additionally, the two DARPins are connected by a $(G_4S)_3$ linker, which spans a maximum distance of 50 Å when fully extended [150]. Functionally, NKp46 is a key component of the NK cell activation machinery, initiating cytotoxic responses upon ligand recognition [32], while CD56 functions predominantly as an adhesion and identification marker, facilitating immune synapse formation and cell–cell interactions [29]. These distinct biological roles are reflected in their membrane localization patterns, which are likely to be spatially separated to some extent on the NK cell surface. Consequently, it is improbable that both CD56 and

NKp46 are positioned closely enough to allow simultaneous engagement by a single VP1 capsid subunit displaying both DARPins. This is further supported by the lack of evidence for physical interactions between CD56 and NKp46 [151,152].

This contrasts with the previously published CD32a-CD4-AAV, which achieved improved transduction through a proposed tetra- or trivalent interaction with the cell surface, resulting in increased avidity [112]. The superior performance of CD32a-CD4-AAV on double-positive cells likely results from receptor biology, density, co-localization, and the inherent properties of the targeted cell type. In contrast, the functional and spatial separation of CD56 and NKp46 likely limits the efficacy of a CD56-NKp46-AAV.

Understanding the precise interaction between 43H3 and CD56 is essential for interpreting the performance of the CD56-AAV and 2xCD56-AAV. When further investigating the 43H3 DARPin and its corresponding CD56-AAVs, particularly the 2xCD56-AAV, it was crucial to identify and evaluate the DARPin's binding epitope. To narrow down potential binding mechanisms and target sites, AI-driven tools have become increasingly valuable solutions, playing an ever-growing role in life sciences.

The significance of reliable protein folding and docking prediction tools was further underscored by the 2024 Nobel Prize in Chemistry, awarded to David Baker, Demis Hassabis, and John M. Jumper for their contributions to "computational protein design" and "protein structure prediction" [153]. In the case of 43H3 and CD56, this computational approach generated high-confidence structural models for both the DARPin and the CD56 ectodomain. The binding predictions indicated that the DARPin specifically interacts with the distal Ig-like C2-type-2 domain of CD56, precisely within the region spanning amino acids 154–165.

Interestingly, AlphaFold3 predicted that the key amino acids of both 43H3 DARPins interact with the same CD56 domain in a clamp-like manner (Table 3, Figure 24, Figure 30A). However, this may reflect a limitation of the algorithm rather than the true binding mechanism. Because the docking simulation was performed using only a single CD56 molecule as the target, the model predicted the most probable interactions for both DARPin copies simultaneously. This could create the misleading assumption that both copies of 43H3 bind to the same CD56 molecule at the same time, whereas in reality, they likely bind sequentially to two separate CD56 molecules.

While it is theoretically possible to perform docking simulations using two CD56 molecules, the accuracy of such results would depend on whether CD56 is correctly modelled in its native dimeric state—either before or after its initial interaction with one of the DARPins. Without an accurately structured dimer, the simulation may fail to fully capture the

sequential binding process, leading to potential misinterpretations of the true interaction dynamics. Conversely, docking simulations using only a single copy of the CD56 DARPin failed to yield reliable binding predictions, possibly because the algorithm deemed the interaction too weak to be biologically relevant. This highlights the limitations of *in silico* modelling when studying complex protein-protein interactions and underscores the necessity of integrating computational predictions with experimental validation to refine structural models and enhance their biological accuracy.

The experimental confirmation that the Ig-like C2-type-2 domain of CD56 is essential for the binding of the 43H3 DARPin and its corresponding DART-AAVs provided at least partial validation of the AlphaFold3 predictions. Interestingly, these experiments not only demonstrated that amino acids 1–212 of CD56 are sufficient for 2xCD56-AAV-mediated transduction but also revealed that binding of the purified 2x43H3 protein was clearly detectable. More importantly, binding was significantly stronger for the 2x43H3 construct compared to the single DARPin, strongly indicating an avidity effect resulting from the dual DARPin configuration.

It is worth noting that the chimeric CD56-CD4 receptors used in these experiments may or may not form dimers. Although CD4 primarily functions as a monomer, CD4 dimers have been observed under certain conditions. Notably, the essential cysteines in the cytoplasmic tail of CD4, which are implicated in dimer formation, were present in both fusion constructs [154]. Additionally, the Ig-like C2-type-2 domain of CD56 plays a key role in the dimerization of CD56, contributing to its function in cell-cell adhesion and signalling [155]. Given that both parental proteins—CD56 and CD4—exhibit dimerization potential, it is possible that these artificial fusion proteins may also form dimers, mimicking their naturally occurring counterparts. However, without direct structural confirmation, the exact dimerization behaviour of these fusion constructs remains uncertain.

These findings suggest that doubling the CD56 DARPins may indeed facilitate the simultaneous engagement of multiple CD56 molecules on the cell surface, a mechanism further supported by the presence of CD56 homodimers. Homodimers provide paired receptor sites that are ideally suited for bivalent binding, which enhances avidity, receptor clustering, and vector uptake—key factors in improving transduction efficiency.

The design of two identical CD56 DARPins, connected by a flexible linker, ensures optimal reach and spatial alignment, allowing both DARPins to effectively engage nearby CD56 molecules on the same cell. This dual binding strategy may increase local receptor density and enhance receptor-mediated endocytosis, improving vector internalization (Figure ***30***B).

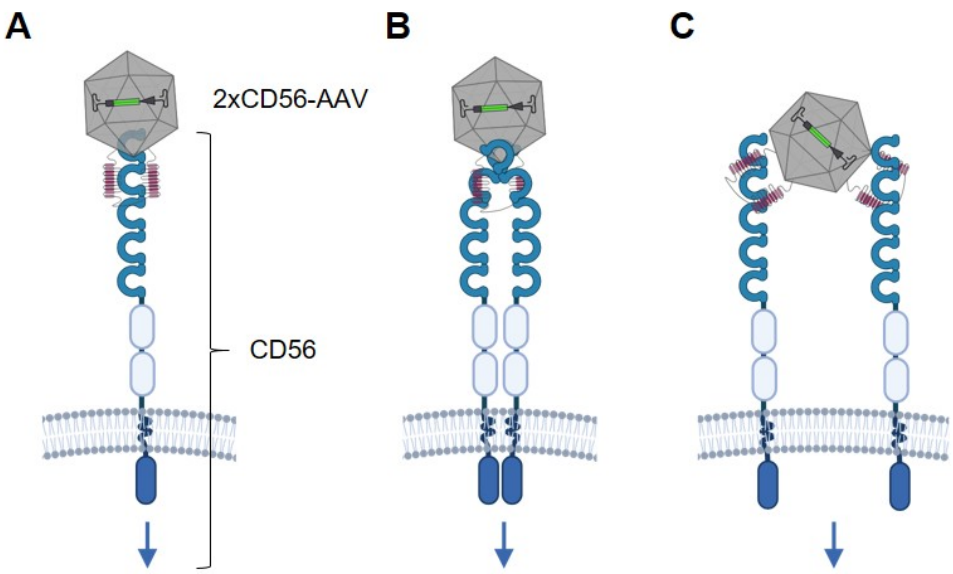



**Figure 30: Possible Binding Modes of 2xCD56-AAV to CD56. (A)** Model of 2xCD56-AAV (grey capsid surface) displaying two 43H3 DARPins in tandem configuration (red) bound to a single CD56 monomer (blue), as predicted by AlphaFold3**. (B)** Possible interaction of 2xCD56-AAV with a CD56 cis-homodimer, enabling bivalent engagement of adjacent CD56 molecules. **(C)** Hypothetical binding of 2xCD56-AAV to two spatially separated CD56 monomers, potentially across adjacent membrane regions.

This mechanism is reminiscent of previous findings with CD20-targeting antibodies, where receptor homodimerization was shown to significantly enhance binding affinity and functional activity [156]. Similarly, in the case of CD56, receptor clustering induced by DARPin engagement may contribute to increased transduction efficiency.

Alternatively, as shown for the previously published CD32a-CD4-AAV construct [112], multivalent interactions via spatially separated CD56 monomers could also enhance avidity through simultaneous binding of several DARPins (Figure 30C). However, this scenario appears less likely, as the purified tandem 43H3 DARPin—expressed and tested as a soluble protein, without any capsid display—already showed increased binding (Figure 25D).

Together, these factors suggest that CD56 homodimerization may be a key contributor to the enhanced transduction efficiency observed with 2xCD56-AAV. However, it is unlikely to be the sole explanation. Interestingly, preliminary results with 2xCD8-AAV and 2xNKp46-AAV did not show any significant increase in transduction rates compared to their mono-DARPin counterparts, despite both CD8 and NKp46 also forming homodimers [101,157]. This suggests that receptor dimerization alone is insufficient to account for the superior performance of 2xCD56-AAV.

The unique improvement seen with 2xCD56-AAV implies that additional target-specific factors must also be at play. These could include differences in receptor density, clustering dynamics, or intracellular trafficking pathways specific to CD56. It is also possible that the spatial organization of CD56 homodimers on the cell surface is more favourable for bivalent DARPin binding compared to CD8 or NKp46, leading to stronger avidity effects.

Overall, while homodimerization may contribute to the superior performance of 2xCD56-AAV, it appears to stem from a combination of target-specific properties that warrant further investigation.

### 3.7 CD56-AAVs: Major Challenges Remain

Whereas CD8-AAVs have been well validated for *in vivo* applications targeting T cells, as demonstrated in this thesis and beyond, CD56-AAV requires further optimization for *in vivo* approaches targeting primary NK cells. There are two major challenges that need to be addressed.

First, although 2xCD56-AAV demonstrated promising transduction efficiency in the clinically relevant NK-92 cell line, even outperforming unmodified AAV6, these results were not reproducible in primary NK cells. Surprisingly, in contrast to the findings in NK-92 cells, unmodified AAV6 significantly outperformed 2xCD56-AAV in primary NK cells, with transduction by 2xCD56-AAV being barely detectable. Since CD56 receptor availability is unlikely to be the limiting factor—given that both NK-92 and primary NK cells express high levels of CD56—the logical next step was to enhance post-entry transduction.

However, the addition of the TBK1/IKKε inhibitor BX795, which has previously been shown to enhance lentiviral transduction in NK cells [113,114], not only failed to improve transduction for both 2xCD56-AAV and unmodified AAV6, but paradoxically reduced transduction rates by more than two-fold. This discrepancy can be explained by the fundamentally different intracellular processing mechanisms of lentiviral vectors and AAVs. Lentiviral vectors require reverse transcription and nuclear import, and BX795 enhances their transduction by suppressing innate immune defences, which otherwise induce interferon-stimulated genes and restriction factors that inhibit viral entry and genome integration [113]. However, AAV transduction is not primarily restricted by innate immune responses but rather by intracellular trafficking, nuclear entry, and second-strand synthesis [158]. Inhibiting TBK1/IKKε may have even hindered AAV processing, possibly by disrupting endosomal trafficking. This underscores the need for vector-specific transduction enhancement strategies when optimizing AAV delivery to NK cells.

The second major challenge in developing CD56-AAV for *in vivo* applications is tissue specificity. While CD56 is highly abundant on NK cells, serving as a defining marker, it is

also strongly expressed in neurons, glia, and skeletal muscle [35,37]. This broad expression profile raises concerns about off-target transduction when using systemic IV administration, making it unlikely that only NK cells would be transduced. This, in turn, increases the required vector dose and elevates the risk of adverse effects.

One potential solution is the use of tissue-specific promoters, such as the NKp46 (*NCR1*) promoter to restrict expression to NK cells [38], or Perforin (*PRF1*) and Granzyme B (*GZMB*) promoters to target activated, cytotoxic NK cells [39,159]. However, this broad CD56 expression may also expand the possible applications of CD56-AAV, allowing local administration for specific therapeutic targets, such as direct intramuscular, CNS, or intramyocardial delivery. One potential application could be the generation of biological pacemakers by delivering therapeutic transgenes to cardiomyocytes [160,161].

For precise NK cell-specific transduction, the vector system requires further refinement. Unfortunately, the approach of combining the CD56 DARPin with the more tissue-specific NKp46 DARPin did not yield the desired improvement. However, the proof-of-principle for dual-targeting AAVs has already been demonstrated, as seen in CD32a-CD4-AAV and other unpublished candidates in the Buchholz lab [112]. This justifies further exploration of alternative DARPin combinations.

In conclusion, while CD56-targeted AAVs present a promising approach for NK cell-specific gene delivery, several key challenges remain, particularly in enhancing primary NK cell transduction efficiency and ensuring precise *in vivo* tissue specificity. Moving forward, further optimization and refinement of these targeting approaches will be crucial to fully harness the therapeutic potential of DART-AAVs in NK cell-based gene therapies.

### 3.8 CD8-AAVs Outperform Unmodified AAVs in Preclinical Settings

CD8-targeted AAVs enabled efficient and selective gene transfer into $CD8^+$ cells, even within complex cell mixtures such as human blood or partially humanized mouse models, and despite the presence of serum factors and off-target cells. In contrast, systemic *in vivo* administration of unmodified AAV2 and AAV6 in these settings resulted in minimal transduction activity. Following a single injection of CD8-targeted AAV2 into immunodeficient mice transplanted with human PBMCs, approximately 40 % of all human $CD8^+$ T cells in the blood were successfully transduced, with nearly absolute selectivity. This demonstrates the potential for *in vivo* gene delivery to T cells. It is important to note, however, that the human T cells in this mouse model were strongly activated.

An additional experiment utilizing non-activated T cells revealed substantially reduced transduction rates, with an average of around 2.2 % of $CD8^+$ target cells being transduced.

Despite these lower rates, transduced cells remained consistently detectable, confirming the ability of the vector to target $CD8^+$ cells under these challenging conditions. Remarkably, conditioning of these animals with huIL-2 and a low dose of anti-huCD3 antibody dramatically improved transduction efficiency. This intervention restored on-target transduction rates to as high as 80 % in the spleen, demonstrating the critical role of T-cell activation in enhancing vector performance.

Notably, this dramatic improvement was not limited to the humanized mouse model. The same effect was observed in a syngeneic mouse model, further validating the impact of *in vivo* T-cell conditioning as a universal strategy to optimize AAV transduction efficiency [96].

The enhanced susceptibility of activated T cells to AAV transduction is likely due to a combination of increased receptor expression, improved endocytic and trafficking pathways [162], and a more favourable intracellular environment for AAV genome processing and transgene expression [163]. Systematic studies investigating restriction factors for AAV transduction have primarily focused on cell lines, such as HeLa, or on hepatocytes [164,165]. These studies have identified restriction factors that impair AAV uncoating, nuclear entry, and double-stranded DNA synthesis [166]. While it is plausible that some of these restriction mechanisms may apply to non-activated T cells, no systematic studies have specifically addressed restriction factors in human or murine T lymphocytes.

These findings emphasize the need for further exploration of the molecular barriers in non-activated T cells and highlight the importance of immune conditioning to maximize the therapeutic potential of DART-AAVs.

## 3.9 Implications on AAV-based Gene Therapy of Lymphocytes

The next generation of *in vivo* gene therapies is expected to heavily depend on advanced vector-targeting strategies to achieve precise, cell-specific delivery while minimizing off-target effects. In this context, DART-AAVs represent a promising approach, not only showcasing the modularity of a DARPin-AAV-based platform but also emphasizing the critical role of receptor compatibility in AAV transduction efficiency and specificity.

Thanks to their modular design, DART-AAVs offer a highly adaptable system that can be tailored to different cellular targets, as presented in this thesis and analysed in detail for CD8 and CD56, making them valuable for both fundamental research and preclinical evaluation of innovative gene therapy strategies. Their ability to selectively transduce specific cell types allows for detailed mechanistic studies, while their compatibility with humanized mouse models facilitates thorough preclinical validation under conditions that closely mimic human physiology.

### 3.9.1 CD8-AAVs: Precision Targeting for Next-Generation T Cell Gene Therapy

By utilizing cross-reactive DARPins or switching to species-specific DARPins targeting the same receptor (e.g., murine CD8), DART-AAVs can be easily adapted for use in syngeneic mouse models [96]. This provides a significant advantage over AAV variants selected from capsid libraries, which are often highly adapted to the specific experimental conditions under which they were generated, limiting their broader applicability.

A key example of this limitation is Ark313, an AAV variant specifically targeting mouse T cells. Ark313 preferentially transduces T lymphocytes from C57BL/6 mice due to the strain-specific expression of the host factor QA2, an MHC class I component that likely serves as its primary attachment receptor [167]. This strain dependency restricts its use in other genetic backgrounds and complicates broader preclinical applications.

In contrast to traditional AAV selection methods, the rational design approach of DART-AAVs allows for the selection of optimal target proteins, defining unique cell subsets with conserved expression across different mouse strains and even species. This strategic flexibility enhances the translatability of preclinical models to clinical applications, making DART-AAVs a highly versatile and scalable gene delivery platform. By enabling precise receptor-mediated targeting, DART-AAVs pave the way for more efficient, safe, and personalized gene therapies, driving advancements in next-generation gene delivery technologies.

However, there are certain limitations of the presented CD8- and CD56-DART-AAVs that require further research and optimization to overcome. One major challenge is the strong dependence of CD8-AAV transduction efficiency on T-cell activation. As demonstrated, transduction rates dramatically increased when T cells were activated. Although numerous physiological and pathological conditions naturally lead to T-cell activation and proliferation, expanding the applicability of CD8-AAVs to non-activated $CD8^+$ T cells would significantly broaden their potential clinical use.

Currently, several scenarios where T-cell activation plays a central role appear well-suited for targeted DART-AAV-based interventions. These include T-cell differentiation and maturation in the thymus during development, immune responses to infections, and clonal expansion of $CD8^+$ T cells in autoimmune diseases such as multiple sclerosis [168,169]. Additionally, excessive $CD8^+$ T-cell activation is a known contributor to severe adverse reactions in immune checkpoint inhibition therapy for cancer [170].

Targeted modification or elimination of these T-cell populations through DART-AAV-mediated transduction could provide novel treatment strategies for these often difficult-to-treat conditions. By selectively intervening in disease-driving T-cell subsets, this approach has the potential to restore immune balance, mitigate autoimmune pathology, and reduce immune-related complications in cancer therapy.

Developing strategies to enable efficient transduction of non-activated $CD8^+$ T cells would significantly broaden the therapeutic potential of CD8-DART-AAVs. In combination with *in vivo* T-cell conditioning, these vectors already provide a promising platform for direct *in vivo* CAR or TCR gene delivery, which could revolutionize adoptive T-cell therapies. By integrating insights from established CAR T-cell therapies with advancing *in vivo* gene delivery technologies, immune cell-targeted gene therapy presents the potential for more accessible, cost-effective, and scalable treatment strategies. For the *in vivo* generation of CAR T cells, both lentiviral vectors and AAVs offer distinct and complementary advantages [6].

Lentiviral vectors stably integrate transgenes into the host genome, ensuring long-term expression. However, because this integration occurs almost randomly, it carries an inherent risk of insertional mutagenesis. In contrast, AAVs typically remain episomal, significantly reducing the risk of genomic disruption [171]. Yet, in rapidly dividing cells such as activated T lymphocytes, episomal transgenes may become diluted or lost over time, potentially limiting therapeutic durability.

To address this limitation, AAVs can be used in conjunction with CRISPR-Cas technologies to enable site-specific, stable integration of CAR transgenes. This is typically achieved by delivering the CAR construct as a homologous repair template via AAV, while CRISPR-Cas9 induces a double-strand break at a predefined genomic location—such as the T-cell receptor alpha constant (TRAC) locus—thereby facilitating precise integration through homology-directed repair [172]. This has already been demonstrated in preclinical models using Ark313 AAV [173]. Future adaptations of CD8-DART-AAVs could incorporate similar strategies to enhance transgene persistence and therapeutic durability.

Additionally, AAVs benefit from well-established, scalable manufacturing pipelines and an expanding body of clinical experience, further supporting their suitability for *in vivo* therapeutic applications.

Ultimately, the CD8-AAVs described here represent a crucial first step toward next-generation *in vivo* gene therapies. By improving targeted T-cell modification, this platform holds the potential to increase accessibility, reduce manufacturing costs, and expand the reach of gene and cell therapies worldwide.

# 4 Material and Methods

## 4.1 CD56 Expression, Purification and Pre-Tests for Ribosome Display

### 4.1.1 Molecular Cloning

The generate a plasmid encoding mCD56 and suitable for expression in mammalian cells the coding sequence of the mCD56 ectodomain (UniProt P13595; aa 20-711) was synthesized *de novo* (GeneArt; Thermo Fisher Scientific) and genetically fused with an huFc-Avi tag. Thus, the mCD56 coding sequence was inserted into the phuFc-$G_4$S-Avi plasmid (kindly provided by Jessica Hartmann, PEI) via SfiI/NotI cloning, generating pmCD56-huFc-Avi. All restriction enzymes were obtained from New England Biolabs.

### 4.1.2 CD56 Protein Expression and Purification

Biotinylated and unbiotinylated mCD56-huFc-Avi protein was expressed in 250 ml cultures of 293F suspension cells (ThermoFischer, A35347) cultivated in BalanCD HEK293 (FUJIFILM Irvine Scientific, 91165) supplemented with 2 % L-glutamine (Sigma-Aldrich, G5792) and 10 µM biotin (Sigma-Aldrich, B4639). For the expression of unbiotinylated protein 5 x $10^8$ cells were transfected with 350 µg of pmCD56-huFc-Avi DNA and 1085 µg polyethyleneimine (Sigma-Aldrich, 408727). For the expression of biotinylated protein 5 x $10^8$ cells were transfected with 175 µg of pmCD56-huFc-Avi, plasmids for both soluble and membrane bound biotin ligase, 87.50 µg pBirA-ER and 87.50 µg psBirA (kindly provided by Jessica Hartmann, PEI) and 1085 µg polyethyleneimine, respectively. After 48 h, cells were centrifuged at 1300 x *g* for 3 min, the supernatants with the recombinant proteins were filtered with a 0.45 µm filter (Thermo Scientific, 10706971) and collected. Both supernatants were incubated with 500 µl of Protein A sepharose beads (Zymed/Invitrogen, 10-1142) at 4 °C O/N while shaking. On the next day, the bead containing supernatants were centrifuged at 800 x *g* for one min and the supernatant was carefully aspirated. The beads were resuspended in PBS (ZMB PEI, ZMBLL052) and transferred to Poly-Prep® Chromatography Columns (BioRad, 7311550). The settled beads were washed twice with 5 ml of PBS. The proteins were eluted with 5 mL 100 mM glycine (Merck, 104201) buffer pH 2.7 into 500 µL 1 M Tris-HCl pH 8.8 (ZMB PEI, ZMBLL098). The eluted proteins were further purified via SEC in a HiLoad® 26/600 Superdex® 200 prep grade column (Cytiva, GE28-9893-36) under the supervision of Gerrit Praefcke (PEI). The target protein-containing fractions were combined and concentrated with Amicon™ Ultra-15 Centrifugal Filter Units 30 kDa MWCO (Merck, UFC9030). To determine protein concentrations the Quick Start™ Bradford Protein Assay (BioRad, 23225) was performed. The purified proteins were aliquoted and stored at 70-80 °C supplemented with 5 % glycerol (Sigma-Aldrich, G7757), 0.1 % BSA (Sigma-Aldrich, A3294) and cOmplete ULTRA protease inhibitor cocktail (Roche, 05 892 970 001).

### 4.1.3 SDS-PAGE Analysis of Purified Protein

To check for remaining impurities after SEC an SDS-PAGE was performed following a Coomassie staining and a western blot. The protein samples were heated for 10 min at 95 °C in a Tris-HCl pH 6.8 (ZMB-PEI, ZMBLL095) buffer containing 2 % SDS (ZMB PEI, ZMBLL072) 10 % Glycerol, 0.05 % bromophenol blue (Merck, 108122), 2.5 % β-mercaptoethanol (Sigma-Aldrich, M3148) and 300 mM DTT (Sigma-Aldrich, D-9779) before loaded on a 10 % Tris-glycine gel. For the Coomassie the gel was stained with PageBlue™ Protein Staining Solution (Thermo Scientific, 24620) and for the western blot the proteins were transferred on a nitrocellulose membrane (Amersham, 10600004). The membrane was incubated with a 1:10,000 dilution of the anti-huIgG-Fc-HRP (Sigma Aldrich, A0170) antibody at 4 °C O/N and developed with Pierce™ ECL Western Blotting Substrate (Thermo Scientific, 32109). All antibodies were diluted in TBS-T (ZMB PEI, ZMBLL085) containing 5 % milk powder (Carl Roth, T145).

### 4.1.4 Determination of Biotinylation

To determine the extent of the biotinylation of the CD56 protein, a plate based colorimetric assay was performed. Both the biotinylated CD56 and the unbiotinylated CD56 as well as a biotinylated and unbiotinylated standard protein (Avidity, BIS-300) were coated in 1:1 serial dilutions ranging from 55.74 fmol/per well to 0.44 fmol/per well on a 96-well plate (Thermo Scientific, 442404) for 1 h at room temperature (RT). All wells were washed four times with PBS-T (ZMB PEI, ZMBLL059) and blocked with 0.5 % BSA in PBS for 1 h at RT. After four additional washing steps with PBS-T 50 µL of a 1:500 dilution of Streptavidin-HRP (Jackson, 016-030-084) was pipetted to each well. At the end of the 1 h incubation time at RT, all wells were washed 4 times with PBS-T again, followed by one additional washing step with PBS. 100 µL of SureBlue™ TMB 1-Component Microwell Peroxidase Substrate (KPL, 52-00-02) was added to each well and developed for 5-15 min. The reaction was terminated with 100 µl 1 N $H_2SO_4$ (Carl Roth, K027.1) and absorbance measured at 450 nm with a reference wavelength of 650 nm with an EMax® Plus microplate reader (Molecular Devices).

### 4.1.5 Determination of Protein Amount Needed for Panning

To determine the ideal amount for panning of the CD56 protein for ribosome display, an ELISA was performed. For capturing 100 µl of 66 nM neutravidin (Thermo Scientific, 31000) solution was put on a 96-well plate and incubated for 1 h at RT for coating. All wells were washed with 250 µL TBS-T trice and blocked with 300 µL TBS-T + 0.5 % BSA for 1 h at RT. The blocking solution was removed and both the biotinylated CD56 and the unbiotinylated CD56 as well as a biotinylated and unbiotinylated huFc-Avi standard protein (kindly provided by Jessica Hartmann, PEI) were pipetted in 1:1 serial dilutions ranging from

160 nM to 0.08 nM on the 96-well plate for panning. After panning for 1 h at 4 °C while shaking, the plate was washed with 250 µL TBS-T and incubated with an anti-huIgG-Fc-HRP (Sigma Aldrich, A0170) antibody for 1 h at RT. The plate was developed with 100 µL SureBlue™ TMB 1-Component Microwell Peroxidase Substrate after three final washing steps with TBS-T. The substrate turnover was terminated after 10 min with 100 µL 1 N $H_2SO_4$ and absorbance measured at 450 nm with a reference wavelength of 650 nm with the EMax® Plus microplate reader.

### 4.1.6 CD56 Antibody Binding

To confirm that plated CD56 can be recognized by an antibody or DARPin, an ELISA was performed. Either 10 nM of biotinylated or unbiotinylated CD56 was directly coated onto a 96-well plate or immobilized on neutravidin, as described in section 4.1.5. For detection, a 1:1,000 dilution of rabbit anti-mCD56 (Abcam, ab220360) was used, followed by a 1:1,000 dilution of anti-rabbit IgG-Fc-HRP (Dako, P0448) as the secondary antibody. The remaining steps followed the protocol outlined in section 4.1.5.

## 4.2 Ribosome Display

The ribosome display was performed in five selection rounds with 2-3 different DARPin DNA libraries. Two DARPin libraries consisting of two (N2C) and three (N3C) diversified repeats in-between the constant N- and C-capping modules (kindly provided by Jessica Hartmann, PEI [105]) and one additional N3C library (kindly provided by Markus A. Seeger, University of Zürich [107]) referred to as Seeger library. For each selection rounds, the DNA DARPin libraries were *in vitro* transcripted and translated with the PURExpress® *In Vitro* Protein Synthesis Kit (NEB, E6800). For the selection rounds 1-3 a 96-well plate was coated with a 66 nM solution of neutravidin (round one and three) or streptavidin (round 2) and a 10 nM solution of biotinylated CD56 was panned for target protein binding and 80 nM biotinylated huFc-Avi protein for negative selection. In the selection rounds four and five, the biotinylated CD56 protein of a 1.95 pmol solution was immobilized on magnetic Dynabeads™ MyOne™ Streptavidin T1 (Invitrogen, 65601). Pre-selection was performed with a 2.7 pmol solution of unbiotinylated huFc-Avi protein followed by off-rate selection upon incubation with a 195 pmol solution of unbiotinylated CD56 for 5 min (fourth round) or 60 min (fifth round) at 4 °C. In every round the unbound DARPin-mRNA-ternary complexes were washed away and the bound ones eluted with elution buffer pH 7.5 containing 50 mM Tris-Acetate, 150 mM NaCl, 25 mM EDTA (ZMB, PEI, ZMBLL019) and 50 µg/mL *Saccharomyces cerevisiae* RNA (Sigma-Aldrich, R6750). The RNA was recovered with the RNeasy Mini Kit (Qiagen, 74104) and reverse transcripted with primer 1 for the N2C/N3C libraries or primer 2 for the Seeger library (Table 5), using AffinityScript reverse transcriptase (Agilent, 600107). After that, cDNA was amplified with primers 3 and 4 for the N2C/N3C libraries or

5 and 6 for the Seeger library (Table 5) using Vent® DNA Polymerase (NEB, M0254). The amplified cDNA was cut with SfiI/DraIII and ligated into the library pRDV backbone (kindly provided by Jessica Hartmann, PEI) after every cycle. After the fourth and fifth round of the ribosomal display the output library was additionally cloned into the pQE-HisHA bacterial expression backbone (kindly provided by Jessica Hartmann, PEI) with SfiI/DraIII cloning as well. Before starting a new cycle, the DNA template for the *In Vitro* Protein Synthesis was amplified with primers 7 and 8 (Table 5) and Vent® DNA Polymerase from the pRDV backbone. PCR products were purified via 1 % agarose gel electrophoreses followed by gel extraction (Thermo Scientific, K0691).

## 4.3 Crude Lysate Screening After Ribosome Display

### 4.3.1 Production of *E. coli* Crude Lysates

For DARPin crude lysates, the preselected pQE-HisHA-DARPin libraries from selection round four and five were transferred into *E. coli* XL1-Blue bacteria (Stratagene, 200236). The bacterial cultures were plated on LB + Amp agar-plates (ZMB-PEI, ZMBAP2) and single clones were picked to inoculate a pre-culture of 600 µL 2xYT-medium (ZMB-PEI, ZMBNB021) containing 1 % Glucose (VWR Chemicals, 24379.294) and 100 mg/L ampicillin (Roche, 10 835 269 001) in a 96 deep well plate (Axygen, P-2ML-SQ-C-S). This pre-culture was cultivated at 28 °C O/N while shaking and used to create a bacterial suspension with an $OD_{600}$ = 0.1, which was incubated for 1 h at 37 °C. The rest of the O/N culture was used to prepare glycerol stocks with 15 % glycerol for long time storage at -70 to -80 °C. After 1 h, the DARPin expression was induced by addition of IPTG (Thermo scientific, R0392) to a final concentration of 0.5 mM and the bacterial suspension was cultivated for another 5 h. Bacteria were harvested by centrifugation at 4400 rpm for 15 min. The supernatants were discarded and the bacterial pellets lysed with 50 µl B-PER™ II Bacterial Protein Extraction Reagent (Thermo Scientific, 78260) for 2 h at RT. The lysed pellets were vortexed and diluted 1:10 with TBS (ZMB PEI, ZMBLL083). To remove cell debris, the diluted crude lysates were centrifuged at 4000 x *g* for 20 min. The supernatant was then aliquoted and stored at -70 to -80 °C.

### 4.3.2 Validation of DARPin Expression

To validate DARPin expression in the crude lysates, a plate-based colorimetric assay was performed. Each crude lysate (5 µL) was diluted in 100 µL TBS and coated on a 96-well plate O/N at 4 °C while shaking. The wells were then washed three times with TBS-T and blocked with TBS-T + 0.5 % BSA for 1 h at 4 °C. The DARPins were detected using an anti-Penta His biotin-conjugated antibody (Qiagen, 34440) diluted 1:2,000 and incubated for 1 h at RT. After three additional washes with TBS-T, a 1:500 dilution of streptavidin-HRP was added to each well and incubated for 2 h at RT, followed by another three washes. For

detection, 100 µl SureBlue™ TMB 1-Component Microwell Peroxidase Substrate was added to each well and developed for 5-30 min. The reaction was stopped with 100 µl 1 N $H_2SO_4$ and absorbance measured at 450 nm with a reference wavelength of 650 nm with the EMax® Plus microplate reader.

### 4.3.3 Cellular Binding with Crude Lysates

For crude lysate binding assay GL-261 (DSMZ, ACC 802) cells were cultivated in DMEM (Sigma-Aldrich, D6546) supplemented with 10 % FBS (Sigma-Aldrich, F7524) and 1 % L-glutamine (Sigma-Aldrich, G5792). Before staining, cells were detached with trypsin (ZMB PEI, ZMBLL104) in PBS and washed twice with washing buffer containing PBS, 2 % FBS and 0.1 % NaN3 (ZMB-PEI, ZMBLL038). They were then incubated with 50 µL crude DARPin extracts for 1 h at 4 °C or with the anti-CD56 antibody Anti-NCAM1 antibody [EPR21827] (Abcam, ab220360) as an expression control. Following two additional washing steps, the cells were stained for 30 min at 4 °C using a 1:100 dilution of a PE-labeled anti-HA antibody (Miltenyi Biotec, 130-092-257) or an anti-rabbit Alexa 647 antibody (Dianova, 111-605-046). Additionally, a 1:500 dilution of eFluor780 live-dead stain (eBioscience, 65-0865-18) was included to assess cell viability. Cells were washed again twice and fixed with 0.5 % formaldehyde (Sigma-Aldrich, F1635). Measurements were acquired on MACSQuant 10 (Miltenyi Biotec) and results were analysed with FCS Express 6 (De Novo Software).

## 4.4 DARPin Expression and Purification

### 4.4.1 Molecular Cloning

The DARPin sequences of promising binders were PCR amplified using primers 9 and 10 and inserted into the pET32a backbone (kindly provided by Angeliki Chroni, demokritos) with XbaI/XhoI cloning generating pET32a-His.TEV.HA-DARPin plasmids.

To generate the pET32a-His.TEV.HA plasmid harbouring copies of the 43H3 DARPin, the coding sequence of both DARPin copies was excised from pRC22 2xCD56.43H3 (4.8.1) using SfiI and NotI restriction sites and subsequently inserted into the pET32a-His.TEV.HA vector. This process resulted in the construct pET32a-His.TEV.HA-2xCD56.43H3.

### 4.4.2 DARPin Expression in *E. coli*

DARPin expressions were performed with the pET32a-His.TEV.HA-DARPin plasmids in Rosetta™ 2(DE3) BL21 (Sigma Aldrich, 71400) *E. coli* bacteria. 25 mL O/N cultures were either inoculated with single clones or glycerol stocks. On the next day, grown O/N culture was added in a 1:10 ratio to 100 or 200 mL 37 °C pre-warmed LB medium (ZMB PEI, ZMBNB006) supplemented with 1 % glucose and 100 mg/l ampicillin. Protein expression was induced by adding IPTG to a final concentration of 0.5 mM when the $OD_{600}$ reached

between 0.4 and 0.7. After cultivation for 4 h at 37 °C, bacteria were pelleted by centrifugation at 4600 rpm for 20 min and stored at -70 to -80 °C.

### 4.4.3 DARPin Purification

Bacterial pellets were thawed and resuspended in HBS pH 7.4 (ZMB PEI, ZMBLL023) containing 2 µL/mL Pierce™ Universal-Nuclease (Thermo Fisher Scientific, 88702). The cells were lysed with a cell disrupter (Constant Systems Ltd., CF1) at 1300 bar pressure or by sonication (BANDELIN electronic, SONOPULS HD 4100). Cell debris was removed by centrifugation at 4600 rpm for 40 min. DARPins were purified by gravity flow affinity chromatography on Ni-NTA agarose columns (Qiagen, 30210). For surface plasmon resonance measurements, the elution fraction was purified further by SEC using a HiLoad™ 16/60 Superdex™ 200 prep grade column (GE Healthcare Bio-Sciences, 17-1069-01), an ÄKTA™ start (GE Healthcare Life Sciences) system and PBS-T buffer. The DARPin-containing fractions were combined and concentrated with Amicon™ Ultra-15 Centrifugal Filter Units 10 kDa MWCO (Merck, UFC901008). DARPin concentration was measured via Pierce™ BCA Protein Assay (Thermo Fisher Scientific, 23225) and a SpectraMax® iD3 plate reader (Molecular Devices). For long time storage at -70 to −80°C, DARPins were supplemented with 5 % glycerol and cOmplete ULTRA protease inhibitor cocktail (Roche, 05 892 970 001).

### 4.4.4 DARPin SDS-PAGE and Western Blots

To check for remaining impurities after SEC an SDS-PAGE was performed similar to 4.1.3. In short, the protein samples were heated for 10 min at 95 °C in a Tris-HCl pH 6.8 buffer containing 2 % SDS, 10 % Glycerol, 0.05 % bromophenol blue, 2.5 % β-mercaptoethanol and 300 mM DTT before loaded on a 12 %Tris-glycine gel. For the Coomassie the gel was stained with GelCode™ Blue Safe Protein Stain (Thermo Scientific, 24596) and for the Western blot, the proteins were transferred on a nitrocellulose membrane. The membrane was incubated with a 1:1000 dilution of the anti-RGS-His (Qiagen, 34650) antibody at 4 °C O/N followed by the anti-mouse-Fc-HRP (DAKO, P0260) secondary antibody for 2 h at RT. The blots were developed with Pierce™ ECL Western Blotting Substrate and luminescent signals were detected on the chemiluminescence reader MicroChemi (DNR Bio-Imaging Systems). All antibodies were diluted in TBS-T containing 5 % horse serum (Sigma-Aldrich, H1138). Brightness and contrast of images were adjusted using ImageJ retaining relative signal strength.

## 4.5 Generation of CD56$^+$ HeLa Cell Lines

### 4.5.1 Molecular Cloning

To create a lentiviral transfer plasmid encoding full-length hCD56, the complete coding sequence of hCD56 (NCBI Reference Sequence NM_000615.7) was synthesized *de novo* (GeneArt, Thermo Fisher Scientific) and inserted into the pSEWsn-IP backbone (kindly provided by Jessica Hartmann, PEI) via BamHI/SpeI restriction sites, resulting in the plasmid pSEW-hCD56-IP.

To generate a lentiviral transfer plasmid encoding the first 212 amino acids of hCD56, the corresponding sequence was amplified from the pSEW-hCD56-IP plasmid using primers 11 and 12. The resulting PCR product was digested with BamHI and BsiWI and subsequently cloned into the pSEW-X-CD4-TM-IP and pSEW-X-CD4-IP plasmids (kindly provided by Ruben Bender, PEI). This process yielded the constructs pSEW-1-CD56-212-CD4-TM-IP and pSEW-1-CD56-212-CD4-IP, respectively.

### 4.5.2 Generation of a VSV-Pseudotyped Lentiviral Vector

VSV-pseudotyped lentiviral vectors encoding either full-length hCD56 or the 1-CD56-212-CD4 fusion proteins were produced using 293F suspension cells cultured in BalanCD HEK293 medium supplemented with 2 % L-glutamine. A total of 2 × 10$^6$ cells were transfected with the following plasmids: pMD2.G (encoding the VSV glycoprotein), pCMV-dR8.9 (encoding second-generation HIV gag/pol/rev; both kindly provided by Jessica Hartmann, PEI), and one of the pSEW IP plasmids. Transfection was performed using polyethyleneimine. 4-6 h after transfection, the medium was changed completely, and 5 mM sodium butyrate (Sigma-Aldrich, 303410) was added as transfection enhancer. The vectors were harvested 48 h post transfection by centrifugation of the producer cells at 1000 x *g* for 3 min and collection of the supernatant.

### 4.5.3 Transduction and Selection of CD56$^+$ HeLa Cells

HeLa (ATCC, CCL-2) cells were cultivated in DMEM complete, which is DMEM supplemented with 10 % FBS and 1 % L-glutamine, and seeded at a density of 2 × 10$^5$ cells per well in a 12-well plate one day before lentiviral transduction. The next day, the medium was removed from the cells and 1.5; 1; 0.5; or 0.25 ml of vector containing supernatant from 4.5.2 was added. The cells together with the vector were incubated for 2 h at 37 °C, before the medium was replaced with 2 ml DMEM complete. For the selection of CD56$^+$ HeLa cells, the medium was replaced with DMEM complete supplemented with 0.3 µg/mL puromycin (Thermo Fisher Scientific, J67236.XF) 72 h after transduction and maintained thereafter. The cells were split regularly and the dead cells removed. Two weeks after transduction CD56 expression was validated by flow cytometry.

## 4.6 DARPin Binding Assays

### 4.6.1 Isolation und Cultivation of NK Cells

Human PBMCs were isolated from buffy coats derived from healthy and anonymous donors acquired from DRK Blutspendedienst (Frankfurt am Main, Germany) in agreement with the ethics committee of the Frankfurt University Hospital. PBMCs were separated from plasma and the majority of red blood cells using density gradient centrifugation with Histopaque-1077 (Sigma-Aldrich, 10771). The remaining red blood cells were lysed with 0.86 % ammonium chloride (ZMB PEI, ZMBLL005). NK cells were isolated with a NK Cell Isolation Kit (Miltenyi Biotec, 130-092-657) and cultured for 3-13 days in NK MACS® (Miltenyi Biotec, 130-114-429) supplemented with 5 % human AB serum (Sigma-Aldrich, H4522), 120-140 U/mL IL-15 (Miltenyi Biotec, 130 095 765) and 500 U/mL IL-2 (Miltenyi Biotec, 130-097-746).

### 4.6.2 NK Cell and HeLa DARPin Binding Assay

The DARPin binding assay with purified DARPins was performed similar to the crude lysate binding assay described in 4.3.3. In short, the HeLa cells were detached with PBS pH 7.1 + 1 mM EDTA (ZMB PEI, ZMBLL057) and were washed twice with washing buffer containing PBS, 2 % FBS and 0.1 % $NaN_3$. For the NK cells this step was performed likewise without detachment. Cells were then incubated with 2.5 or 5 µg purified DARPins in 150 µL washing buffer for 1 h at 4 °C. After two washing steps, cells were stained with a 1:100 dilution of a PE-labelled anti-HA antibody and a 1:500 dilution of an eFluor780 life-dead-staining for 30 min at 4 °C. Cells were washed again twice and fixed with 0.5 % formaldehyde. Measurements were acquired on MACSQuant 10 and results were analysed with FCS Express 6.

## 4.7 Surface Plasmon Resonance Affinity Measurement

Binding affinity of the CD56.43H3 DARPin against huCD56 was assessed using SPR on a Biacore T200 system with a CM5 sensor chip (Cytiva, 29104988). The DARPins were immobilized directly onto the sensor chip via the amine coupling kit (Cytiva, BR100050), and binding analysis was conducted with huCD56 as the analyte. All experiments were performed in PBS-T at 25 °C. The DARPins (40 µg/ml in 10 mM acetate pH 5.5) were immobilized to a level of 570 response units (RU). To determine the dissociation constant ($K_D$), huCD56-hFc (Sino Biological, 10673 H02H) was injected in serial dilutions (ranging from 0.5 µM to 0.98 nM) using a multi-cycle approach. Each cycle was conducted at a flow rate of 30 µL/min, with a 280-second association phase followed by a 480-second dissociation phase. Regeneration was carried out performed with 10 mM NaOH at a flow rate of 30 µL/min and 30 s contact time. Kinetic parameters were calculated using a 1:1

binding model and the data were analysed with Biacore Evaluation Software v3.2 (Cytiva). The SPR measurements were performed in-house at PEI by Sascha Hein.

## 4.8 DART-AAV Generation

### 4.8.1 Molecular Cloning

To generate a plasmid encoding hCD8-targeted AAV2 VP1 capsid proteins with a $(G_4S)_5$ linker, an intermediate plasmid was required. The coding sequence for the human CD8-specific DARPin 63A4 was amplified by PCR using primers 13 and 14 and the pCG-NiV-$G_{mut}$-L3-huCD8.63A4 plasmid (kindly provided by Annika Frank, PEI). The PCR product was then digested with SpeI/SfiI, and inserted into the pAAV2-9.29-GH2_GH3-$VP2_{mut}$ plasmid (kindly provided by Dorothee Günther, ESI/PEI), resulting in the creation of the intermediate plasmid pAAV2-huCD8.63A4_GH2_GH3-$VP2_{mut}$. The DARPin sequence from the intermediate plasmid was inserted between residues 453 and 459 of VP1 via BsiWI/XcmI cloning into the pRCRRVP1r1c3 plasmid (Addgene, 65724), resulting in the construct pRC22-hCD8.63A4.

The plasmid encoding hCD4-targeted AAV2 VP1 capsid proteins with a $(G_4S)_5$ linker was generated as described for CD8. The coding sequence for the human CD4 specific DARPin 55.2 was amplified by PCR using primers 15 and 16 and the peGFP-CD4-DARPin55.2-$HSPG_{mut}$ (kindly provided by Robert Münch, PEI).

For engineering of AAV6, the coding sequences for the unmodified AAV6 VPs, the AAV6 VP1 with the DARPin 63A4 replacing seven amino acids at positions 453–459 and the blinding mutations (V473D, K531E) and AAV6 only with the blinding mutations were synthesized de novo (GeneArt; Thermo Fisher Scientific, Regensburg, Germany). The synthetic sequences were inserted into pRC22-hCD8.63A4 by SwaI/NotI generating pRC26 encoding unmodified AAV6 capsid, pRC26-hCD8.63A4 and $pRC26_{mut}$. To create the $pRC26_{mut}$ plasmid encoding only VP2 and VP3, the VP1 start codon was modified using primers 17 and 18. The resulting PCR product was then inserted into $pRC26_{mut}$ via SwaI/KasI cloning, generating the plasmid $pRC26_{mut}$-VP1-KO.

Short linker variants of the pRC22-hCD8.63A4, pRC22-hCD4.55.2 and pRC26-hCD8.63A4 plasmids, incorporating a $(G_4S)_2$ linker, were generated by Fabian John (PEI) and Muhammed Burak Demircan (PEI), following the method outlined by *Demircan, Zinser et al.* [96].

The plasmid encoding the CD56.43H3 DARPin with a $(G_4S)_5$ linker in the AAV2 VP1, was generated via SfiI/NotI cloning with the plasmids pQE-HisHA-CD56.43H3 (4.2) and pRC22-HH-$VP1_{mut}$ (kindly provided by Julia Brynza, PEI), resulting in pRC22-CD56.43H3.

Dual-DARPin AAV plasmids, containing the CD56.43H3 DARPin combined with NKp46.3A8, were generated by PCR amplification of the NKp46.3A8 coding sequence from the pRC22 HH VP1mut-NKp46.3A8 plasmid (kindly provided by Julia Brynza, PEI) using primer pairs 19/20 and 21/22 (Table 5). The amplified sequence was then inserted via NheI/SpeI and SfiI/NheI cloning into the pRC22-CD56.43H3-X and pRC22-X-CD56.43H3 plasmids (kindly provided by Fabian John, PEI), resulting in the constructs pRC22-CD56.43H3-NKp46.3A8 and pRC22-NKp46.3A8-CD56.43H3.

Dual-DARPin AAV plasmids containing two copies of the CD56.43H3 DARPin were generated using SfiI/NheI cloning with the plasmids pRC22-CD56.43H3-X and pRC22-X-CD56.43H3, resulting in the construct pRC22-2xCD56.43H3.

### 4.8.2 Production of AAVs

For the generation of unmodified AAV2 and AAV6, HEK293T cells were transiently co-transfected with the helper plasmid pXX6-80 [174], packaging plasmid pRC22 (Addgene: 104963) for AAV2 or pRC26 for AAV6, and the transfer vector pscAAV-SFFV-GFP [93] in a 3:1:1 ratio. For the production of DART-AAVs, pXX6-80, the transfer vector, and the complementary capsid plasmids—either pRC22-VP1-KO (for DART-AAV2, kindly provided by Dorothee Günther, ESI/PEI) or pRC26-VP1-KO (for DART-AAV6)—were combined with the according pRC22/6-VP1-DARPin constructs in a 3:1:1:1 ratio. For large-scale productions of AAV2, cells were harvested by scraping, pelleted, and then lysed in AAV lysis buffer (ZMB PEI, ZMBLL135) using five freeze-thaw cycles alternating between liquid nitrogen and a 37 °C water bath, 48 hours after transfection. For AAV6, supernatants and cell lysates were collected 3 days post-transfection. Remaining plasmid DNA was digested with 50 U/mL Benzonase (Merck Millipore, E1014) for 30 min at 37 °C. All vector types were purified by ultracentrifugation through an iodixanol density gradient followed by buffer exchange to 0.01 % pluronic (Sigma-Aldrich, P5556) in PBS via a 50 kDa Amicon™ Ultra-4 centrifugal filter unit (Merck, UFC805024). For small-scale productions, AAV particles were either concentrated from the cell culture supernatant of transfected HEK293T cells or the unprocessed supernatant was directly used to transduce target cells after a brief centrifugation at 1,000 × *g* for 3 min to remove debris. For AAV concentration, the supernatant was filtered through a 0.45 µm filter (Sartorius, 17829) and treated with 50 U/mL Benzonase for 30 min at 37 °C. This was followed by ultracentrifugation at 100,000 × g for 16 h through a 40 % sucrose cushion (Sigma-Aldrich, 84097) in TNE buffer

(TBS with 0.1 mM EDTA). The concentrated AAV particles were resuspended in PBS and stored in aliquots at -70 °C to -80 °C for long-term storage.

## 4.9 Biophysicochemical Characterization of DART-AAVs

### 4.9.1 AAV Genome Copy qPCR

For all stocks produced and used in this thesis, AAV genome copy quantification by qPCR was routinely performed by Manuela Gallet. The DNA was isolated from 3 µL AAV stock following the DNeasy® Blood and Tissue kit (Qiagen, 69506) protocol. Genomic AAV titers were determined via inverted terminal repeat (ITR)-specific qPCR using primers 23, 24, and probe 25 (Table 5) with the LightCycler 480 Probes Master mix (Roche, 04707494001) on a Light-Cycler 480 Real-Time PCR System (Roche).

### 4.9.2 Western Blotting

Western blotting was performed as described in 4.1.3 and 4.4.4 with some adjustments. In brief, AAV2 stocks were incubated in a Tris-HCl pH 8 buffer (ZMB PEI, ZMBLL097) containing 5 % SDS, 8 M urea (Sigma-Aldrich, U5378), 0.03 % bromophenol blue, 0.1 mM EDTA and 2.5 % DTT at 95 °C for 10 min and loaded on 8 % Tris-glycine gel. After electrophoretic separation, proteins were blotted on a nitrocellulose or PVDF membrane (Amersham, 10600023). Membranes were incubated O/N at 4 °C with either anti-AAV VP1/VP2/VP3 rabbit polyclonal VP51 serum (Progen, 61084) at a 1:200 dilution or anti-AAV VP1/VP2/VP3 mouse monoclonal (B1) antibody (Progen, 65158) at a 1:50 dilution. This was followed by a 2 h incubation at RT with either polyclonal goat anti-rabbit HRP conjugate (Agilent, P0448) or anti-mouse-HRP, both diluted 1:2,000. After application of chemiluminescent peroxidase substrate-1 (Sigma-Aldrich, CPS160) or Pierce™ ECL Western Blotting Substrate, luminescent signals were detected on the chemiluminescence reader MicroChemi. All antibodies were diluted in TBS-T containing 5 % horse serum. Brightness and contrast of images were adjusted using ImageJ retaining relative signal strength.

### 4.9.3 Transmission and Cryo Electron Microscopy

Transmission and cryo-electron microscopy were performed as described in *Demircan, Zinser et al.* [96]. Technical supervision for the transmission electron microscopy was provided in-house at PEI by Julia Brynza and Regina Eberle. Cryo-electron microscopy was conducted by Petr Chlanda at BioQuant, Heidelberg University.

## 4.10 Functional Characterisation of DART-AAVs

### 4.10.1 *In Vitro* Transduction Experiments with Cell Lines

For the CD8-DART-AAV transduction experiments, SUP-T1 cells (DSMZ, ACC 140) were cultured in Advanced RPMI 1640 (Gibco, 12633-012) supplemented with 4 % FBS and 1 % L-glutamine. NALM-6 (DSMZ, ACC 128), MOLT4/8 and A3.01 (kindly provided by Jessica Hartmann, PEI) cells were cultured in RPMI 1640 (VWR, 392-0429) with 10 % FBS and 2 mM L-glutamine. All cell lines—SUP-T1, NALM-6, MOLT4/8 and A3.01—were subcultured twice a week at a 1:10 dilution ratio. For the transduction described in Figure 12 and Figure 13, 1 x $10^4$ MOLT4/8 or A3.01 cells were seeded in 75 µL of RPMI 1640 with 10 % FBS and 2 mM L-glutamine per 96-well and were mixed with 75 µL/ well of unprocessed vector supernatant. 6 h after vector addition, 100 µL fresh medium was added to each well. For the titration described in Figure 16, 4 × $10^4$ NALM-6 or SUP-T1 cells were seeded in 50 µL of RPMI 1640 with 10 % FBS and 2 mM L-glutamine, or in Advanced RPMI 1640 with 4 % FBS and 1 % L-glutamine, respectively, per well of a 96-well plate. Cells were then treated with 100 µL per well of sucrose-concentrated vector dilutions in respective medium. 6 h after vector addition, 100 µL of fresh medium was added to each well.

For the CD56-DART-AAV transduction experiments, HeLa cells overexpressing CD56 were generated as described in section 4.5 and cultivated in DMEM complete supplemented with 0.3 µg/mL puromycin at least for two weeks before transduction. For the titration described in Figure 14, 1 × $10^4$ HeLa or $CD56^+$ HeLa cells were seeded in DMEM complete, with or without 0.3 µg/mL puromycin, per well of a 96-well plate one day prior to transduction. On the next day, the medium was removed and 100 µL of iodixanol-purified vector dilutions were added to each well. Six hours after vector addition, 100 µL of fresh medium was added.

The transduction described in Figure 25 was conducted with minor modifications. A fixed vector concentration of 1 × $10^5$ vg/cell was used, and the medium was completely replaced 2–3 h post-transduction.

For the NKp46-DART-AAV transduction experiments, HT-1080 cells overexpressing NKp46 (kindly provided by Jessica Hartmann, PEI) were cultivated in DMEM complete supplemented with 400 µg/mL Zeocin™ (Thermo Fisher Scientific, R25001). For the titration described in Figure 20, 8 × $10^3$ NKp46-HT-1080 cells were seeded in DMEM complete with 400 µg/mL Zeocin™, per well of a 96-well plate one day prior to transduction. On the next day, the medium was removed and 50 µL of non-purified vector supernatant dilutions were added to each well. Two h after vector addition, 150 µL of fresh medium was added.

For the dual-DART-AAV transductions experiments, KHYG-1 (DSMZ, ACC 725) cells were cultured in RPMI-1640 supplemented with 10 % FBS, 1 % L-glutamine and 10 ng/mL IL-2. NK-92 (DSMZ, ACC 488) were cultured in X VIVO 10™ (Lonza, 04-380Q) supplemented with 10 % human AB serum and 100 U/mL IL-2. KHYG-1 were seeded into 96-well plates at 4 × $10^4$ cells/well in 50 µL RPMI-1640 supplemented with 10 % FBS, 1 % L-glutamine and 10 ng/mL IL-2. Vector stocks were diluted in 50 µL of the same medium and added to the cells. After 4-6 h at 37 °C, 100 µL/well of fresh-medium was added. NK-92 were seeded into 96-well plates at 4 × $10^4$ cells/well in 50 µL X VIVO 10™ w/o supplements. Vector stocks were diluted in 50 µL of the same medium and added to the cells. After 2 h at 37 °C, 100 µL/well of fresh X VIVO 10™ supplemented with 20 % human AB serum and 200 U/mL IL-2 was added.

### 4.10.2 *Ex Vivo* Transduction of Primary NK Cells

For the *ex vivo* transduction of NK cells, cells were isolated and cultured as described in section 4.6.1. Three days post-isolation, NK cells were seeded at a density of 1 × $10^5$ cells per well in 50 µL NK MACS medium supplemented with 500 U/mL IL-2, 140 U/mL IL-15, and optionally 10 mM BX795 (Sigma-Aldrich, 204001). Vector stocks were added in 50 µL of the same medium at a concentration of 5 × $10^{10}$ vg/well. 1-2 h after transduction, 100 µL of fresh medium supplemented with 10 % human AB serum was added to each well. Transduction efficiency was evaluated three days post-transduction using flow cytometry.

### 4.10.3 *Ex Vivo* Transduction of Activated Human PBMC

For the transduction of human T-cells with CD8-DART-AAV, human PBMCs were isolated from buffy coats as described in section 4.6.1. The cells were cryopreserved in aliquots of 2 × $10^6$ or 1 × $10^7$ cells/mL in a solution of 90 % FBS and 10 % DMSO (AppliChem, A3672). For the activation of PBMCs, wells were prepared by coating either one well of a 6-well plate (for 1 × $10^7$ cells) or a 24-well plate (for 2 × $10^6$ cells) with a 1 µg/mL solution of anti-CD3 antibody (Miltenyi Biotec, 130-093-387). The PBMCs were thawed at 37 °C, washed twice, and seeded in NutriT-medium (Sartorius, 05-11F2001) containing 0.5 % penicillin/streptomycin (ZMB PEI, ZMBLL075), 3 µg/mL anti-CD28 antibody (Miltenyi Biotec, 130-093-375), 25 U/mL IL-7 (Miltenyi Biotec, 130-095-361) and 50 U/mL IL-15. Three days after activation, medium was replaced with NutriT-medium only containing IL-7 and IL-15. Activated human PBMCs were seeded into 96-well plates at 4–8 × $10^4$ cells/well in 50 µL NutriT-medium containing IL-7 and IL-15. Vector stocks were diluted in 50 µL NutriT-medium supplemented with cytokines and added to the cells. After 4-6 h at 37 °C, 100 µL/well of supplemented NutriT-medium was added or complete medium was exchanged on the next day.

### 4.10.4 Transduction Inhibition by Human Serum

To evaluate the impact of human serum on the transduction efficiency of CD8-AAV6 and unmodified AAV6, vector stocks were pre-incubated for 1 h at 37 °C with 1:3 serial dilutions of human off-the-clot serum (PAN-Biotech, P30-2701), ranging from 30 % to 0.04 %. SUP-T1 cells were then transduced at a dose of 4 × $10^4$ vg/cell for 6 h at 37 °C. Following transduction, the medium was replaced with Advanced RPMI 1640 supplemented with 4 % FBS and 1 % L-glutamine. Transduction efficiency was assessed by flow cytometry three days post-transduction.

### 4.10.5 Whole Blood Transduction Assay

For the CD8-DART-AAV whole blood transduction assay, blood was collected with sodium citrate anticoagulant tubes (Sarstedt, 02.1067.001) from healthy anonymous in-house donors in accordance with the ethics committee guidelines. A total of 400 µL whole blood was mixed with 1 × $10^{11}$ vg of DART-AAVs and incubated in a 48-well plate for 6 h at 37 °C with continuous shaking at 400 rpm in a cell culture incubator. Subsequently, blood was diluted with 600 µL PBS and overlaid on 800 µL cold Histopaque. After centrifugation at 1,800 × *g* for 30 min at RT, PBMCs were isolated and washed with PBS to remove platelets. The cells were then resuspended in 200 µL of NutriT medium supplemented with IL-7, IL-15, and anti-CD28, and transferred to anti-CD3-coated 48-well plates as outlined in section 4.10.3.

### 4.10.6 *In Vivo* Gene Transfer of CD8-DART-AAVs

To assess the *in vivo* transduction efficiency and specificity of the CD8-DART-AAVs, male NSG mice (Charles River Laboratories, 614NSG) were IV injected with 1 × $10^7$ activated human PBMCs, as outlined in section 4.10.3, in 100 µL lukewarm PBS. This was followed by an IV injection of 1.7 × $10^{11}$ vg AAVs per animal the next day.

For *in vivo* transduction of non-activated T cells using CD8-DART-AAVs, 2 × $10^7$ human PBMCs from two donors were IV into male NSG-MHCI/II DKO mice (The Jackson Laboratory, #025216). For T-cell conditioning, 15 µg of IL-2 was administered intraperitoneally (i.p.) one day after PBMC transplantation, followed by 4 µg of anti-CD3 antibody co-administered with the AAV vectors. The vectors were delivered at a dose of 1 × $10^{12}$ vg per animal in 150 µL of lukewarm PBS on day two post-transplantation.

Three days after vector application, mice were sedated with isoflurane, bled via retroorbital sampling, and subsequently euthanized by cervical dislocation. Nucleated cells were isolated from spleens and blood for analysis via flow cytometry as described by *Weidner et al.* [175].

For the retrieval of PBMCs, blood samples were transferred to 15 mL tubes and diluted with 1 mL of PBS. After centrifugation at 400 × *g* for 5 min at RT, the plasma was discarded. The cell pellets were then resuspended in 3 mL of red blood cell lysis buffer (BD Biosciences, 555899) and incubated for 15 min at RT. Following this, the samples were diluted with 10 mL of PBS and centrifuged again under the same conditions. The resulting pellets were washed with 13 mL of PBS and resuspended in 1 mL of PBS per tube. Finally, the cells were counted using the automated staining function of a MACSQuant Analyzer 10 with propidium iodide solution (Miltenyi Biotec, 130-093-233).

Mouse spleens were extracted and transferred into 1 mL of NutriT-medium. To generate cell suspensions, the spleens were ground through 70 µm cell strainers (Thermo Scientific, 352350) into a 50 mL tube using the back sides of syringe plungers. The strainers were rinsed several times with NutriT-medium, and the cell suspensions were homogenized by pipetting. Afterward, strainers were flushed with 12 mL of PBS, and the suspensions were centrifuged at 400 × *g* for 5 min at RT. Subsequent processing followed the same steps as for PBMCs, with a final resuspension in 3 mL of PBS.

All mice were kept in individually ventilated cages under pathogen-free conditions on standard diet. The animal experiment was conducted in accordance with the regulations of the German animal protection law and the respective European Union guidelines.

### 4.11 Immunostaining and Flow Cytometry

Immunostaining for flow cytometry was performed in 1.4 mL round-bottom tubes (Micronic, MP32022). Cells were washed twice with wash buffer (PBS with 2 % FBS and 0.1 % sodium azide) by centrifuging at 300–350 × *g* for 3–5 min. All washing steps were done with 500 µL of wash buffer. Prior to staining, the cells were resuspended in 100–150 µL of wash buffer, and 20 µL of the staining mixture, consisting of the antibody cocktails (Table 6) and eFluor780 viability dye in wash buffer, was added to each tube. After staining, cells were washed twice and fixed in 0.5 % formaldehyde in PBS. Samples were kept in the dark at 4 °C until measurement. Single stain controls were prepared with UltraComp eBeads Plus (Thermo, 01-3333) to enable compensation. Measurements were acquired on MACSQuant 10 and results were analysed with FCS Express 6.

For the CD56-DART-AAV binding assay, KHYG-1 cells were seeded at a density of $1 \times 10^5$ cells in 50 µL per tube. AAVs diluted in 50 µL of RPMI were added to the cells, followed by incubation for 2 h at 37 °C. After washing three times, FcR blocking reagent (Miltenyi Biotec, 130-092-575) diluted 1:100 was added and incubated for 15 min at 37 °C. After two additional washes, anti-AAV2 A20 antibody (Progen, 61555) diluted 1:50 was added and incubated for 30 min at 37 °C. Unbound antibodies were removed with two further washes.

Finally, cells were incubated with secondary anti-mouse IgG (Jackson ImmunoResearch, 115-605-164) diluted 1:200 for 30 min at 4 °C. Subsequent processing was carried out as described above.

## 4.12 ZMB Products Used

The Zentrallabor für Medienherstellung und Biomonitoring (ZMB) service platform is available to all PEI employees in regulation and research. In addition to the provision and production of media, culture media, buffers and solutions, the services also include biomonitoring of water systems and autoclaves. All products provided by the ZMB and used in this thesis are listed in the following table.

**Table 4: ZMB products.**

| Order No. | Short description | Long description |
|---|---|---|
| ZMBAP2 | LB + Amp | LB-Agar + Ampicillin |
| ZMBNB006 | LB pH 7.0 | LB-medium pH 7.0 |
| ZMBNB021 | 2xYT | 2xYT-mediume |
| ZMBLL005 | Ammonium chloride 0.86 % | Ammonium chloride 0.86 % |
| ZMBLL019 | EDTA 0.5 M pH 8.0 | EDTA 0.5 M pH 8.0 |
| ZMBLL023 | HBS pH 7.4 | HBS buffer pH 7.4 |
| ZMBLL038 | Sodium azide | Sodium azide 10 % |
| ZMBLL045 | NaCl 5 M | NaCl 5 M |
| ZMBLL052 | PBS pH 7.1 | PBS w/o $Ca^{2+}$ $Mg^{2+}$ pH 7.1 |
| ZMBLL057 | PBS pH 7.1 + 1 mM EDTA | PBS w/o $Ca^{2+}$ $Mg^{2+}$ pH 7.1 + 1 mM EDTA |
| ZMBLL059 | PBS pH 7.1 + 0.05% Tween | PBS w/o $Ca^{2+}$ $Mg^{2+}$ pH 7.1 + 0.05 % Tween 20 |
| ZMBLL072 | SDS 10 % | SDS 10 % |
| ZMBLL075 | SP | Streptomycin/Penicillin solution |
| ZMBLL083 | TBS pH 7.4 | TBS-buffer 1x pH 7.4 |
| ZMBLL085 | TBS pH 7.4 + 0.1% Tween | TBS-buffer pH 7.4 + 0.1% Tween 20 |
| ZMBLL095 | Tris-HCl 1 M pH 6.8 | Tris-HCl 1 M pH 6.8 |
| ZMBLL097 | Tris-HCl 1 M pH 8 | Tris-HCl 1 M pH 8 |
| ZMBLL098 | Tris-HCl 1 M pH 8.8 | Tris-HCl 1 M pH 8.8 |
| ZMBLL104 | Trypsin 2.5 % | Trypsin 2.5 % |
| ZMBLL135 | AAV Lysis | AAV lysis buffer |

## 4.13 Primers

All primers were ordered at Merck/Sigma-Aldrich, HPLC purified and dissolved in water at a concentration of 100 µM.

**Table 5: Primers.**

| Primer No. | Sequence (5' to 3') |
|---|---|
| 1 | CGGAATTCCACTGCGTGAGCGGCCGCTGCTGCTTTCTGCAGCACTTC GGCAATATCTTCATTGCCATTATCAATGGCCAGATCAAACGGTG |
| 2 | CTTTTGCAGCACTTCAGCAATG |
| 3 | AAATTTGGCCCAGCCGGCCGACCTGGGTAAAAAAC |
| 4 | CCGGAATTCCACTGCGTGAGCGGCCGCTGCTGCTTTCTGCAGCACTT CGGCAATATC |
| 5 | ATCTTATGGCCCAGCCGGCCTAAGGTGAAGCTGGTGGAATCT |
| 6 | AAATTTGGCCCAGCCGGCCAAGGACCTGGGTAAGAAACTG |
| 7 | ATACGAAATTAATACGACTCACTATAGGGAGACCACAACGG |
| 8 | CCGCACACCAGTAAGGTGTGCGGTTTCAGTTGCCGCTTTCTTTCT |
| 9 | TAATGCTCGAGTCACACTGCGTGAGC |
| 10 | CGATCTAGAAAAGAGGAGAAATTAACTATGAGAGGATC |
| 11 | TAAGGATCCACCGGTCGCCA |
| 12 | CGCGGCCGCCGTACGAGGTGGGACGTTCACGATC |
| 13 | CTCTGGAGGTGGTGGATCAG |
| 14 | GCATACTAGTCGCACCGCCACCGCCCGCTGCTGCTTTCTGC |
| 15 | CATTGGCCAAGCTGGCCGGTATGGACCTGGGTAAG |
| 16 | GAAGACTAGTCGCACCGCCACCGCCATTAAGCTTTTGCAG |
| 17 | GAAATTTAAATCAGGTAAGGCTGCC |
| 18 | CAATGCCAATTTCCTGAGG |
| 19 | ATAAATGGCCCAGCCGGCC |
| 20 | TAGCTAGCGCCACCGCCGCTACCGCCACCGCCCGAACCGCCACCGC CCGCTGCTGCTTTCTGCAGC |
| 21 | AGAAAACTGAAGACTAGTCGCACCGCCAC |
| 22 | ATAAGCTAGCGACCTGGGTAAAAAACTGCTGGAAGC |
| 23 | GGAACCCCTAGTGATGGAGTT |
| 24 | CGGCCTCAGTGAGCGA |
| 25 | 6-FAM*-CACTCCCTCTCTGCGCGCTCG |

* 6-carboxyfluorescein

## 4.14 Flow Cytometry Antibodies

**Table 6: Antibodies used for flow-cytometry.**

| Figure | Antibody | Manufacturer, ID | Dilution |
|---|---|---|---|
| 19; 25; 27; 28 | anti-CD8-VioBlue | Mitlenyi Biotec, 130-110-683 | 1:200 |
| 17; 18 | anti-CD8-APC | Mitlenyi Biotec, 130-113-154 | 1:200 |
| 17; 18; 19; 23; 25; 27 | anti-CD3-PerCP | Mitlenyi Biotec, 130-113-131 | 1:200 |
| 28 | anti-CD3-PE-Vio770 | Mitlenyi Biotec. 130-113-130 | 1:200 |
| 28 | anti-CD45-APC | Mitlenyi Biotec. 130-113-114 | 1:200 |
| 18; 25; 27 | anti-CD4-PE-Vio770 | Mitlenyi Biotec, 130-113-216 | 1:200 |
| 17; 18 | anti-CD4-VioBlue | Mitlenyi Biotec, 130-113-219 | 1.200 |
| 14; 23; 24 | anti-CD56-PE | Miltenyi Biotec, 130-113-307 | 1:200 |

## 4.15 Analysis, Visualization and Writing

### 4.15.1 Illustrations

Illustrations like workflows, literature summaries and other visitations are mainly created with BioRender.com.

### 4.15.2 Writing

An AI language model (ChatGPT-4 by OpenAI) was used for assistance with language clarity, grammar, and structure. All intellectual content, conclusions, and interpretations are entirely my own.

### 4.15.3 DARPin Sequence Alignment

DARPin sequences were aligned utilizing Clustal Omega [176].

### 4.15.4 Protein and AAV Structure Visualizations

Capsid structures were approximated using the VP3 PDB structures 1LP3 for AAV2 and 3OAH for AAV6, UCSF Chimera 1.13.1. and AlphaFold 3 [177]. AlphaFold 3 predicted the structure of VP3 carrying the CD8.63A4 or (2x)-CD56.43H3 DARPin. *In silico* folded proteins were aligned with VP3 of an unmodified electron density AAV2/6 capsid structure and inserted into five random positions of the AAV particle using UCSF ChimeraX-1.9.

For the structural predictions and docking simulations of 2x43H3 and the CD56 ectodomain, the entire huCD56 protein (UniProt P13591) was modeled, along with the 2x43H3 DARPin,

both with and without its VP1 fusion. These models were generated and analyzed *in silico* using AlphaFold3. To focus on the interaction between the DARPin and the extracellular portion of huCD56, the transmembrane domain and cytoplasmic tail of huCD56 were omitted from visual representations, improving clarity and simplifying the analysis.

### 4.15.5 Statistics and Graphing

All Statistical analyses were carried out in GraphPad Prism version 9.5.0. Statistical tests were performed as indicated and flow cytometry data were assumed to be normally distributed.

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# 6 Abbreviations

| | |
|---|---|
| ® | Registered trademark |
| °C | Degree Celsius |
| µg | Micro gram |
| µL | Micro litre |
| µm | Micro meter |
| AAV | Adeno-Associated Viral vector |
| AAVR | Adeno-Associated Virus receptor |
| ADA | Adenosine deaminase |
| AI | Artificial Intelligence |
| Amp | Ampicillin |
| APC | Allophycocyanin |
| ARSA | Arylsulfatase A |
| ATCC | American Type Culture Collection |
| ATMP | Advanced Therapy Medicinal Product |
| BBB | Blood-Brain Barrier |
| BCMA | B-Cell Maturation Antigen |
| BirA | Biotin ligase |
| BSA | Bovine Serum Albumin |
| CAR | Chimeric Antigen Receptor |
| CD | Cluster of Differentiation |
| cDNA | Complementary DNA |
| CI | Confidence interval |
| CLIC | Clathrin-Independent Carrier |
| CNS | Central Nervous System |
| CRISPR | Clustered Regularly Interspaced Short Palindromic Repeats |
| CRP | C-Reactive Protein |
| Cryo-EM | Cryogenic Electron Microscopy |
| DARPin | Designed Ankyrin Repeat Protein |
| DART-AAV | Designed Ankyrin Repeat Protein-Targeted Adeno-Associated Viral vector |
| DDC | DOPA-Decarboxylase |
| DKO | Double knock-out |
| DMEM | Dulbecco's modified Eagle's medium |
| DMSO | Dimethyl sulfoxide |
| DNA | Deoxyribonucleic Acid |
| dpt | Days post transduction |
| DRK | Deutsches Rotes Kreuz |
| DTT | Dithiothreitol |
| *E. coli* | *Escherichia coli* |
| ECD | Ectodomain |
| ECL | Enhanced Chemiluminescence |
| EDTA | Ethylenediaminetetraacetic acid |
| ELISA | Enzyme-linked Immunosorbent Assay |
| EMA | European Medicines Agency |
| ER | Endoplasmic Reticulum |
| ESI | Ernst Strüngmann Institut |
| FBS | Fetal bovine serum |
| Fc | Fragment crystallizable |
| FcR | Fragment crystallisable Receptor |
| FIX | Factor IX |
| fmol | Femtomole |
| FVIII | Factor VIII |

| | |
|---|---|
| *g* | gravitational force equivalent |
| g | gram |
| G3BP | Galectin 3 Binding Protein |
| GEEC | Glycosylphosphatidylinositol -anchored protein-enriched early endosomal compartment |
| GFP | Green Fluorescent Protein |
| GM-CSF | Granulocyte-macrophage colony-stimulating factor |
| GPI | Glycosylphosphatidylinositol |
| GTMP | Gene Therapy Medicinal Product |
| GZMB | Granzyme B |
| h | hour |
| HA | hemagglutinin |
| HBS | 4-(2-hydroxyethyl)-1-piperazineethanesulfonic acid buffered saline |
| HIV | Human Immunodeficiency Virus |
| HPLC | High-Performance Liquid Chromatography |
| HRP | Horseradish Peroxidase |
| hRPE65 | human retinal pigment epithelium-specific 65-kDa protein |
| HSC | Hematopoietic Stem Cells |
| HSV | Herpes Simplex Virus |
| hu | human |
| HVR | Hypervariable Regions |
| i.p. | intraperitoneally |
| IgG | Immunoglobulin G |
| IKKε | IkappaB kinase-epsilon |
| IL | Interleukin |
| IPTG | Isopropyl β-D-1-thiogalactopyranoside |
| ITR | Inverted Terminal Repeat |
| IV | Intravenously |
| $K_D$ | dissociation constant |
| kDa | Kilodalton |
| L | Litre |
| LB | Lysogeny Broth |
| LNP | Lipid Nanoparticle |
| m | murine |
| M | Molar |
| MHC | Major Histocompatibility Complex |
| min | Minute(s) |
| mL | Millilitre |
| mM | Millimolar |
| MOI | Multiplicity of Infection |
| MWCO | Molecular Weight Cut-Off |
| N | Equivalent concentration/Normality |
| NCBI | National Center for Biotechnology Information |
| NCAM1 | Neural Cell Adhesion Molecule 1 |
| NCR1 | Natural cytotoxicity triggering receptor 1 |
| NEB | New England Biolabs |
| NK cell | Natural killer cell |
| nm | nanometre |
| nM | nanomolar |

| | |
|---|---|
| No. | Number |
| NSG | Non-obese diabetic Scid Gamma |
| NTA | Nitrilotriacetic acid |
| O/N | Overnight |
| $OD_{600}$ | Optical density 600 nanometres |
| p | Plasmid |
| PAGE | Polyacrylamide Gel Electrophoresis |
| PBMC | Peripheral Blood Mononuclear Cell |
| PBS | Phosphate Buffered Saline |
| PBS-T | Phosphate Buffered Saline-Tween 20 |
| PCR | Polymerase Chain Reaction |
| PE | Phycoerythrin |
| PEI | Paul-Ehrlich-Institut |
| PerCP | Peridinin-Chlorophyll-Protein |
| pH | potentia Hydrogenii |
| PLA2 | Phospholipase A2 |
| pmol | picomole |
| PRF1 | Perforin 1 |
| qPCR | quantitative Polymerase Chain Reaction |
| RNA | Ribonucleic Acid |
| rpm | Revolutions per minute |
| RT | Room Temperate |
| RU | Response Unit |
| s | soluble |
| SCID | Severe Combined Immunodeficiency |
| SD | Standard Deviation |
| SDS | Sodium Dodecyl Sulfate |
| SEC | Size-Exclusion Chromatography |
| SFFV | Spleen Focus-Forming Virus |
| SMN | Survival of Motor Neuron |
| SPR | Surface Plasmon Resonance |
| TBK1 | TANK-binding kinase 1 |
| TBS-T | Tris(hydroxymethyl)aminomethane-buffered saline – Tween 20 |
| TCR | T Cell Receptor |
| TEV | Tobacco Etch Virus |
| TMB | Tetramethylbenzidin |
| TNE | Tris(hydroxymethyl)aminomethane NaCl Ethylenediaminetetraacetic acid |
| TRAC | T cell receptor alpha constant |
| Tregs | regulatory T cells |
| Tris | Tris(hydroxymethyl)aminomethane |
| ™ | Trademark |
| U | Units |
| vg | Viral genomes |
| VP | Viral capsid protein |
| VSV | Vesicular stomatitis virus |
| w/o | without |
| ZMB | Zentrallabor für Medienherstellung und Biomonitoring |

# 7 List of Figures and Tables

## 7.1 Figures

Figure 1: Schematic representation of the haematopoietic system highlighting key T cell subtypes and their surface markers.

Figure 2: Schematic representation of the AAV transduction process.

Figure 3: CD56 purity and biotinylation.

Figure 4: Determination of CD56 protein amount needed for panning and antibody binding.

Figure 5: Ribosome display.

Figure 6: Crude lysate screening after ribosome display.

Figure 7: DARPin sequences and structures.

Figure 8: DARPin purification.

Figure 9: Cellular binding of purified DARPins and surface plasmon resonance analysis.

Figure 10: Structural prediction of DART-AAVs and confirmation of DARPin integration.

Figure 11:

Figure 12: Transduction efficiency of CD8-AAV2 compared to AAV2mut and AAV2.

Figure 13: Transduction efficiency of CD4-AAV2 compared to AAV2mut and AAV2.

Figure 14: CD56-AAVs selectively transduce CD56+ cells.

Figure 15: CD8-AAV6 structure and assembly.

Figure 16: Gene transfer activity of CD8-AAV6 on cell lines.

Figure 17: Selective gene transfer of CD8-AAV2 into human T cells.

Figure 18: Selective gene transfer by CD4-AAV2 into human T cells.

Figure 19: Selective gene transfer of CD8-AAV6 into human T cells.

Figure 20: NKp46-AAV screening.

Figure 21: Characterization of CD56-NKp46-AAVs.

Figure 22: Increased transduction of CD56+ cells with 2xCD56-AAVs.

Figure 23: Isolation and transduction of human NK cells.

Figure 24: AlphaFold3-based binding prediction of the 2x43H3 DARPin interaction with the ectodomain of human CD56.

Figure 25: Determination of the binding domain of the 43H3 DARPin.

Figure 26: Transduction of CD8+ T cells in whole blood.

Figure 27: Influence of human serum on gene transfer of CD8-AAV6 and unmodified AAV6.

Figure 28: *In vivo* gene transfer into activated human CD8+ T cells.

Figure 29: *In vivo* gene transfer into non-activated human CD8+ T cells. **(**

Figure 30: Possible Binding Modes of 2xCD56-AAV to CD56.

## 7.2 Tables

Table 1: Market authorized GTMPs in the EU.

Table 2: Tested binding models and values of SPR measurement.

Table 3: Predicted interactions between 43H3 DARPin and CD56.

Table 4: ZMB products.

Table 5: Primers.

Table 6: Antibodies used for flow-cytometry.

# 11 Declaration of Collaboration

Unless otherwise indicated by reference or acknowledgment, the work presented in this thesis was conducted by me or under my direct supervision. All contributions from colleagues are explicitly cited within the thesis. The following data was generated as part of collaborative research efforts.

Figure 3: CD56 purity and biotinylation.

| | |
|---|---|
| Collaboration partners: | Jessica Hartmann and Julia Brynza, Paul-Ehrlich-Institut |
| Contribution: | Protein production and western blot analysis |
| My contribution: | Protein purification and biotinylation analysis |

Figure 6: Crude lysate screening after ribosome display.

| | |
|---|---|
| Collaboration partner: | Lea Knapp, Paul-Ehrlich-Institut |
| Contribution: | Crude lysate screening and cellular binding assay of 43E9, 43E10 and 43H3. |
| My contribution: | Providing the source library and protocols. |

Figure 7: DARPin sequences and structures.

| | |
|---|---|
| Collaboration partner: | Lea Knapp, Paul-Ehrlich-Institut |
| Contribution: | Alphafold analysis and sequencing of 43E9, 43E10 and 43H3. |
| My contribution: | Visualization and correction. |

Figure 9: Cellular binding of purified DARPins and surface plasmon resonance analysis.

| | |
|---|---|
| Collaboration partner: | Sascha Hein, Paul-Ehrlich-Institut |
| Contribution: | SPR Analysis |
| My contribution: | Sample preparation and visualization. |

Figure 11: Cryogenic electron microscopy of CD8-AAV2.

| | |
|---|---|
| Collaboration partner: | Petr Chlanda, University of Heidelberg |
| Contribution: | Cryo-electron microscopy analysis of AAVs. |
| My contribution: | Providing AAV stocks and visualization of the results. |

Figure 17: Selective gene transfer of CD8-AAV2 into human T cells.

| | |
|---|---|
| Collaboration partners: | Muhammed Burak Demircan and Fabian John, Paul-Ehrlich-Institut |
| Contribution: | Production of CD8-AAV$^{SL}$ and titration of AAVs on activated PBMC |
| My contribution: | Providing CD8-AAV$^{IL}$ stocks and initial cloning plasmids |

Figure 18: Selective gene transfer by CD4-AAV2 into human T cells.

| | |
|---|---|
| Collaboration partners: | Muhammed Burak Demircan and Fabian John, Paul-Ehrlich-Institut |
| Contribution: | Production of CD8-AAV$^{SL}$ and titration of AAVs on activated PBMC |
| My contribution: | Providing CD8-AAV$^{IL}$ stocks and initial cloning plasmids |

Figure 20: NKp46-AAV screening.

| | |
|---|---|
| Collaboration partners: | Jessica Hartmann and Julia Brynza, Paul-Ehrlich-Institut |
| Contribution: | Cloning and AAV titration. |
| My contribution: | Analysis and visualization. |

Figure 29: *In vivo* gene transfer into non-activated human CD8+ T cells.

| | |
|---|---|
| Collaboration partners: | Muhammed Burak Demircan and Johanna Gorol, Paul-Ehrlich-Institut |
| Contribution: | Planning and conducting animal experiment |
| My contribution: | Providing AVV stocks, analysis and visualization |

Some figures shown in the result are modified from *Demircan and Zinser et al.* [96]. Each figure legend indicates the publication reference.

Frankfurt am Main, .....(Date)…….(Signature)

# 12 Declaration of an Oath

D E C L A R A T I O N

I herewith declare that I have not previously participated in any doctoral examination procedure in a mathematics or natural science discipline.

Frankfurt am Main, .....(Date)……(Signature)

Author's Declaration

I herewith declare that I have produced my doctoral dissertation on the topic of

**Engineering AAV Tropism:**

**DARPin-Mediated Retargeting Enables Specific Gene Delivery to Lymphocyte Subtypes**

independently and using only the tools indicated therein. In particular, all references borrowed from external sources are clearly acknowledged and identified.

I confirm that I have respected the principles of good scientific practice and have not made use of the services of any commercial agency in respect of my doctorate.

Frankfurt am Main, .....(Date)……(Signature)